# 18.786: Automorphic Forms

Nir Elber

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# Contents

*How strange to actually have to see the path of your journey in order to make it.*

—Neal Shusterman, [Shu16]


**Contents** 2

**1 Tate's Thesis** 5
- 1.1 February 2 . . . 5
  - 1.1.1 Logistic Notes . . . 5
  - 1.1.2 Places of Global Fields . . . 5
  - 1.1.3 Adéles . . . 7
  - 1.1.4 Characters on the Adéles . . . 8
- 1.2 February 4 . . . 9
  - 1.2.1 Adelic Quotients . . . 9
  - 1.2.2 Idelic Quotients . . . 11
  - 1.2.3 Pontryagin Duality . . . 12
  - 1.2.4 Fourier Theory . . . 14
- 1.3 February 9 . . . 15
  - 1.3.1 Duality for Local Fields . . . 15
  - 1.3.2 Duality for the Adéles . . . 17
  - 1.3.3 Duality for Function Fields . . . 19
- 1.4 February 11 . . . 20
  - 1.4.1 Multiplicative Measures . . . 20
  - 1.4.2 Our $L$-functions . . . 22
  - 1.4.3 Functional Equations . . . 23
- 1.5 February 17 . . . 25
  - 1.5.1 More General $L$-functions . . . 25
  - 1.5.2 Proof of the Global Functional Equation . . . 26
  - 1.5.3 A Little Geometric Class Field Theory . . . 27
  - 1.5.4 Local Theory . . . 28

**2 Nonarchimedean Representation Theory** 30
- 2.1 February 18 . . . 30
  - 2.1.1 Overview . . . 30

2.1.2 Totally Disconnected Groups . . . 30
2.1.3 Smooth Representations . . . 31
2.1.4 Categorical Properties . . . 32
2.1.5 Hecke Algebra . . . 34
2.2 February 25 . . . 35
2.2.1 Representations of the Hecke Algebra . . . 35
2.2.2 Parabolic Induction . . . 37
2.3 March 2 . . . 39
2.3.1 The Modular Character . . . 39
2.3.2 Back to Induction . . . 41
2.3.3 Spherical Representations . . . 42
2.4 March 4 . . . 43
2.4.1 The Satake Isomorphism . . . 43
2.5 March 9 . . . 45
2.5.1 The Bruhat–Tits Tree . . . 45
2.5.2 The Spherical Hecke Algebra . . . 46
2.5.3 The Satake Isomorphism, Again . . . 47
2.6 March 11 . . . 48
2.6.1 The Langlands Dual . . . 49

**3 Archimedean Representation Theory** **51**
3.1 March 11 . . . 51
3.1.1 Real Lie Groups . . . 51
3.1.2 Smooth Representations . . . 52
3.2 March 16 . . . 54
3.2.1 $(\mathfrak{g}, K)$-Modules . . . 54
3.2.2 The Harish-Chandra Isomorphism . . . 56
3.3 March 18 . . . 57
3.3.1 Admissible Representations . . . 57
3.3.2 The Infinitesimal Character . . . 58
3.3.3 The Discrete Series . . . 59

**4 Automorphic Forms** **61**
4.1 March 30 . . . 61
4.1.1 Modular Forms . . . 61
4.1.2 Double Quotients . . . 62
4.1.3 Automorphic Forms . . . 63
4.2 April 1 . . . 65
4.2.1 Back to Modular Forms . . . 65
4.2.2 Suddenly Shimura Varieties . . . 66
4.2.3 Siegel Domains . . . 67
4.3 April 6 . . . 68
4.3.1 Cusp Forms . . . 68
4.3.2 Automorphic Forms for Function Fields . . . 70
4.4 April 13 . . . 72
4.4.1 The Central Character . . . 72
4.4.2 Cusp Forms for Function Fields . . . 74
4.4.3 Reduction Theory for Function Fields . . . 75
4.4.4 The Cuspidal Subspace . . . 76
4.5 April 15 . . . 76
4.5.1 Some Hecke Algebras . . . 77
4.5.2 The Tensor Product Theorem . . . 78
4.6 April 22 . . . 80
4.6.1 Hecke Operators . . . 80

4.6.2 Hecke Operators on Modular Forms . . . . . 82

**Bibliography** **85**

**List of Definitions** **86**

# Theme 1
# Tate's Thesis

## 1.1 February 2

Here we go.

### 1.1.1 Logistic Notes

Here are some logistic notes.

- The book *Automorphic Forms and Representations* by Bump [Bum97] will be our main reference. We will focus on the third chapter.
- There are two quizzes, which will count as about 30% of the final grade. The rest of the grade will come from the homework.
- The course requires knowledge of number theory and some representation theory. Most notably, we need control of Lie groups and Lie algebras.

The story of automorphic forms begins with modular forms. Roughly speaking, a modular form is a function on the upper-half plane which is symmetric for $\mathrm{SL}_2(\mathbb{C})$. There is an exposition in the last chapter of [Ser12]. However, our story will start with $\mathrm{GL}_1$ instead of $\mathrm{GL}_2$. The perspective we take is Tate's thesis, who used Fourier analysis to reprove the analytic properties of the relevant (automorphic) $L$-functions.

### 1.1.2 Places of Global Fields

To do our Fourier analysis, we need to decompose our number field at each place, for which we will need the ring of adéles.

**Definition 1.1** (global field)**.** A *global field* $F$ is either a number field or a function field. Here, a number field is a finite extension of $\mathbb{Q}$, and a function field refers to the function field of a smooth, projective, geometrically connected curve $X$ over a finite field $\mathbb{F}_q$.

**Example 1.2.** The field $\mathbb{F}_q(t)$ is a global field.

**Definition 1.3** (place)**.** Fix a global field $F$. Then a *place* is an equivalence class of multiplicative absolute values of $F$.

- If $F$ is a number field, then the *finite* (or *nonarchimedean*) places are those in bijection with $\mathcal{O}_F$, and the *infinite places* (or *archimedean*) are those in bijection with the embeddings $F \hookrightarrow \mathbb{C}$ (up to conjugation).
- If $F$ is the function field of a curve $X$, then the places are in bijection with the closed points of the curve $X$.

We let $V(F)$ be the set of all places, and we let $V(F)_\infty$ denote the set of infinite places.

**Example 1.4.** For $F = \mathbb{Q}$, the place at a finite prime $p$ is represented by $|q|_p := p^{-\nu_p(q)}$, where $\nu_p(q)$ is the number of $p$s appearing in the prime factorization of $q$.

**Example 1.5.** For $F = \mathbb{F}_q(t)$, we have the curve $X = \mathbb{P}^1$, so there is one place at infinity, and the rest of the points come from $\mathbb{A}^1$. The places in $\mathbb{A}^1$ are parameterized by the monic irreducible polynomials of $\mathbb{F}_q[t]$.

**Notation 1.6.** Fix a global field $F$. For each place $v$, we let $F_v$ be the completion of $F$ along a norm represented by $v$. We let $\mathcal{O}_v$ denote the elements with norm at most $1$; we let $\mathcal{O}_v$ denote the elements with norm $1$, and we let $\mathfrak{p}_v$ denote the elements with norm less than !.

**Remark 1.7.** If $v$ is nonarchimedean, then it turns out that $\mathcal{O}_v$ is a discrete valuation ring with maximal ideal $\mathfrak{p}_v$. It also turns out that there is an exact sequence

$$1 \to \mathcal{O}_v^\times \to F_v^\times \to \mathbb{Z} \to 0,$$

where the map $F_v^\times \to \mathbb{Z}$ is the valuation map.

It is helpful to normalize our absolute values. Let's start with the global fields.

**Notation 1.8.** Fix a place $v$ of a global field $F$. We normalize a choice of absolute value $|\cdot|_v$ as follows.

- For $F = \mathbb{Q}$, each prime $p$ produces the absolute value $|q|_p := p^{-\nu_p(q)}$. The infinite place $\infty$ produces the absolute value $|x|_\infty$ which is the usual one (in $\mathbb{R}$).
- For a finite extension $F$ of $\mathbb{Q}$, say that $v$ lies over $v_0$ of $\mathbb{Q}$, and we define

$$|x|_v := \left|\mathrm{N}_{F_v/\mathbb{Q}_{v_0}}(x)\right|_{v_0}.$$

**Example 1.9.** For $F_v = \mathbb{C}$, we see that $|x|_v = |x\overline{x}|_\mathbb{R}$ is the square of the usual absolute value on $\mathbb{C}$. Note that this norm does not obey the triangle inequality.

For a function field $\mathbb{F}_q(X)$, there is not a canonical embedding $\mathbb{F}_q(t)$ into $\mathbb{F}_q(X)$, so it does not seem suitable to proceed as above by taking norms. Instead, we normalize directly.

**Notation 1.10.** Fix a function field $F$ of a smooth, projective, geometrically connected curve $X$ over $\mathbb{F}_q$, and choose a place $v \in X$. Then the completion $F_v$ is isomorphic to $k_v((t))$, where $t \in \mathcal{O}_v$ is a choice of uniformizer and $k_v/\mathbb{F}_q$ is a finite extension. Then we normalize our norm $|\cdot|_v$ by $|t|_v := (\#k_v)^{-1}$.

These choices of normalization obey a product formula.

**Proposition 1.11.** Fix a global field $F$. For each $x \in F$,

$$\prod_{v \in V(F)} |x|_v = 1.$$

*Sketch.* This is included in a standard first course in number theory, so we will be brief. For number fields, this is checked directly by passing to $\mathbb{Q}$, where it is a consequence of unique prime factorization. For function fields $\mathbb{F}_q(X)$, we may think of $f \in \mathbb{F}_q(X)$ as a rational function $X$, and $|f|_v = q^{\deg(v) \cdot - \operatorname{ord}_v(f)}$, where $\operatorname{ord}_v$ is the order of vanishing. Thus, the product formula more or less amounts to the statement that the sum of the zeroes and poles of $f$ all cancel out (over the algebraic closure). ■

### 1.1.3 Adéles

We now define the adéles by gluing together our localizations.

**Definition 1.12** (adéles)**.** Fix a global field $F$. Then the ring of *adéles* $\mathbb{A}_F$ is defined as the restricted product

$$\mathbb{A}_F := \prod_{v \in V(F)} (F_v, \mathcal{O}_v),$$

meaning that $\mathbb{A}_F$ consists of sequences of elements in $F_v$ which are in $\mathcal{O}_v$ for all but finitely many $v$.

**Remark 1.13.** By construction, we see that

$$\mathbb{A}_F = \bigcup_{\substack{\text{finite } S \subseteq V(F) \\ S \supseteq V(F)_\infty}} \left( \prod_{v \notin S} \mathcal{O}_v \times \prod_{v \in S} F_v \right).$$

Thus, $\mathbb{A}_F$ is a colimit of (product) topological rings, so $\mathbb{A}_F$ is a topological ring.

**Remark 1.14.** A basis neighborhood basis of $0 \in \mathbb{A}_F$ is given as follows: for any choice of finite $S \subseteq V(F)$ containing $V(F)_\infty$, choose open neighborhoods $U_v \subseteq \mathcal{O}_v$ of $0$, and then we have the open subset

$$\prod_{v \notin S} \mathcal{O}_v \times \prod_{v \in S} U_v.$$

One can further require that the open subsets $U_v$ take the form $\mathfrak{p}_v^{m_v}$, where $m_v$ is some integer.

Tate's thesis is about $\mathrm{GL}_1(\mathbb{A}_F)$, so the following group will be important to us.

**Definition 1.15.** Fix a global field $F$. Then the group of *idéles* $\mathbb{A}_F^\times$ is defined as the restricted product

$$\mathbb{A}_F^\times := \prod_{v \in V(F)} (F_v^\times, \mathcal{O}_v^\times),$$

meaning that $\mathbb{A}_F^\times$ consists of sequences of elements in $F_v^\times$ which are in $\mathcal{O}_v^\times$ for all but finitely many $v$.

Notably, $\mathrm{GL}_1(\mathbb{A}_F) = \mathbb{A}_F^\times$.

**Remark 1.16.** This is not the set of nonzero elements in $\mathbb{A}_F$ because we require the inverse to also be an adéle!

**Remark 1.17.** One should not give the subset $\mathbb{A}_F^\times \subseteq \mathbb{A}_F$ the subspace topology. Instead, the topology should be given by the restricted product, whose open subsets can be smaller. Thus, an element of the neighborhood basis of $1 \in \mathbb{A}_F^\times$ can be described as follows: for any choice of finite $S \subseteq V(F)$ containing $V(F)_\infty$, choose open neighborhoods $U_v \subseteq \mathcal{O}_v$ of $0$, and then we have the open subset

$$\prod_{v \notin S} \mathcal{O}_v^\times \times \prod_{v \in S} U_v.$$

One can further require that the open subsets $U_v$ take the form $\mathfrak{p}_v^{m_v}$, where $m_v$ is some integer.

Later in the course, we will even want to study groups like $\mathrm{GL}_2(\mathbb{A}_F)$ or $\mathrm{GL}_n(\mathbb{A}_F)$. Let's be explicit about what this notation means.

**Definition 1.18** (general linear group)**.** Fix a ring $R$. Then we define $\mathrm{GL}_n(R)$ to be the group of invertible $n \times n$ matrices. Explicitly, this can be described as the group of $n \times n$ matrices with entries in $R$ whose determinant is invertible.

**Remark 1.19.** Fix a global field $F$. One can check that

$$\mathrm{GL}_n(\mathbb{A}_F) = \prod_{v \in V(F)} (\mathrm{GL}_n(F_v), \mathrm{GL}_n(\mathcal{O}_v)),$$

which also tells us what the topology should be.

**Remark 1.20.** Here is another way to construct the topology: $\mathrm{GL}_n$ can embed (as a scheme) as a closed subspace of $(n^2+1)$-dimensional space $A$, where the embedding sends $g \in \mathrm{GL}_n(R)$ to the tuple of coordinates follows by the inverse of the determinant. This is a closed embedding, essentially by definition of $\mathrm{GL}_n$. Then we can give $\mathrm{GL}_n(\mathbb{A}_F)$ the natural topology given as a closed subspace of $A(\mathbb{A}_F)$. It is not too hard (but rather annoying) to check that these definitions agree.

**Example 1.21.** The determinant map $\det\colon \mathrm{GL}_n(\mathbb{A}_F) \to \mathbb{A}_F^\times$ is continuous. One can see this via Remark 1.20 because the determinant and its inverse are both continuous maps to $\mathbb{A}_F$. But the topology on $\mathbb{A}_F^\times$ is given as a closed subspace of $\mathbb{A}_F \times \mathbb{A}_F$ (where the embedding is given by $x \mapsto (x, 1/x)$).

This course is interested in the representation theory of $\mathrm{GL}_n(\mathbb{A}_F)$, focusing on the cases $n \in \{1, 2\}$. If we think about such representations appropriately, it turns out that such a representation $\pi$ will decompose into a tensor product $\bigotimes_v' \pi_v$, where $\pi_v$ is a representation of $\mathrm{GL}_n(F_v)$. More than half of the course will thus be interested in the representation theory of $\mathrm{GL}_n(F_v)$ because we will want to study the finite and infinite places separately.

### 1.1.4 Characters on the Adéles

We will need more structure theory of the adéles.

**Proposition 1.22.** Fix a global field $F$. The diagonal embedding $F \hookrightarrow \mathbb{A}_F$ embeds $F$ as a discrete subgroup.

*Proof.* Fix distinct $a, b \in F$. By examining the open subsets we have access to, we need to show that $|a-b|_v \geq 1$ for some $v$, which follows from Proposition 1.11. ■

**Corollary 1.23.** Fix a global field $F$. The diagonal embedding $F^\times \hookrightarrow \mathbb{A}_F^\times$ embeds $F$ as a discrete subgroup.

*Proof.* For each $a \in F^\times$, we need to know that there is an open subset $U$ of $\mathbb{A}_F^\times$ for which $U \cap F^\times = \{a\}$. But there is such an open subset of $\mathbb{A}_F$, which continues to be open in $\mathbb{A}_F^\times$. ■

We will be interested in characters on $\mathbb{A}_F^\times$.

**Notation 1.24.** Fix a place $v$ of a global field $F$. Given a continuous character $\chi\colon \mathbb{A}_F^\times \to \mathbb{C}^\times$, we let $\chi_v\colon F_v^\times \to \mathbb{C}^\times$ denote the induced character.

**Remark 1.25.** The continuity of $\chi$ forces $\chi_v|_{\mathcal{O}_v^\times} = 1$ for all but finitely many $v$: indeed, $\prod_{v \nmid \infty} \mathcal{O}_v^\times$ is a profinite group, so a "no small subgroups" argument shows that $\ker \chi$ must contain an open subgroup of $\prod_{v \nmid \infty} \mathcal{O}_v^\times$. Conversely, given a family $\{\chi_v\}_{v \in V(F)}$ of continuous characters for which $\chi_v|_{\mathcal{O}_v^\times} = 1$ for all but finitely many $v$, the product character $\chi := \prod_v \chi_v$ on $\mathbb{A}_F^\times$ makes sense, and it is continuous because the kernel contains an open subgroup of $\prod_{v \nmid \infty} \mathcal{O}_v^\times$ (and hence factors through a finite quotient of that group).

The previous remark motivates the following definition.

**Definition 1.26** (unramified)**.** Fix a place $v$ of a global field $F$. Then a character $\chi_v\colon F_v^\times \to \mathbb{C}^\times$ is *unramified* if and only if $\chi_v|_{\mathcal{O}_v^\times} = 1$.

**Example 1.27.** By definition, $\chi_v$ factors through $F_v^\times / \mathcal{O}_v^\times \cong \mathbb{Z}$. Thus, $\chi_v$ can be described as $\chi_v = |\cdot|_v^s$ for some $s \in \mathbb{C}$.

Not all characters are interesting to us because we want our characters $\chi_v$ to talk to each other.

**Definition 1.28** (Hecke character)**.** Fix a global field $F$. A *Hecke character* is a continuous character $\chi\colon \mathbb{A}_F^\times \to \mathbb{C}^\times$ which vanishes on $F^\times$.

**Remark 1.29.** It is equivalent to ask for $\chi$ to be continuous on $F^\times \backslash \mathbb{A}_F^\times$ by Corollary 1.23.

## 1.2 February 4

Here we go.

### 1.2.1 Adelic Quotients

Thus, it will be worthwhile to know something about the quotient $F^\times \backslash \mathbb{A}_F^\times$. Let's start with the additive group.

**Theorem 1.30** (approximation)**.** Fix a number field $F$. Then

$$\mathbb{A}_F = F + \prod_{v \notin V(F)_\infty} \mathcal{O}_v + \prod_{v \in V(F)_\infty} F_v.$$

*Proof.* Given an adéle $(a_v)_v \in \mathbb{A}_F$, we see that we may ignore the infinite places. Then we are asked to find $a \in F$ for which $a \equiv a_v \pmod{\mathcal{O}_v}$ for all $v$. After multiplying out some denominators, this amounts to the Chinese remainder theorem for $\mathcal{O}_F$. ■

Here is an analog for function fields.

**Proposition 1.31.** Fix a function field $F = \mathbb{F}_q(X)$. Then one has

$$F \backslash \mathbb{A}_F \Big/ \prod_{v \in V(F)} \mathcal{O}_v \cong \mathrm{H}^1(X; \mathcal{O}_X).$$

*Proof.* The idea is to use the "two-step complex" $F \to \mathbb{A}_F / \prod_v \mathcal{O}_v$ to compute the cohomology of $\mathcal{O}_X$. Note that $\mathbb{A}_F / \prod_v \mathcal{O}_v$ is the restricted product

$$\prod_{v \in V(F)} \left( \frac{F_v}{\mathcal{O}_v}, \frac{\mathcal{O}_v}{\mathcal{O}_v} \right) = \bigoplus_{v \in V(F)} F_v / \mathcal{O}_v.$$

Now, there is a short exact sequence

$$0 \to \mathcal{O}_X \to \mathcal{K} \to \bigoplus_{v \in V(F)} i_{v*}(F_v/\mathcal{O}_v) \to 0,$$

where $\mathcal{K}$ is the constant sheaf of rational functions. Taking the long exact sequence in cohomology produces an exact sequence

$$F \to \bigoplus_{v \in V(F)} F_v/\mathcal{O}_v \to \mathrm{H}^1(X; \mathcal{O}_X) \to \mathrm{H}^1(X; \mathcal{K}).$$

Because $X$ is irreducible, the constant sheaf $\mathcal{K}$ is flasque, so $\mathrm{H}^1(X; \mathcal{K}) = 0$. The result now follows. ■

**Remark 1.32.** As an application, the right-hand side will frequently have more than one element: it has dimension over $\mathbb{F}_q$ equal to the genus of $X$, so the cohomology group has one element if and only if $X$ is $\mathbb{P}^1_{\mathbb{F}_q}$!

**Remark 1.33.** If one expands one of the $\mathcal{O}_v$s to $F_v$s, then it turns out that the quotient is trivial.

**Remark 1.34.** One can check that the stabilizer of a double coset of the $F$-action on a double coset is exactly $\mathbb{F}_q = \mathrm{H}^0(X; \mathcal{O}_X)$.

Returning to number fields, we see that Theorem 1.30 grants us a surjection

$$F \otimes_{\mathbb{Q}} \mathbb{R} \twoheadrightarrow F \backslash \mathbb{A}_F \Big/ \prod_{v \notin V(F)} \mathcal{O}_v.$$

The kernel is exactly given by the elements $x \in F$ for which $v \in \mathcal{O}_v$ for all $v$, which is exactly $\mathcal{O}_F$. Thus, there is an isomorphism

$$\frac{F \otimes_{\mathbb{Q}} \mathbb{R}}{\mathcal{O}_F} \to F \backslash \mathbb{A}_F \Big/ \prod_{v \notin V(F)} \mathcal{O}_v$$

of topological groups. Here, the left-hand side is a torus of dimension $[F : \mathbb{Q}]$: it is isomorphic as a topological group to $\mathbb{R}^d / \mathbb{Z}^d$.

### 1.2.2 Idelic Quotients

Of course, we are more interested in $\mathbb{A}_F^\times$, so let's turn our attention there. As usual, arguing with function fields is easier.

**Proposition 1.35.** Fix a function field $F := \mathbb{F}_q(X)$. Then one has

$$F^\times \backslash \mathbb{A}_F^\times \Big/ \prod_{v \in V(F)} \mathcal{O}_v^\times \cong \operatorname{Pic} X.$$

*Proof.* Once again, we use the "two-step complex" $F^\times \to \mathbb{A}_F^\times / \prod_v \mathcal{O}_v^\times$. Here, $\mathbb{A}_F^\times / \prod_{v \in V(F)} \mathcal{O}_v^\times$ is the restricted product

$$\prod_{v \in V(F)} \left( \frac{F_v^\times}{\mathcal{O}_v^\times}, \frac{\mathcal{O}_v^\times}{\mathcal{O}_v^\times} \right) \cong \bigoplus_{v \in V(F)} \mathbb{Z}.$$

Here, the last isomorphism occurs by taking valuations. Now, this latter group is isomorphic to $\operatorname{Div}(X)$, and we can see that $F^\times$ embeds via these isomorphisms as the principal divisors. The result follows. ■

**Remark 1.36.** It turns out that $\operatorname{Pic}$ upgrades into a group scheme $\operatorname{Pic}_X$ with a connected component $\operatorname{Pic}_X^n$ for each degree. The Jacobian $\operatorname{Jac} X$ is exactly $\operatorname{Pic}_X^0$. Thus, $\operatorname{Pic}(X)$ is infinite, but the degree-zero part $\operatorname{Jac} X(\mathbb{F}_q)$ is some group, which has about $q^g$ points by the Weil conjectures.

**Remark 1.37.** The kernel of the map $F^\times \to \operatorname{Div} X$ is exactly the constant functions $\mathbb{F}_q^\times$. This reflects the fact that line bundles have some action by $\mathbb{F}_q^\times$.

And now we move to number fields. Here is a starting result.

**Lemma 1.38.** Fix a number field $F$. The map

$$\prod_{v \mid \infty} F_v^\times \to F^\times \backslash \mathbb{A}_F^\times \Big/ \prod_{v \nmid \infty} \mathcal{O}_v^\times$$

has cokernel isomorphic to the class group of $F$.

*Proof.* The cokernel is

$$F^\times \backslash \mathbb{A}_{F,f}^\times \Big/ \prod_{v \nmid \infty} \mathcal{O}_v^\times,$$

where $\mathbb{A}_{F,f}^\times$ is the ring of finite adéles. The right-hand quotient is $\bigoplus_{v \nmid \infty} F_v^\times / \mathcal{O}_v^\times$, which is isomorphic to the group of fractional ideals (or equivalently, $\operatorname{Div}(\operatorname{Spec} \mathcal{O}_F)$). Taking a further quotient by $F^\times$ shows that the cokernel is the class group. ■

**Remark 1.39.** The kernel of the map is exactly the elements of $F^\times$ which are units at every place, which is exactly $\mathcal{O}_F^\times$. It follows that we have an exact sequence

$$\mathcal{O}_F^\times \to (F \otimes_{\mathbb{Q}} \mathbb{R})^\times \to F^\times \backslash \mathbb{A}_F^\times \Big/ \prod_{v \nmid \infty} \mathcal{O}_v^\times \to \operatorname{Cl} F \to 0.$$

To continue cutting down the size of the quotient, note that both $F^\times$ and $\prod_v \mathcal{O}_v^\times$ have global norm in $\mathbb{A}_F^\times \to \mathbb{R}^+$ equal to $1$.

**Notation 1.40.** Fix a number field $F$. Then we define $\mathbb{A}_F^{\times,1}$ to be the subset of elements with global norm $1$.

**Remark 1.41.** The map

$$F^\times \backslash \mathbb{A}_F^\times \Big/ \prod_{v \nmid \infty} \mathcal{O}_v^\times \to F^\times \backslash \mathbb{A}_{F,f}^\times \Big/ \prod_{v \nmid \infty} \mathcal{O}_v^\times$$

continues to be surjective because we can always choose the archimedean part of an adéle so that the adéle has norm $1$.

Thus, our exact sequence now looks like

$$1 \to \mu(F) \to \mathcal{O}_F^\times \to (F \otimes_{\mathbb{Q}} \mathbb{R})^{\times,1} \to F^\times \backslash \mathbb{A}_F^{\times,1} \Big/ \prod_{v \nmid \infty} \mathcal{O}_v^\times \to \operatorname{Cl} F \to 0,$$

where $(F \otimes_{\mathbb{Q}} \mathbb{R})_1$ refers to the subgroup whose product is $1$. Taking $\log |\cdot|_v$ (with $|\cdot|_v$ chosen as before) maps $(F \otimes_{\mathbb{Q}} \mathbb{R})^{\times,1}$ into a Euclidean space isomorphic to $\mathbb{R}^{r_1+r_2-1}$, where $(r_1, r_2)$ is the signature of $F$. Note that the kernel of $\log |\cdot|_v$ is given by the elements of archimedean norm $1$, which when restricted to $\mathcal{O}_F^\times$ is exactly the group $\mu(F)$ of roots of unity.

Now, by Dirichlet's unit theorem, we see that $\mathcal{O}_F^\times$ embeds as a lattice of full rank into $\mathbb{R}^{r_1+r_2-1}$, so the quotient is a compact torus. We have thus proven the following result.

**Theorem 1.42.** Fix a number field $F$ with signature $(r_1, r_2)$ The double quotient

$$F^\times \backslash \mathbb{A}_F^{\times,1} \Big/ \prod_{v \nmid \infty} \mathcal{O}_v^\times$$

is isomorphic to an extension of $(\mathbb{R}/\mathbb{Z})^{r_1+r_2-1}$ by the class group $\operatorname{Cl} F$. In particular, it is compact.

**Remark 1.43.** Thus, we see that

$$F^\times \backslash \mathbb{A}_F^\times \Big/ \prod_{v \nmid \infty} \mathcal{O}_v^\times$$

is not compact, but it is an extension of a compact abelian group by $\mathbb{R}^+$.

### 1.2.3 Pontryagin Duality

Our next task is to do some Fourier analysis on $\mathbb{A}_F$ and $\mathbb{A}_F^\times$. Let's first recall generalities of Fourier analysis on locally compact abelian topological groups.

**Definition 1.44** (Pontryagin dual)**.** Fix a locally compact abelian group $X$. Then its *Pontryagin dual* $X^*$ is the set of homomorphisms $X \to S^1$, equipped with the compact open topology.

**Remark 1.45.** There is a functoriality as follows: for any homomorphism $f\colon X \to Y$, we have a homomorphism $f^*\colon Y^* \to X^*$ given by pre-composition.

Here are some theorems about this construction.

**Theorem 1.46** (Duality)**.** There is a natural isomorphism $\mathrm{id} \Rightarrow (-)^{**}$. For a given group $G$, it is given by sending $g \in G$ to the character $\mathrm{ev}_g\colon G^* \to S^1$ defined by $\mathrm{ev}_g\colon \chi \mapsto \chi(g)$.

**Theorem 1.47** (Exact)**.** The functor $(-)^*$ is exact.

Let's see some examples.

**Example 1.48.** If $X = \mathbb{Z}$, then its Pontryagin dual is just $S^1$. On the other hand, all continuous homomorphisms $S^1 \to S^1$ take the form $z \mapsto z^n$, so $\left(S^1\right)^* = \mathbb{Z}$.

**Example 1.49.** Homomorphisms $\mathbb{R} \to S^1$ all take the form $\chi_\xi\colon t \mapsto e^{i\xi t}$, where $\xi \in \mathbb{R}$ is some real number. Thus, $\mathbb{R}^* = \mathbb{R}$.

**Example 1.50.** In general, given a finite-dimensional real vector space $V$, we may identify the dual $V^*$ with the Pontryagin dual, where one sends $\varphi\colon V \to \mathbb{R}$ to the character $v \mapsto e^{i\varphi(v)}$.

**Example 1.51.** Homomorphisms $\mathbb{Z}/n\mathbb{Z} \to S^1$ are uniquely determined by where they send $1$, so

$$(\mathbb{Z}/n\mathbb{Z})^* \cong \mu_n.$$

Conversely, all homomorphisms $\mu_n \to \mu_n$ are given by $z \mapsto z^k$ for some $k$, so $\mu_n^* \cong \mathbb{Z}/n\mathbb{Z}$. In particular, we see that $(\mathbb{Z}/n\mathbb{Z})^*$ is non-canonically isomorphic to $\mathbb{Z}/n\mathbb{Z}$, but the isomorphism $\mathbb{Z}/n\mathbb{Z} \to (\mathbb{Z}/n\mathbb{Z})^{**}$ is canonical! In general, for any finite abelian group $G$, we see that $G^*$ is non-canonically isomorphic to $G$.

**Example 1.52.** Exactness of the functor $(-)^*$ implies that

$$\mathbb{Z}_p^* = (\lim \mathbb{Z}/p^\bullet\mathbb{Z})^* = \operatorname{colim} \mu_{p^\bullet} = \mu_{p^\infty}.$$

**Example 1.53.** Once again, exactness of the functor $(-)^*$ implies that

$$\mathbb{Q}_p^* = \left(\operatorname{colim}\left(\mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_p \xrightarrow{p} \cdots\right)\right)^* = \lim\left(\mu_{p^\infty} \xrightarrow{p} \mu_{p^\infty} \xrightarrow{p} \cdots\right).$$

Thus, this limit is some kind of coherent sequence of taking $p$th roots, which is then isomorphic to $\mathbb{Q}_p$. Indeed, $\mu_{p^\infty}$ is isomorphic to $\mathbb{Q}_p/\mathbb{Z}_p$, which we see by sending $a/p^n \in \mathbb{Q}_p/\mathbb{Z}_p$ to $\exp(2\pi i a/p^n)$. In fact, it turns out that the exact sequence

$$0 \to \mathbb{Z}_p \to \mathbb{Q}_p \to \mathbb{Q}_p/\mathbb{Z}_p \to 0$$

dualizes to an isomorphism

$$\begin{tikzcd}
0 \arrow[r] & \mathbb{Z}_p \arrow[r] \arrow[d] & \mathbb{Q}_p \arrow[r] \arrow[d] & \mathbb{Q}_p/\mathbb{Z}_p \arrow[r] \arrow[d] & 0 \\
1 \arrow[r] & \lim \mu_{p^n} \arrow[r] & \mathbb{Q}_p^* \arrow[r] & \mu_{p^\infty} \arrow[r] & 1
\end{tikzcd}$$

sending $x \in \mathbb{Z}_p$ to the sequence $\{\zeta_{p^n}^x\}_n$.

Thus, we see that all local fields are identified with their Pontryagin duals. In fact, all of our constructions amount to identifying a space with its dual upon choosing a single character.

**Remark 1.54.** Explicitly, given a choice of nontrivial character $\psi_p \in \mathbb{Q}_p^*$, there is a map $\mathbb{Q}_p \to \mathbb{Q}_p^*$ given by taking $x$ to the character $y \mapsto \psi_p(xy)$. It turns out that this map is an isomorphism, so we have more or less defined a non-degenerate bilinear form $\mathbb{Q}_p \times \mathbb{Q}_p \to S^1$. This procedure also works for $\mathbb{R}$!

**Remark 1.55.** A similar argument shows that each local field $F_v$ has $F_v^* \cong F_v$, again given by a choice of character $\psi_v \in F_v^*$. Namely, there is a map $F_v \to F_v^*$ given by sending $x \in F_v$ to the character $y \mapsto \psi_p(xy)$.

### 1.2.4 Fourier Theory

To do Fourier analysis, we need a notion of measure.

**Theorem 1.56** (Haar)**.** Fix a locally compact group $X$. Then there is a left-invariant Radon measure $dx$ on $X$ which is unique up to scalar.

Now, here is our Fourier transform.

**Definition 1.57** (Fourier transform)**.** Fix a locally compact abelian group $X$. The *Fourier transform* sends a function $f \in L^1(X)$ to the function $\widehat{f}\colon X^* \to \mathbb{C}$ given by

$$\widehat{f}(\xi) = \int_X f(x)\overline{\xi}(x)\,dx.$$

It is a large theorem that there is an inversion.

**Theorem 1.58** (Fourier inversion)**.** Fix a locally compact abelian group $X$, and let $dx$ be a Haar measure on $X$. Then there is a Haar measure $d\chi$ on $X^*$ such that

$$f(x) = \int_{X^*} \widehat{f}(\chi)\chi(x)\, d\chi$$

for any $f \in L^1(X)$ for which $\widehat{f} \in L^1(X^*)$.

**Remark 1.59.** It turns out that the Fourier transform extends to an isomorphism $L^2(G) \to L^2(G^*)$.

**Remark 1.60.** Equivalently, we see that the double Fourier transform of $f$ is $f(-x)$.

**Remark 1.61.** If $X$ admits an isomorphism $X \cong X^*$, then the Haar measure $d\chi$ is not necessarily equal to the Haar measure $dx$ because it might be off by a scalar: indeed, replacing $dx$ with $c\,dx$ replaces $\widehat{f}$ with $c\widehat{f}$, and so we see that we end up replacing $d\chi$ with $c^{-1}\, d\chi$. Thus, there is a unique measure $dx$ on $X$ (even up to scalar!) which is "Fourier self-dual."

## 1.3 February 9

Here we go.

### 1.3.1 Duality for Local Fields

In light of the previous remark, it is useful to fix some characters.

**Notation 1.62.** Fix a place $v$ of $\mathbb{Q}$.

- If $v = p$ is finite, then we define the character $\psi_p \colon \mathbb{Q}_p \to S^1$ by the composite $\mathbb{Q}_p \to \mathbb{Q}_p/\mathbb{Z}_p \cong \mu_{p^\infty}$, where the second isomorphism sends $a/p^n$ to $e^{2\pi i a/p^n}$.
- If $v = \infty$ is infinite, then we define the character $\psi_\infty \colon \mathbb{R} \to S^1$ by $\psi_\infty(x) := e^{-2\pi i x}$.

In general, for a place $w$ of a number field $F$ lying over place $v$ of $\mathbb{Q}$, we define $\psi_v := \psi_p \circ \operatorname{tr}_{F_v/\mathbb{Q}_p}$.

**Remark 1.63.** The character $\psi_p$ has the property that it is trivial on $\mathbb{Z}_p$ but nontrivial on $p^{-1}\mathbb{Z}_p$. In general, for finite places $v$ of a field $F$, one finds that $\psi_v|_{\mathcal{O}_v} = 1$, but $\mathcal{O}_v$ may not be the largest subgroup with this property, though this is true for all but finitely many $v$.

**Remark 1.64.** The choice of $\psi_\infty$ is done so that the assembled character $\psi_{\mathbb{Q}} \colon \mathbb{A}_{\mathbb{Q}} \to S^1$ vanishes on $\mathbb{Q}$. In fact, the character $\psi_F := \prod_v \psi_v$ also multiplies to $1$ because we can just take a trace of everything to $\mathbb{Q}$.

Self-duality allows us to define Fourier transforms sending functions on $F_v$ to functions on $F_v$. Let's be careful about what sorts of functions we want to integrate.

**Definition 1.65** (Schwartz)**.** Fix a local field $F_v$.

- If $F_v$ is nonarchimedean, then a *Schwartz function* is a locally constant, compactly supported function $F_v \to \mathbb{C}$.
- If $F_v$ is archimedean, then a *Schwartz function* is a smooth function, all of whose derivatives vanish faster than a polynomial at $\infty$.

The space of all Schwartz functions is denoted $\mathcal{S}(F_v)$.

**Notation 1.66.** Fix a Schwartz $f$ on a local field $F_v$. Then we define the *Fourier transform* by

$$\widehat{f}(y) := \int_{F_v} f(x)\psi_v(xy)\, d_v x.$$

**Remark 1.67.** One can check that the Fourier transform of a Schwartz function is Schwartz. We could of course integrate any function in $L^2(F_v)$, but we will not have a reason to.

**Remark 1.68.** By Remark 1.61, there is a unique choice of Haar measure $d_v x$ so that the second Fourier transform of a function $f$ is $f(-x)$. For $F_v = \mathbb{R}$, this turns out to be the Lebesgue measure, and for $F_v = \mathbb{C}$, this turns out to be twice the Lebesgue measure. One can check this by plugging in the function $f$ to be a Gaussian, which turns out to be self-dual for the given measures.

The previous remark (in the case of $\mathbb{C}$) has indicated that the self-dual Haar measure is potentially interesting. Let's explain what we receive for nonarchimedean local fields.

**Example 1.69.** For $\mathbb{Q}_p$, one can check that $\int_{\mathbb{Z}_p} d_p x = 1$. Indeed, suppose that $V = \int_{\mathbb{Z}_p} d_p x$, and consider the indicator function $1_{\mathbb{Z}_p}$. Then the Fourier transform

$$\widehat{1}_{\mathbb{Z}_p}(y) = \int_{\mathbb{Z}_p} \psi_p(xy)\, d_p x.$$

Now, for $y \in \mathbb{Z}_p$, this integral is constant, so we receive $V$. For $y \notin \mathbb{Z}_p$, we see that this character $x \mapsto \psi_p(xy)$ is nontrivial on $\mathbb{Z}_p$, so the integral vanishes. (To be explicit, say $y = a/p^n$ where $\gcd(a,p) = 1$ and $n \geq 1$, and then $\int_{\mathbb{Z}_p} \psi_p(ax/p^n)\, d_p x = \int_{\mathbb{Z}_p} \psi_p(a(x+1)/p^n)\, d_p x = \psi_p(a/p^n) \int_{\mathbb{Z}_p} \psi_p(ax/p^n)\, d_p x$. Because $\psi_p(a/p^n) \neq 0$, so the integral vanishes.) Thus, $\widehat{1}_{\mathbb{Z}_p} = V 1_{\mathbb{Z}_p}$, so the second Fourier transform is $V^2 1_{\mathbb{Z}_p}$. So for $d_p x$ to be self-dual, we are required to have $V = 1$. As a by-product, we note that we have also shown that the function $1_{\mathbb{Z}_p}$ is self-dual!

**Remark 1.70.** On the homework, we will compute $\int_{\mathcal{O}_v} d_v x$ for places $v$ lying over $p$, which is done by a similar procedure. Namely, we find that

$$\widehat{1}_{\mathcal{O}_v} = \int_{\mathbb{Z}_p} \psi_p(xy)\, d_p x$$

is $V$ times an indicator of the inverse different ideal

$$\mathcal{D}_v^{-1} := \{y \in F_v : \operatorname{tr} y\mathcal{O}_v \subseteq \mathbb{Z}_p\}.$$

This contains $\mathcal{O}_v$, but it may in general be bigger. If $F_v/\mathbb{Q}_p$ is unramified, then $\mathcal{D}_v^{-1} = \mathbb{Z}_p$, so in fact our self-dual measure should give $\int_{\mathcal{O}_v} d_v x = 1$. Otherwise, one finds that setting $\int_{\mathcal{O}_v} d_v x = 1$ need not be self-dual: one needs to multiply or divide by some square root of the index of $\mathcal{D}_v$.

### 1.3.2 Duality for the Adéles

We are now ready to put a measure on $\mathbb{A}_F$.

**Definition 1.71** (Schwartz)**.** Fix a number field $F$. A function $f\colon \mathbb{A}_F \to \mathbb{C}$ is *Schwartz* if and only if it lives in the restricted tensor product

$$\mathcal{S}(\mathbb{A}_F) := \bigotimes_v (\mathcal{S}(F_v), 1_{\mathcal{O}_v}),$$

where it is restricted in the sense that almost all factors of a pure tensor are equal to $1_{\mathcal{O}_v}$.

**Definition 1.72.** Fix a number field $F$. Then we define a measure $dx$ on $\mathbb{A}_F$ to be the product $\prod_v d_v x$, where the measurable subsets consist of finite unions of basic open sets (except at the infinite places).

**Remark 1.73.** This measure is well-defined because any subset in the Borel algebra will produce a factor $\mathcal{O}_v$ at all but finitely many places $v$, and all but finitely many of those places have $\int_{\mathcal{O}_v} d_v x = 1$, so the entire product turns out to be finite on any Borel subset.

**Remark 1.74.** Let's be more precise about this construction. There is a unique Haar measure $dx$ on $\mathbb{A}_F$ such that the measure of $\prod_{v\nmid\infty} \mathcal{O}_v \times \prod_{v|\infty} B_v(0,1)$ is the expected product of the given measures. The character $\psi_F\colon \mathbb{A}_F \to S^1$ produces an isomorphism $\mathbb{A}_F \to \mathbb{A}_F^*$ (by gluing together the local isomorphisms), and the corresponding self-dual measure can be checked to be $dx$ by computing the Fourier transform of the indicator of $\bigotimes_v f_v$, where $f_v = 1_{\mathcal{O}_v}$ at finite places and the Gaussian at infinite places.

The reason we want to be able to work globally is that there is a Poisson summation formula for the subgroup $F \subseteq \mathbb{A}_F$.

**Theorem 1.75** (Poisson summation)**.** Fix a number field $F$. For $f \in \mathcal{S}(\mathbb{A}_F)$,

$$\sum_{x\in F} f(x) = \sum_{y\in F} \widehat{f}(y).$$

Let's explain the general story here.

**Definition 1.76** (cocompact)**.** Fix a topological abelian group $X$. A closed subgroup $\Gamma \subseteq X$ is *cocompact* if and only if the quotient $\Gamma\backslash X$ is compact.

**Remark 1.77.** It turns out that $X$ is compact if and only if $X^*$ is discrete. Thus, if $\Gamma \subseteq X$ is discrete and cocompact, then the dual subgroup

$$\Gamma^\perp := \{\chi \in X^* : \chi|_\Gamma = 1\}$$

is a discrete cocompact subgroup of $X^*$.

**Example 1.78.** The subgroup $\mathbb{Z} \subseteq \mathbb{R}$ is discrete and cocompact because $\mathbb{R}/\mathbb{Z}$ is the circle group. Upon identifying $\mathbb{R}^*$ with $\mathbb{R}$ via the character $\psi_\infty$, the dual subgroup $\mathbb{Z}^*$ of $\mathbb{R}$ is exactly $\mathbb{Z}$: indeed, we are asking for $x \in \mathbb{R}$ for which the character $y \mapsto \psi_\infty(xy)$ is trivial on $\mathbb{Z}$, which is equivalent to having $x \in \mathbb{Z}$ because $\psi_\infty(xy) = e^{-2\pi i x y}$.

**Theorem 1.79** (Poisson summation)**.** Fix a locally compact topological group $X$, and let $\Gamma \subseteq X$ be a discrete cocompact subgroup. For any $f \in L^2(X)$, we have

$$\operatorname{vol}(\Gamma\backslash X; dx) \sum_{x \in \Gamma} f(x) = \sum_{y \in \Gamma^\perp} \widehat{f}(y)$$

provided that the left-hand side converges absolutely and uniformly. Here, $X^*$ has been given the dual measure of Theorem 1.58.

*Proof.* The idea is to consider the function

$$\varphi(x) = \sum_{\gamma \in \Gamma} f(x + \gamma).$$

Provided convergence, this function descends to a function on $\Gamma\backslash X$. This then has a Fourier transform $\widehat{\varphi}$, which is a function on $(\Gamma\backslash X)^* = \Gamma^\perp$. Fourier inversion now provides the equality

$$\varphi(0) = \sum_{y \in \Gamma^\perp} \widehat{\varphi}(y).$$

We will be done as soon as we can check that $\widehat{\varphi}(y) = \widehat{f}(y)$, which is some explicit calculation. Namely, the self-duality $X \cong X^*$ comes from a pairing $X \times X \to S^1$, which gives our Fourier transform the form

$$\begin{aligned} \widehat{f}(y) &= \int_X f(x)\langle x, y\rangle \, dx \\ &\overset{*}{=} \int_{\Gamma\backslash X} f(x)\langle x, y\rangle \, dx \\ &= \widehat{\varphi}(y), \end{aligned}$$

where $\overset{*}{=}$ holds because the inner product $\langle x, y\rangle$ only depends on the class in $\Gamma\backslash X$ (because $y \in \Gamma^\perp$). ■

**Remark 1.80.** If we have an isomorphism $X \cong X^*$ which sends $\Gamma$ to $\Gamma^\perp$, and we choose a self-dual Haar measure under the isomorphism, then one can check that the induced volume of $\Gamma\backslash X$ is $1$ by plugging in $f$ and its Fourier transform into the Poisson summation formula!

**Example 1.81.** For the application to $F \subseteq \mathbb{A}_F$, one needs to check that $F^\perp \subseteq \mathbb{A}_F$ is identified with $F$ in the isomorphism $\mathbb{A}_F \cong \mathbb{A}_F^*$ given by the character $\psi_F$. Certainly $F \subseteq F^\perp$ because $\psi_F(a) = 1$ for all $a \in F$. For the other inclusion, we see that $F^\perp \subseteq \mathbb{A}_F$ is a discrete subgroup of $\mathbb{A}_F$ including $F$, which we show must be $F$ on the homework. (Here, $F^\perp$ is discrete because its dual is the compact quotient $\mathbb{A}_F/F$.)

**Remark 1.82.** One should also check that the function

$$y \mapsto \sum_{x \in F} f(x+y)$$

converges absolutely and uniformly for any $f \in \mathcal{S}(\mathbb{A}_F)$. Well, we may descend to a function of the form $\prod_v f_v$, so we are looking at some indicator on a set which is a product of compacts. Using the finite places, we see that we are requiring some bounded valuation at every place, so we are summing over a fractional ideal. But $f$ is Schwartz at the infinite places, so the desired converges follows.

**Remark 1.83.** Having a self-duality means that the Poisson summation formula actually automatically implies $\mathrm{vol}(F \backslash \mathbb{A}_F; dx) = 1$, provided that $dx$ is the self-dual Haar measure. Roughly speaking, $\prod_v \mathcal{O}_v$ includes into this space (up to some controlled kernel) with finite but likely large cokernel. We thus expect that $\prod_v \mathcal{O}_v$ to have volume at most $1$, so we expect each of the $\mathcal{O}_v$s to have volume at most $1$. This is a useful sanity check for when we compute the self-dual Haar measure on the local fields $F_v$.

### 1.3.3 Duality for Function Fields

Let's say something about duality for function fields $F = \mathbb{F}_q(X)$. Then the correct dual object for $\mathbb{A}_F$ turns out to be differential forms.

**Notation 1.84.** Fix a function field $F$. Then we define

$$\mathbb{A}_\omega := \prod_v (\omega_{F_v}, \omega_{\mathcal{O}_v}).$$

Here, $\omega$ is the $F$-bundle of $1$-forms on $X$, $\omega_v$ is the stalk at $v$, $\omega_{\mathcal{O}_v}$ is its completion, and $\omega_{F_v}$ is the base-change $\omega_{\mathcal{O}_v} \otimes_{\mathcal{O}_v} F_v$.

Let's see how this produces duality.

**Definition 1.85** (residue)**.** Fix a closed point $v$ of a smooth projective geometrically connected curve $X$. Then we define the *residue* $\omega_{F_v} \to \mathbb{F}_q(v)$ defined as follows: for a differential form $\theta$, choose a local coordinate $dt$ and expand

$$\theta = \sum_{n \in \mathbb{Z}} a_n t^n \, dt.$$

Then the residue is $\mathrm{res}_v \, \theta := a_{-1}$.

**Remark 1.86.** It turns out that the residue is independent of the choice of coordinate $t$. Indeed, any other coordinate is of the form $s = ut$ for a unit $u$, and we see that $dt/t = ds/s$.

Here is the local duality.

**Proposition 1.87.** Fix a function field $F = \mathbb{F}_q(X)$, and choose a nontrivial character $\psi \colon \mathbb{F}_p \to S^1$. Then the pairing $F_v \times \omega_{F_v} \to S^1$ defined by

$$\langle f, \theta \rangle_v := \psi(\mathrm{tr}_{\mathbb{F}_q(v)/\mathbb{F}_p} f\theta)$$

realizes an isomorphism $F_v^* \cong \omega_{F_v}$. Furthermore, $\mathcal{O}_v^\perp$ is identified with $\omega_{\mathcal{O}_v}$.

Here is the global duality, which we prove on the homework.

**Proposition 1.88.** Fix a function field $F = \mathbb{F}_q(X)$, and choose a nontrivial character $\psi\colon \mathbb{F}_p \to S^1$. Then the product pairing $\mathbb{A}_F \times \mathbb{A}_{\omega_F} \to S^1$ defined by

$$(a, \theta) \mapsto \prod_v \langle a_v, \theta_v \rangle_v$$

produces an isomorphism $\mathbb{A}_F^* \to \mathbb{A}_{\omega_F}$.

**Remark 1.89.** A choice of $\theta \in \omega_F$ allows us to identify $\mathbb{A}_{\omega_F}$ with $\mathbb{A}_F$ and thus identify $\mathbb{A}_F$ with itself. This is analogous to choosing a full character of $\mathbb{Q}_p$ for each $p$.

**Remark 1.90.** On the homework, we will show that $F^\perp = \omega_F$. The inclusion $\omega_F \subseteq F^\perp$ follows from the residue theorem: for any $\theta \in \omega_F$, we have

$$\sum_v \operatorname{tr}_{\mathbb{F}_q(v)/\mathbb{F}_q} \operatorname{res}_v \theta = 0.$$

The other inclusion uses a compactness argument.

Now that we are refusing to write down any self-dualities, our choices of Haar measure now depend on a scalar. Instead, we will take by convention that $\operatorname{vol}(\mathcal{O}_v, d_v x) = 1$ for each $v \in X$, and we take the product to produce a measure on $\mathbb{A}_F$.

**Example 1.91.** Recall that there is an exact sequence

$$\mathbb{F}_q \to \prod_v \mathcal{O}_v \to F \backslash \mathbb{A}_F \to \mathrm{H}^1(X; \mathcal{O}_X) \to 0.$$

This shows that the measure of $F \backslash \mathbb{A}_F$ is $q^{g-1}$.

**Example 1.92.** Analogously, one can show that there is an identification of

$$\omega_F \backslash \mathbb{A}_{\omega_F} \Big/ \prod_v \mathcal{O}_v$$

with $\mathrm{H}^1(X; \mathcal{O}_X)$, and there is a stabilizer of $\mathrm{H}^0(X; \mathcal{O}_X)$. Thus, $\omega_F \backslash \mathbb{A}_{\omega_F}$ has measure $q^{1-g}$.

**Remark 1.93.** It turns out that the corresponding Poisson summation formula is an incarnation of the Riemann–Roch formula, if one plugs in a special choice of function.

## 1.4 February 11

Here we go.

### 1.4.1 Multiplicative Measures

We are now ready to move from adéles to idéles. Let's start by trying to fix a measure.

**Remark 1.94.** Fix some local field $F_v$. By definition of $|\cdot|_v$, we see that $d_v x / |\cdot|_v$ is a Haar measure on $F_v^\times$.

Now, for a global field $F$, one may attempt to put a measure on $\mathbb{A}_F^\times$ by multiplying together all the local measures. However, we are going to want to integrate indicators on basic open subsets of $\mathbb{A}_F^\times$. For example, we could try to integrate $\prod_{v\nmid\infty} 1_{\mathcal{O}_v} \prod_{v|\infty} 1_{B_v(0,1)}$, whose integral will be a scalar times

$$\prod_{v\nmid\infty} \int_{\mathcal{O}_v^\times} d_v^\times x.$$

But this integral is $\mu_v(\mathcal{O}_v) - \mu_v(\varpi_v\mathcal{O}_v) = (1 - |\varpi_v|_v)\,\mu_v(\mathcal{O}_v)$. For all but finitely many $v$, we see that $\mu_v(\mathcal{O}_v) = 1$, so we see that the above product has all but finitely many of its factors not equal to $1$! In fact, it vanishes for $F = \mathbb{Q}$. Thus, this normalization is not suitable for our purposes. Instead, we divide out by this factor $1 - |\varpi_v|_v$.

**Definition 1.95.** Fix a nonarchimedean local field $F_v$. Then we define

$$d_v^\times x := \frac{1}{1 - q_v^{-1}} \cdot \frac{d_v x}{|x|_v},$$

and we define $d^\times x$ on $\mathbb{A}_F^\times$ as the product measure.

**Remark 1.96.** It follows from the preceding calculation that $d^\times x$ is a finite product on basic open subsets.

An interesting question is to calculate the volume of $F^\times\backslash\mathbb{A}_F^{\times,1}$ according to the measure $d^\times x$. For number fields $F$, recall from the proof of Theorem 1.42 that we have an exact sequence

$$1 \to \mu(F) \to \mathcal{O}_F^\times \to (F \otimes_\mathbb{Q} \mathbb{R})^{\times,1} \to F^\times\backslash\mathbb{A}_F^{\times,1} / \prod_{v\nmid\infty} \mathcal{O}_v^\times \to \operatorname{Cl} F \to 0.$$

One eventually finds the following.

**Proposition 1.97.** Fix a number field $F$ with signature $(r_1, r_2)$. Then the volume of $F^\times\backslash\mathbb{A}_F^{\times,1}$ is

$$\frac{h_F \operatorname{Reg}_F}{w} \cdot \frac{2^{r_1}(2\pi)^{r_2}}{\sqrt{|\operatorname{disc}\mathcal{O}_F|}}.$$

Here, $h_F$ is the class number, $\operatorname{Reg}_F$ is the regulator (which is an appropriately measured covolume of $\mathcal{O}_F^\times$ sitting in $(F \otimes_\mathbb{Q} \mathbb{R})^{\times,1}$), and $w$ is the number of roots of unity.

**Remark 1.98.** The second factor basically arises from how we have chosen our additive measures.

We will shortly see that the volume of $F^\times\backslash\mathbb{A}_F^{\times,1}$ is also related to the Dedekind $\zeta_F$-function, thereby proving the analytic class number formula.

**Definition 1.99** (Dedekind $\zeta$-function)**.** Fix a global field $F$. Then we define the *Dedekind $\zeta_F$-function* as

$$\zeta_F(s) := \sum_{I \subseteq \mathcal{O}_F} \frac{1}{\mathrm{N}(I)}.$$

This sum converges absolutely and uniformly on compacts for $\operatorname{Re} s > 1$.

**Remark 1.100.** Unique prime factorization of ideals produces an Euler product

$$\zeta_F(s) = \prod_{v \nmid \infty} \frac{1}{1 - q_v^s}.$$

This is remarkable because it looks like "$\zeta_F(1)$" is the scale factor between the failed "product" measure $dx/|x|$ on $\mathbb{A}_F^\times$ and our successful measure $d^\times x$.

### 1.4.2 Our $L$-functions

The goal of Tate's thesis is to reprove some general results on the functional equations of $L$-functions. For example, we will be able to reprove the functional equation of the Dirichlet $L$-functions, defined by

$$L(s, \chi) := \sum_{\substack{n=1 \\ \gcd(n,N)=1}}^{\infty} \frac{\chi(n)}{n^s},$$

where $\chi\colon (\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$ is some character.

**Example 1.101.** Taking $\chi = 1$ (and $N = 1$) recovers Riemann's $\zeta$-function.

Roughly speaking, our functional equation will equate $L(s, \chi)$ and $L(1-s, \overline{\chi})$, after we "fix" these $L$-functions slightly.

We will be able to work over general global fields. Let's start by relating the discussion of the previous example with our adélic discussion.

**Example 1.102.** It turns out that $\mathbb{A}_\mathbb{Q}^\times = \mathbb{Q}^\times \times \mathbb{R}^+ \times \widehat{\mathbb{Z}}^\times$ (recall $\widehat{\mathbb{Z}} = \prod_p \mathbb{Z}_p$), which one can see by taking some successive quotients and appealing to Theorem 1.42. The moral is that a Dirichlet character $(\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$ can be viewed as a continuous character $\mathbb{A}_\mathbb{Q}^\times \to \mathbb{C}^\times$ which "happens" to factor through $\widehat{\mathbb{Z}}^\times$.

**Definition 1.103** (idéle class character)**.** Fix a global field $F$. Then an *idéle class character* is a character $\chi\colon F^\times \backslash \mathbb{A}_F^\times \to \mathbb{C}^\times$. It is *unitary* if its image is contained in $S^1$.

**Remark 1.104.** The unitary characters (by definition) live in the Pontryagin dual of $F^\times \backslash \mathbb{A}_F^\times$.

**Example 1.105.** Note that there is a norm character $|\cdot| : F^\times \backslash \mathbb{A}_F^\times \to \mathbb{R}^+$. In fact, for any $s \in \mathbb{R}$, we receive a character $|\cdot|^s$.

**Remark 1.106.** Suppose $F$ is a number field. Then $|\cdot| : F^\times \backslash \mathbb{A}_F^\times \to \mathbb{R}^+$ is surjective, so we can take $\mathrm{Hom}(-, \mathbb{C}^\times)$ of the short exact sequence

$$1 \to F^\times \backslash \mathbb{A}_F^{\times,1} \to F^\times \backslash \mathbb{A}_F^\times \to \mathbb{R}^+ \to 1$$

to see that $\left(F^\times \backslash \mathbb{A}_F^\times\right)^*$ is an extension of $\mathrm{Hom}(\mathbb{R}^+, \mathbb{C}^\times) = \mathbb{C}$ by a discrete group $\left(F^\times \backslash \mathbb{A}_F^{\times,1}\right)^*$. (We have used the compactness of $F^\times \backslash \mathbb{A}_F^{\times,1}$ to show that any map to $\mathbb{C}^\times$ factors through $S^1$. Note that exactness holds on the right after the duality because already $(-)^*$ is exact.) We are thus able to conclude that the space of idéle class characters inherits a topology of a complex manifold.

**Remark 1.107.** Suppose $F$ is a function field $\mathbb{F}_q(X)$. Then the norm $|\cdot| : F^\times \backslash \mathbb{A}_F^\times \to \mathbb{R}^+$ surjects instead onto the discrete group $q^{\mathbb{Z}}$. We thus see that $\mathrm{Hom}\left(F^\times \backslash \mathbb{A}_F^\times, \mathbb{C}^\times\right)$ is an extension of $\mathrm{Hom}(\mathbb{Z}, \mathbb{C}^\times) = \mathbb{C}^\times$ by a discrete group. The space of idéle class characters continues to be a complex manifold by the same argument.

**Remark 1.108.** By continuity of $\chi$, we see that $\chi|_{\mathcal{O}_v^\times} = 1$ for almost all $v$. Indeed, this follows from a "no small subgroups" argument applied to continuous maps on the group $\prod_{v \nmid \infty} \mathcal{O}_v^\times$.

**Definition 1.109** (unramified)**.** An idéle class character $\chi$ is *unramified* at a finite place $v$ if and only if $\chi|_{\mathcal{O}_v} = 1$; otherwise, we see that it is *ramified* at $v$.

We are now ready to define our $L$-functions.

**Definition 1.110.** Fix an idéle class character $\chi$ of a global field $F$, and choose a finite place $v$. Then we define

$$L_v(\chi_v) := \begin{cases} \frac{1}{1-\chi_v(\varpi_v)} & \text{if } \chi_v|_{\mathcal{O}_v^\times} = 1, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$L(\chi) := \prod_{v \nmid \infty} L_v(s, \chi_v).$$

We may write $L_v(s, \chi_v)$ and $L(s, \chi)$ for $L_v\left(\chi_v |\cdot|_v^s\right)$ and $L\left(\chi |\cdot|^s\right)$, respectively.

**Remark 1.111.** We can expand $L(\chi)$ out as a sum

$$L(\chi) = \sum_{I \subseteq \mathcal{O}_F} \frac{\chi(I)}{\mathrm{N}(I)},$$

where $\chi(I)$ means $\prod_v \chi(\varpi_v)^{\nu_v(I)}$, and $\chi(\varpi_v)$ in this expression means $0$ if $\chi$ is ramified at $v$.

**Remark 1.112.** If $\chi$ is unitary, then the function $s \mapsto L(s, \chi)$ can be checked to converge absolutely and uniformly on compacts in the region $\mathrm{Re}\, s > 1$.

**Example 1.113.** Taking $\chi = 1$ recovers Dedekind $\zeta$-functions.

### 1.4.3 Functional Equations

Tate's main global result is a duality statement.

**Definition 1.114** (global integral)**.** Fix an idéle class character $\chi$ on a number field $F$. For $f \in \mathcal{S}(\mathbb{A}_F)$, we define

$$Z(\chi, f) := \int_{\mathbb{A}_F^\times} \chi(a) f(a)\, d^\times a.$$

We may also write $Z(s, \chi, f) := Z(\chi |\cdot|^s, f)$.

**Remark 1.115.** Of course, there is redundancy in our notation: indeed, $Z(s+t,\chi,f) = Z(s,\chi\,|\cdot|^t\,,f)$.

**Theorem 1.116** (Tate)**.** Fix a global field $F$ and some $f \in \mathcal{S}(\mathbb{A}_F)$.

(a) The function $\chi \mapsto Z(\chi, f)$ admits a meromorphic continuation and a functional equation

$$Z(\chi,f) = Z\left(\chi^{-1}\,|\cdot|\,,\widehat{f}\right).$$

(b) If $\chi$ is nontrivial on $\mathbb{A}_F^{\times,1}$, then $s \mapsto Z(s,\chi,f)$ is holomorphic.

(c) The function $s \mapsto Z(s,1,f)$ is holomorphic everywhere except for simple poles at $s \in \{0,1\}$ with residue $-f(0)\operatorname{vol}\left(F^\times\backslash\mathbb{A}_F^{\times,1}; d^\times x\right)$ at $s=0$.

Let's explain how this relates to our $L$-functions.

**Theorem 1.117.** Fix a number field $F$. Then there is a functional equation relating $\zeta_F(s)$ and $\zeta_F(1-s)$.

*Proof.* The idea is to make $f$ and $\widehat{f}$ the same at almost all places. Define a function $(f_v)$ in $\mathcal{S}(\mathbb{A}_F)$ as follows.

- For finite $v$, we define $f_v = 1_{\mathcal{O}_v}$. In particular, at unramified places $v$, we recall that $1_{\mathcal{O}_v}$ is self-dual.
- If $F_v = \mathbb{R}$ and $\chi_v = 1$, then the function $e^{-\pi x^2}$ is self-dual. The case where $\chi_v = \operatorname{sgn}$ takes the function $xe^{-\pi x^2}$.
- If $F_v = \mathbb{C}$ and $\chi_v = 1$, then the function $e^{-\pi|z|^2}$ is self-dual. If $\chi_v$ is more general, then some slightly different recipe is used.

One can now compute

$$Z(s,1,f) = \prod_v \int_{F_v^\times} |a_v|^s\, f_v(a_v)\, d_v^\times a.$$

If $v$ is finite, then we are being asked to compute

$$\begin{aligned}\int_{\mathcal{O}_v\backslash\{0\}} |a|^s\ d_v^\times a &= \sum_{n\geq 0}^{\infty} \int_{\varpi^n\mathcal{O}_v} |a|^s\ d_v^\times a \\ &= \sum_{n\geq 0} q_v^{-ns} \operatorname{vol}\left(\mathcal{O}_v^\times\right) \\ &= \frac{\operatorname{vol}\left(\mathcal{O}_v^\times\right)}{1-q_v^{-s}}.\end{aligned}$$

If $v$ is unramified over $\mathbb{Q}_p$, then the volume on top is $1$; if it is ramified, then we are computing some square root of the norm of the different (which is the discriminant). Thus, up to these contributions from rational factors, we find that $Z(s,1,f)$ is

$$\zeta_F(s) \prod_{\text{real } v} \pi^{-s/2}\Gamma(s/2) \prod_{\text{complex } v} 2(2\pi)^{1-s}\Gamma(s).$$

We are thus able to produce a functional equation for $\zeta_F(s)$. We did not work this out in class, and I do not have time to do it on my own currently. ■

For function fields $F = \mathbb{F}_q(X)$, we have been careful to avoid identifying $\mathbb{A}_F$ with itself, so we don't have a self-duality.

**Example 1.118.** In this case, note that the Euler product of $\zeta_F(s)$ expands into

$$\sum_{\text{effective } D \subseteq X} q^{-s \deg D} = \sum_{d \geq 0} q^{-ds} \cdot \# X^{(d)}(\mathbb{F}_q),$$

where $X^{(d)}(\mathbb{F}_q)$ refers to the number of effective divisors on $D$ of degree $d$ defined over $\mathbb{F}_q$. We have used the notation $X^{(d)}$ to indicate that this could be thought of as the stack $X^d/\Sigma_d$.

**Example 1.119.** Suppose that $\chi$ is an unramified idéle class character, meaning that it factors through $F^\times \backslash \mathbb{A}_F^\times / \prod_v \mathcal{O}_v^\times$, which we recall is $\operatorname{Pic} X$. One can calculate as before that

$$L(s, \chi) = \sum_{D \subseteq X} q^{-s \deg D} \chi(\mathcal{O}_X(D)).$$

Further suppose that $\chi|_{\operatorname{Pic}^0 X} \neq 1$. Then we claim that $L(s, \chi)$ is a polynomial in $q^{-s}$. Indeed, in light of the above expansion, it is enough to show that all line bundles of large degree $d$ appear an equal number of times as $\mathcal{O}_X(D)$ as $D$ varies over effective divisors of $d$. But this is true because $d > 2g - 2$ makes $X^{(d)}$ a $\mathbb{P}^{d-g}$-bundle over $\operatorname{Pic}^d X$: by an argument with linear systems (and the Riemann–Roch theorem), the fibers are all copies of $\mathbb{P}^{d-g} = \mathbb{P}\mathrm{H}^0(X; \mathcal{O}_X(D))$.

## 1.5 February 17

Today we prove Tate's theorem.

### 1.5.1 More General $L$-functions

To start us off, let's do a local calculation.

**Definition 1.120** (local integral)**.** Fix a local field $F$ and a continuous character $\chi \colon F^\times \to \mathbb{C}^\times$. For $f \in \mathcal{S}(\mathbb{A}_F)$, we define

$$Z(\chi, f) \coloneqq \int_{F^\times} \chi(a) f(a) \, d^\times a.$$

We may also write $Z(s, \chi, f) \coloneqq Z(\chi |\cdot|^s, f)$.

**Proposition 1.121.** Fix a nonarchimedean local field $F$, and suppose $\chi$ is unramified, in the sense that $\chi|_{\mathcal{O}_v^\times} = 1$. Then

$$Z(s, \chi, 1_{\mathcal{O}}) = \frac{1}{\chi(\varpi) q^{-s}} \cdot \operatorname{vol}(\mathcal{O}^\times; d^\times a).$$

*Proof.* Using the translation-invariance of $d^\times a$, we find that

$$\int_{\mathcal{O}^\times} \chi(a) \, |a|^s \, d^\times a = \sum_{n \geq 0} \chi(\varpi^n a) q^{-ns} \int_{\mathcal{O}^\times} \chi(a) \, d^\times a,$$

so the result follows. ∎

Thus, we see that there is a finite set $S$ for which

$$Z(s, \chi, f) = \prod_{v \notin S} L_v(s, \chi_v) \cdot \prod_{v \in S} Z_v(s, \chi_v, f_v).$$

Indeed, one can just take $S$ to contain the archimedean places, the places where $\mathrm{vol}(\mathcal{O}_v^\times; d_v^\times x) \neq 1$, and the places where $f_v$ is not $1_{\mathcal{O}_v}$. Now, the finite product is relatively easy to understand, and an explicit calculation shows that $Z_v(s, \chi_v, f_v)$ admits a meromorphic calculation. For nonarchimedean places, one can argue as above by turning the integral into a geometric series; for archimedean places, one needs to do some analysis with the Schwartz hypothesis.

The point is that meromorphic continuation for $Z(s, \chi, f)$ directly implies meromorphic continuation for the $L$-function $L(s, \chi)$. In fact, one can argue as in Theorem 1.117 to see that Theorem 1.116 implies that $L(s, \chi)$ admits meromorphic continuation, functional equation, and it has prescribed poles.

### 1.5.2 Proof of the Global Functional Equation

We are now ready to prove Theorem 1.116.

*Proof of Theorem 1.116.* Define

$$Z_\pm(s, \chi, f) := \int_{|a|^{\pm 1} > 1} f(a)\chi(a)\,|a|^s \; d^\times a.$$

Because $|a| > 1$, we see that $Z_+(s, \chi, f)$ converges everywhere: both $Z_+$ and $Z_-$ already converges for $\operatorname{Re} s > 1$, and $Z_+$ will only get smaller as $\operatorname{Re} s$ gets smaller.

The idea to prove (a) is to relate $Z_-(s, \chi, f)$ with $Z_+(1-s, \chi^{-1}, \widehat{f})$. Indeed, because $\chi\,|\cdot|^s$ factors through $F^\times \backslash \mathbb{A}_F^\times$, we may "unfold" our integral, writing

$$\begin{aligned} Z_-(s, \chi, f) &= \int_{\substack{a \in \mathbb{A}_F^\times \\ |a| < 1}} f(a)\chi(a)\,|a|^s \; d^\times a \\ &= \int_{\substack{a \in F^\times \backslash \mathbb{A}_F^\times \\ |a| < 1}} \left( \sum_{\gamma \in F^\times} f(\gamma a) \right) \chi(a)\,|a|^s \; d^\times a. \end{aligned}$$

We would like to apply Poisson summation, but we need to add back in $0 \in F$ for this to make sense. To this end, define $f_a \in \mathcal{S}(\mathbb{A})$ by $f_a(x) := f(ax)$. Then

$$Z_-(s, \chi, f) = \int_{\substack{a \in F^\times \backslash \mathbb{A}_F^\times \\ |a| < 1}} \left( \sum_{\gamma \in F} f_a(\gamma) \right) \chi(a)\,|a|^s \; d^\times a - f(0) \int_{\substack{a \in F^\times \backslash \mathbb{A}_F^\times \\ |a| < 1}} \chi(a)\,|a|^s \; d^\times a.$$

We now apply Theorem 1.75. Note $\widehat{f_a}(x) = |a|^{-1}\,\widehat{f}(x/a)$, so we see

$$Z_-(s, \chi, f) = \int_{\substack{a \in F^\times \backslash \mathbb{A}_F^\times \\ |a| < 1}} \left( \sum_{\gamma \in F} \widehat{f}(\gamma/a) \right) \chi(a)\,|a|^{s-1} \; d^\times a - f(0) \int_{\substack{a \in F^\times \backslash \mathbb{A}_F^\times \\ |a| < 1}} \chi(a)\,|a|^s \; d^\times a,$$

which by sending $a$ to $a^{-1}$ gives

$$Z_-(s, \chi, f) = \int_{\substack{a \in F^\times \backslash \mathbb{A}_F^\times \\ |a| > 1}} \left( \sum_{\gamma \in F} \widehat{f}(\gamma a) \right) \chi^{-1}(a)\,|a|^{1-s} \; d^\times a - f(0) \int_{\substack{a \in F^\times \backslash \mathbb{A}_F^\times \\ |a| < 1}} \chi(a)\,|a|^s \; d^\times a.$$

It may look like we should send $d^\times a$ to $-d^\times a$, but this sign is absorbed into the orientation: we start integrating $(0, 1]$ and want to end integrating $[1, \infty)$.

Let's spend a moment to simplify the right-hand term. By fixing $|a|$, this integral is

$$\int_0^1 \left( \int_{F^\times \backslash \mathbb{A}_F^{\times,1}} \chi(a t_\infty) t^s \; d^\times a \right) d^\times t.$$

Here, $t_\infty$ is some idéle supported at a single place, chosen so that $|t_\infty| = t$; it is found basically by splitting the norm map $|\cdot| : \mathbb{A}_F^\times \to \mathbb{R}^+$.[1] Now, if $\chi|_{\mathbb{A}_F^{\times,1}} \neq 1$, then the internal integral vanishes; otherwise, if $\chi = |\cdot|^t$, then the inner integral is $\operatorname{vol}\left(F^\times \backslash \mathbb{A}_F^{\times,1}; d^\times x\right)$, and the outer integral then gives $\int_0^1 t^{s+t}\,\frac{dt}{t} = \frac{1}{s+t}$. Thus,

$$Z_-(s,\chi,f) = \int_{\substack{a \in F^\times \backslash \mathbb{A}_F^\times \\ |a|>1}} \left( \sum_{\gamma \in F} \widehat{f}(\gamma a) \right) \chi^{-1}(a)\,|a|^{1-s}\,d^\times a - \frac{f(0)\operatorname{vol}\left(F^\times \backslash \mathbb{A}_F^{\times,1}\right)}{s+t} 1_{\chi(\mathbb{A}_F^{\times,1})=1}.$$

We now take out the $\gamma = 0$ term from the internal sum and repeat the procedure of the previous paragraph. (There are some nasty sign problems here. The difficulty is that the case of $\chi|_{\mathbb{A}_F^{\times,1}} = 1$ receives an integral of $\int_0^1 t^{1-s-t}\,dt/t = -1/(1-s-t)$.) Being careful with our factors, we are left with

$$Z_-(s,\chi,f) + \frac{f(0)\operatorname{vol}\left(F^\times \backslash \mathbb{A}_F^{\times,1}\right)}{s+t} 1_{\chi(\mathbb{A}_F^{\times,1})=1} = Z_+(1-s,\chi^{-1},\widehat{f}) + \frac{\widehat{f}(0)\operatorname{vol}\left(F^\times \backslash \mathbb{A}_F^{\times,1}\right)}{s-1-t} 1_{\chi(\mathbb{A}_F^{\times,1})=1}.$$

The right-hand side now has good analytic properties, so we receive our meromorphic continuation, and the location (and type) of the poles follows from the expression as well. Sending $f \mapsto \widehat{f}$ and $s \mapsto (1-s)$ and $\chi \mapsto \chi^{-1}$ and summing completes the proof of the functional equation. Indeed, we find

$$Z_+(s,\chi,f) - \frac{f(0)\operatorname{vol}\left(F^\times \backslash \mathbb{A}_F^{\times,1}\right)}{s+t} 1_{\chi(\mathbb{A}_F^{\times,1})=1} = Z_-(1-s,\chi^{-1},\widehat{f}) - \frac{\widehat{f}(0)\operatorname{vol}\left(F^\times \backslash \mathbb{A}_F^{\times,1}\right)}{s-1-t} 1_{\chi(\mathbb{A}_F^{\times,1})=1},$$

so summing gives functional equation. ■

**Remark 1.122.** Basically the same proof works for function fields as soon as we choose an isomorphism $\mathbb{A}_F \to \mathbb{A}_F^*$, which amounts to the data of an isomorphism $\mathbb{A}_F \to \mathbb{A}_{\omega_F}$, which is the data of a nonzero meromorphic differential form $\theta$. Even though this identification depends on the choice of $\theta$, it turns out that the self-dual Haar measure does not. One checks this by comparing the self-dual Haar measure for $\theta$ and some $f\theta$, for any $f \in K(X)^\times$. Alternatively, one can show that the volume of $\prod_v \mathcal{O}_v^\times$ with respect to the self-dual Haar measure only depends on the genus of $X$.

### 1.5.3 A Little Geometric Class Field Theory

Let's give a few remarks about the argument for function fields.

**Remark 1.123.** One can show that Theorem 1.116 for $\chi = 1$ and function fields $\mathbb{F}_q(X)$ amounts to the functional equation for $\zeta_X$. This is on the homework.

One may be interested in what happens for nontrivial $\chi$. For example, if $\chi$ is unramified, then it factors through $F^\times \backslash \mathbb{A}_F^\times / \prod_v \mathcal{O}_v^\times = \operatorname{Pic} X$. To think about such characters geometrically, we need some class field theory. Indeed, class field theory grants a reciprocity map $\operatorname{Art}_F \colon F^\times \backslash \mathbb{A}_F^\times \to \operatorname{Gal}(F^{\mathrm{sep}}/F)^{\mathrm{ab}}$, which turns out to fit into a pullback square as follows.

$$\begin{tikzcd} F^\times \backslash \mathbb{A}_F^\times \arrow[r, "\operatorname{Art}_F"] \arrow[d, "\deg"'] & \operatorname{Gal}(F^{\mathrm{sep}}/F)^{\mathrm{ab}} \arrow[d, two heads] \\ \mathbb{Z} \arrow[r, "\operatorname{Frob}_q"] & \operatorname{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q) \end{tikzcd}$$

Now, note that finite étale covers of $X$ correspond to everywhere unramified field extensions of $F$, so we see that $\operatorname{Gal}(F^{\mathrm{sep}}/F) = \pi_1^{\text{ét}}(X)$. The quotient on the other side is $F^\times \backslash \mathbb{A}_F^\times / \prod_v \mathcal{O}_v^\times = \operatorname{Pic} X$. Taking the kernel

[1] For number fields, we can split by choosing any archimedean place. For function fields, I think something similar is possible.

with respect to degree, we may track around the following diagram.

$$\begin{tikzcd}
 & \pi_1^{\text{ét}}(X_{\overline{\mathbb{F}}_q})^{\text{ab}} \arrow[d, two heads] \\
F^\times \backslash \mathbb{A}_F^{\times,1} / \prod_v \mathcal{O}_v^\times \arrow[r] \arrow[d, hook] & \ker \arrow[d, hook] \\
F^\times \backslash \mathbb{A}_F^{\times} / \prod_v \mathcal{O}_v^\times \arrow[r] \arrow[d] & \pi_1^{\text{ét}}(X)^{\text{ab}} \arrow[d] \\
\operatorname{Pic} X \arrow[r, "\deg"] & \widehat{\mathbb{Z}}
\end{tikzcd}$$

Indeed, we see that $\operatorname{Pic}^0 X$ maps to the kernel of $\pi_1^{\text{ét}}(X) \to \widehat{\mathbb{Z}}$, which at least admits a surjection from $\pi_1^{\text{ét}}(X_{\overline{\mathbb{F}}_q})^{\text{ab}}$.[2] Thus, $\chi$ produces a homomorphism

$$\pi_1^{\text{ét}}(X_{\overline{\mathbb{F}}_q}) \to \mathbb{C}^\times,$$

which is the data of a local system $\mathcal{L}_\chi$ on $X_{\overline{\mathbb{F}}_q}$ of rank $1$. (This map even extends to $\pi_1^{\text{ét}}(X)$ by the same argument, so we receive a local system on $X$.)[3] It now turns out that

$$L(s,\chi) = \prod_{i=0}^{2} \det\left(1 - \operatorname{Frob}_q q^{-s}; \mathrm{H}^i(X_{\overline{\mathbb{F}_q}}; \mathcal{L}_\chi)\right)^{(-1)^{i+1}},$$

which basically follows from the Lefschetz trace formula. For example, if $\chi$ vanishes on $\operatorname{Pic}^0 X$, then the $i = 0$ term is $1 - \chi(a)q^{-s}$ (for some $a \in \mathbb{A}_F^\times$ with $\deg a = 1$), and the $i = 2$ term is $1 - \chi(a)q^{1-s}$. But if $\chi$ is nontrivial on $\operatorname{Pic}^0 X$, then $\mathrm{H}^0(X; \mathcal{L}_\chi)$ vanishes, so $\mathrm{H}^2$ also vanishes by duality, so we only have $\mathrm{H}^1$. The total degree should remain the same no matter what $\chi$ is (by some Euler characteristic calculation), so we see that

$$L(s,\chi) = \det\left(1 - \operatorname{Frob}_q q^{-s}; \mathrm{H}^1(X_{\overline{\mathbb{F}}_q}; \mathcal{L}_\chi)\right)$$

is some polynomial of degree $2g - 2$. Then one can see that Theorem 1.116 yields

$$L(1-s,\chi^{-1}) = q^{(2g-2)s} \det\left(\operatorname{Frob}_q; \mathrm{H}^*(X_{\overline{\mathbb{F}_q}}; \mathcal{L}_\chi)\right) L(s,\chi).$$

### 1.5.4 Local Theory

Let's say a few sentences about the local theory.

**Definition 1.124** (local integral)**.** Fix a local field $F$. For any $f \in \mathcal{S}(F)$ and continuous $\chi\colon F^\times \to \mathbb{C}^\times$, we define

$$Z(s,\chi,f) := \int_{F^\times} f(a)\chi(a)\,|a|^s \; d^\times a.$$

**Remark 1.125.** A direct calculation shows that $Z(s,\chi,f)$ is some Laurent polynomial in $q^{-s}$ (which depends on $f$) with the controlled denominator $L(s,\chi)$, which is the local $L$-factor.

**Theorem 1.126.** Fix a local field $F$ and a nontrivial character $\psi\colon F \to \mathbb{C}^\times$. Then

$$Z\left(1-s,\chi^{-1},\widehat{f}\right) = \gamma(s,\chi,\psi) Z(s,\chi,f)$$

for some function $\gamma(s,\chi,\psi)$ which is independent of $f$.

[2] It turns out that the kernel is exactly the Frobenius co-invariants of $\pi_1^{\text{ét}}(X_{\overline{\mathbb{F}}_q})$, which is a finite group.

[3] It is occasionally convenient to pass from $\pi_1^{\text{ét}}(X)$ to the Weil group.

*Proof.* Omitted. ■

**Remark 1.127.** One can decompose $\gamma$ further into an $\varepsilon$-factor, which is

$$\frac{Z\left(1-s,\chi^{-1},\widehat{f}\right)/L\left(1-s,\chi^{-1}\right)}{Z(s,\chi,f)/L(s,\chi)}.$$

**Remark 1.128.** Combining the local and global functional equations reveals that there is some $\varepsilon(s,\chi)$ for which $L(s,\chi)=\varepsilon(s,\chi)L\left(1-s,\chi^{-1}\right)$, and

$$\varepsilon(s,\chi)=\prod_v \varepsilon_v(s,\chi_v,\psi_v).$$

This factorization is fairly surprising! For example, in the function field situation, we have factored some determinant of Frobenius action on cohomology into a product of local terms.

# Theme 2
# Nonarchimedean Representation Theory

## 2.1 February 18

Today we say something about representation theory.

### 2.1.1 Overview

Approximately speaking, given a reductive group $G$ over a global field $F$, we will be interested in "automorphic representations," which are roughly speaking those representations of $G(\mathbb{A}_F)$ which appear in $L^2(G(F)\backslash G(\mathbb{A}_F))$. Such representations turn out to have poor categorical properties, so there are some technicalities involved in the definition of "automorphic."

Because $G$ is an affine group scheme over $F$, one finds that $G(\mathbb{A}_F)$ is the restricted direct product

$$G(\mathbb{A}_F) = \prod_v (G(F_v), G(\mathcal{O}_v)),$$

where we have given $G$ an integral model which extends over all but finitely many places of $F$. Thus, we should start by discussing some "local" representation theory. We'll start with $p$-adic groups, whose representations are understood via a Hecke algebra; then we will turn to real groups, whose representations are understood via $(\mathfrak{g}, K)$-modules.

Once we have control over local representation theory, we can find automorphic representations of $G(\mathbb{A}_F)$ to be given as a (restricted) tensor product of local representations. Eventually, for $G = \mathrm{GL}_2$, we will be able to relate all this to the classical theory of modular forms.

### 2.1.2 Totally Disconnected Groups

For now, $F$ will be a nonarchimedean local field with ring of integers $\mathcal{O}$ and residue field $k$.

**Definition 2.1.** An *algebraic group* $G$ over a field $F$ is an affine group scheme of finite type over $F$.

*Proof.* It turns out that any such $G$ admits a closed embedding into $\mathrm{GL}_n$ for some $n$. This amounts to the statement that $G$ admits a faithful representation, which can be found by taking a suitable regular representation. ■

When $F$ is a local field (or more generally, a Hausdorff topological ring), one can give $G(F)$ a topology so that closed embeddings of schemes (over $F$) go to closed embeddings of topological spaces, and the topology of $\mathbb{A}^n(F) = F^n$ is the obvious one.

**Example 2.2.** The group $\mathrm{GL}_n$ admits a closed embedding into $\mathbb{A}^{n+1}$ by $g \mapsto \big((g_{ij}), \det g^{-1}\big)$.

**Example 2.3.** If $F$ is a nonarchimedean local field, then $\det\colon \mathrm{GL}_n(F) \to F^\times$ is already continuous, so in fact the map

$$\mathrm{GL}_n(F) \to M_n(F)$$

is an open embedding. Indeed, by construction of the topology, an open neighborhood basis of $\mathrm{GL}_n(F)$ are the set

$$K_{a,b} \coloneqq \{A \in \mathrm{GL}_n(F) : A - 1 \in \mathfrak{p}^a M_n(\mathcal{O}), \det A \in (1 + \mathfrak{p}^b)\},$$

but $\det^{-1}$ is continuous as a map from the subspace of $n \times n$ invertible matrices to $F^\times$ (indeed, $\det^{-1}$ is a rational function), so we may always expand $a$ to satisfy the $b$ condition.

**Remark 2.4.** Set

$$K_m \coloneqq \{A \in \mathrm{GL}_n(\mathcal{O}_F) : (A - 1) \in \mathfrak{p}^n M_n(\mathcal{O})\}.$$

These are compact open subgroups of $\mathrm{GL}_n(\mathcal{O}_F)$. By definition, $K_m$ is the kernel fitting in the exact sequence

$$1 \to K_m \to \mathrm{GL}_n(\mathcal{O}_F) \to \mathrm{GL}_n(\mathcal{O}/\mathfrak{p}^m).$$

**Example 2.5.** Fix a quadratic space $V$ over a field $F$. One can now describe the topology on a classical group $\mathrm{O}(V)(F)$ as the closed subspace topology from $\mathrm{GL}(V)$.

Having so many compact open subgroups is a remarkable property.

**Definition 2.6** (totally disconnected)**.** A topological group $G$ is *totally disconnected* if and only if $G$ is Hausdorff, second countable, and admits an open neighborhood basis of the identity consisting of compact open subgroups.

**Example 2.7.** For any affine algebraic group $G$ over a local field $F$, we can embed $G(F) \subseteq \mathrm{GL}_n(F)$. Then the countable open compact neighborhood basis for $\mathrm{GL}_n(F)$ makes $G(F)$ totally disconnected.

### 2.1.3 Smooth Representations

We will be able to do quite a bit with general totally disconnected groups.

**Definition 2.8** (smooth)**.** Fix a totally disconnected group $G$. Then a complex representation $V$ of $G$ is *smooth* if and only if the action map

$$G \times V \to V$$

is continuous, where $V$ has been given the discrete topology. A *homomorphism* of smooth representations is a morphism of the underlying representations of $G$.

**Remark 2.9.** Equivalently, we are asking for each $v \in V$ to have open pre-image in $G \times V$. Because $V$ has been given the discrete topology, this simply means that $\mathrm{Stab}_G(v) \subseteq G$ is open for all $v \in V$. Note that it is enough for the subgroup to merely contain an open neighborhood $U$ of $1$ because then $\mathrm{Stab}_G(v)$

**Example 2.10.** Fix a nonarchimedean local field $F$. Any continuous homomorphism $F^\times \to \mathbb{C}^\times$ admits some open subgroup $1 + \ker \mathfrak{p}^\bullet$ in its kernel, by a "no small subgroups" argument.

Sometimes our representations are not smooth, but we can find smooth representations within.

**Definition 2.11** (smooth)**.** Fix a complex representation $V$ of a totally disconnected group $G$. A vector $v \in V$ is *smooth* if and only if $\mathrm{Stab}_G(v)$ is open. We let $V^{\mathrm{sm}}$ denote the collection of smooth vectors.

**Remark 2.12.** By taking intersections of open subgroups, we see that linear combinations of smooth vectors continue to be smooth. It follows that $V^{\mathrm{sm}}$ is a subspace, and we can see that it is even $G$-invaraint.

**Example 2.13** (smooth functions)**.** Let $V$ be the space of all functions $G \to \mathbb{C}$, and let $G$ act on $V$ by $(gf)(x) := f(xg)$. (One could also restrict to those functions with compact support.) However, functions $f\colon G \to \mathbb{C}$ are rarely smooth: this amounts to requiring that $f(xg) = f(x)$ for $g$ contained in an open subgroup of $G$, meaning that $f$ descends to a quotient $G/K$ for an open subgroup $K$ of $G$. Equivalently, we may write this representation as

$$\bigcup_{m \geq 1} \mathrm{Mor}(G/K_m, \mathbb{C}),$$

where $\{K_m\}$ is some countable compact open neighborhood basis of the identity.

These smooth functions will be useful to us.

**Notation 2.14.** Fix a totally disconnected group $G$. We let $C_c^\infty(G)$ denote the collection of locally constant functions on $G$ of compact support.

**Remark 2.15.** It turns out that functions in $f \in C_c^\infty(G)$ admit an open neighborhood $K$ for which

$$f(kxk') = f(x)$$

for any $k, k' \in K$. Indeed, $f$ can be written as a finite linear combination of indicators of compact open subsets. By translating, we may further reduce to the case where $f$ is the indicator of some compact open subgroup $K'$, for which we can take $K := K'$.

**Example 2.16** (induction)**.** Given a subgroup $H \subseteq G$, we can let $G$ act by right translation on functions $H\backslash G \to \mathbb{C}$. Then $\mathrm{Mor}(H\backslash G, \mathbb{C})$ is the induction of the trivial representation from $H$ to $G$.

### 2.1.4 Categorical Properties

Let's make a few remarks about functoriality.

**Notation 2.17.** Fix a totally disconnected group $G$. The category of smooth representations of $G$ will be denoted $\mathrm{Rep}_{\mathbb{C}}^{\mathrm{sm}}(G)$.

**Remark 2.18.** By definition, $\mathrm{Rep}_{\mathbb{C}}^{\mathrm{sm}}(G)$ is a full subcategory of $\mathrm{Rep}_{\mathbb{C}}(G)$.

**Remark 2.19.** One can check that the direct sum, subrepresentation, and quotients of smooth representations all continue to be smooth. It follows that $\mathrm{Rep}_{\mathbb{C}}^{\mathrm{sm}}(G)$ is an abelian category, and it admits limits and colimits.

There is also a dual, but we must be careful because functionals have no reason to be smooth.

**Definition 2.20** (contragredient)**.** Fix a smooth representation $V$ of a totally disconnected group $G$. Then the *contragredient* $V^\vee$ consists of the smooth vectors in $V^*$.

**Remark 2.21.** Explicitly, $V^*$ consists of those functionals $\varphi\colon V \to \mathbb{C}$ for which there is an open subgroup $K \subseteq G$ for which

$$\varphi(kv) = \varphi(v)$$

for all $v \in V$ and $k \in K$.

**Remark 2.22.** By construction, we see that $(V^\vee)^K = \mathrm{Mor}(V_K, \mathbb{C})$, where $V_K$ denotes the "coinvariants."

The category of smooth representations is quite flexible, but smooth representations will occasionally be "too big" for our purposes. We don't want to require that our representations are fully finite-dimensional, but it will be nice to have some finiteness.

**Definition 2.23** (admissible)**.** Fix a smooth representation $V$ of a totally disconnected group $G$. Then $V$ is *admissible* if and only if

$$\dim V^K < \infty$$

for any compact open subgroup $K \subseteq G$.

**Remark 2.24.** Note that $V$ being smooth means that $V = \bigcup_{K \subseteq G} V^K$, so admissibility means that $V$ is a union of finite-dimensional vector spaces.

**Non-Example 2.25.** If $G$ fails to be compact, then the right regular representation $\mathrm{Mor}(G, \mathbb{C})^{\mathrm{sm}}$ is not admissible. Indeed, for any open compact subgroup $K \subseteq G$, we see that $\mathrm{Mor}(G, \mathbb{C})^K = \mathrm{Mor}(K\backslash G, \mathbb{C})$, which is infinite-dimensional because $K\backslash G$ is not compact.

**Example 2.26.** Even if $G$ fails to be compact, we may be able to find a closed subgroup $H \subseteq G$ such that $H\backslash G$ is compact. Then for any compact open subgroup $K \subseteq G$, we see that $H\backslash G/K$ is finite (because $K$ is open), so

$$\dim \mathrm{Mor}(H\backslash G, \mathbb{C})^K < \infty.$$

It follows that the induction $\mathrm{Mor}(H\backslash G, \mathbb{C})^{\mathrm{sm}}$ is admissible.

**Remark 2.27.** One can check that the natural map $V \to (V^\vee)^\vee$ is an isomorphism if $V$ is admissible. This is shown on the homework.

**Remark 2.28.** For a fixed compact open subgroup $K \subseteq G$, we will show on the homework that there is a canonical isomorphism

$$\bigoplus_{\rho \in \mathrm{Irr}\, K} V_\rho \otimes \mathrm{Hom}_K(V_\rho, V) \to V$$

for any smooth $V$. It turns out that $V$ is admissible if and only if the multiplicity spaces $\mathrm{Hom}_K(V_\rho, V)$ are finite-dimensional.

### 2.1.5 Hecke Algebra

Our next task is to realize our category of representations as modules over a convenient algebra.

**Example 2.29.** For any group $G$, we know that $\mathrm{Rep}_{\mathbb{C}}(G) \cong \mathrm{Mod}(\mathbb{C}[G])$: indeed, a $G$-action on a vector space extends uniquely (linearly) to an action by $\mathbb{C}[G]$.

**Remark 2.30.** It turns out to be more convenient to view $\mathbb{C}[G]$ (dually) as the functions $G \to \mathbb{C}$ with finite support. In this case, the multiplication is given by convolution: one has

$$(f_1 * f_2)(x) := \sum_{y \in G} f_1(y) f_2\left(y^{-1}x\right).$$

Indeed, we can see that $1_{g_1} * 1_{g_2} = 1_{g_1 g_2}$, so this is the multiplication that we expect.

Motivated by the above remark, we find the following.

**Definition 2.31** (Hecke algebra)**.** Fix a totally disconnected group $G$. Then we define the *Hecke algebra* $\mathcal{H}_G$ to be $C_c^\infty(G)$. If no confusion is possible, we will write $\mathcal{H}$ for $\mathcal{H}_G$. Upon fixing a left Haar measure $dg$, we may define a convolution operation by

$$(f_1 * f_2)(x) := \int_G f_1(g) f_2\left(g^{-1}x\right)\, dg.$$

The convolution operation is distributive. To see that it is associative, note that

$$((f_1 * f_2) * f_3)(x) = \int_G \int_G f_1(h) f_2\left(h^{-1}g\right) f_3\left(g^{-1}x\right)\, dh\, dg,$$

but

$$(f_1 * (f_2 * f_3))(x) = \int_G \int_G f_1(g) f_2(h) f_3\left(h^{-1}g^{-1}x\right)\, dh\, dg.$$

These integrals are seen to be the same by sending $h \mapsto g^{-1}h$.

**Remark 2.32.** There was some concern about if $dg$ should be left- or right-invariant. If $G(F)$ is the $F$-points of a reductive algebraic group, then it turns out that $G(F)$ is unimodular (which one sees by an explicit construction of this measure by choosing a trivialization of $\wedge^{\dim G} TG$), so this doesn't matter!

However, the algebra is not unital.

**Remark 2.33.** If $G$ is discrete, then we can normalize $dg$ to be the counting measure. Then $\mathcal{H}$ admits a unit $\delta_1$ given by the indicator at the identity: one can directly compute $f * \delta_1 = f$. As such, we do not expect $\mathcal{H}$ to be unital.

**Remark 2.34.** The algebra $\mathcal{H}$ admits many idempotents: for any compact open subgroup $K \subseteq G$, note that $1_K * 1_K = \mathrm{vol}(K; dg) 1_K$, so $e_K := \mathrm{vol}(K; dg)^{-1} 1_K$ is an idempotent element! In fact, these elements form an approximate identity: for any function $f$, one can find $K$ small enough so that $e_K * f = f$.

It is somewhat inconvenient that the convolution structure depends on the choice of $dg$. Let's fix this.

**Remark 2.35.** It is actually more convenient to view $\mathcal{H}$ as locally constant, compactly supported measures on $G$, which have an easier time acting on representations. In other words, we are looking at measures of the form $f(g)\,dg$, where $f \in C_c^\infty(G)$. The reason this is convenient is that the multiplication now becomes canonical: given two measures $\mu_1$ and $\mu_2$, note that the product measure $\mu_1 \boxtimes \mu_2$, which continues to have compact support and be locally constant. Thus, there is a pushforward measure $m_*(\mu_1 \boxtimes \mu_2)$, which is again locally constant of compact support. One can unwind the definitions to see that $m_*(f_1\,dg \boxtimes f_2\,dg) = (f_1 * f_2)\,dg$, so this operation agrees with the previous one.

**Example 2.36.** For a compact open subgroup $K \subseteq G$, the elements $e_K := \operatorname{vol}(K; dg)^{-1} 1_K\,dg$ are idempotent: indeed, one can check that $e_K * e_K = e_K$ directly. In fact, if $K' \supseteq K$, then $e_K * e_{K'} = e_{K'} * e_K = e_{K'}$. For example,

$$(e_K * e_{K'})(x) = \frac{1}{\operatorname{vol}(K; dg)\operatorname{vol}(K'; dg)} \int_G e_K(g) e_{K'}\left(g^{-1}x\right)\,dg$$

restricts immediately to those $g \in G$ with $g \in K$ and $g^{-1}x \in K'$. In other words, we are integrating over $K \cap xK'$, which is simply an indicator for $K$ times the measure of $K'$.

**Warning 2.37.** In the sequel, we may work with the Hecke algebra $\mathcal{H}$ as $C_c^\infty(G)\,dg$ instead of $C_c^\infty(G)$.

## 2.2 February 25

Today we continue our nonarchimedean representation theory.

### 2.2.1 Representations of the Hecke Algebra

Let's take a moment to explain how $\mathcal{H}$ acts on representations.

**Lemma 2.38.** Fix a totally disconnected group $G$. There is a fully faithful embedding $\operatorname{Rep}_{\mathbb{C}}^{\mathrm{sm}}(G) \to \operatorname{Mod}(\mathcal{H})$.

*Proof.* For (a), the action is given by

$$\mu \cdot v := \int_G gv\,d\mu(g).$$

Note that the integral always converges because $\mu$ has compact support. By an explicit calculation with $f\,dg$s, one can check that this action makes $V$ into an $\mathcal{H}$-module. This functor is of course faithful (because linear maps are just sent to themselves).

It remains to check that our functor is full. Fix a linear map $\varphi$ between two smooth representations $V$ and $W$, and suppose that $\varphi$ preserves the $\mathcal{H}$-module structure. We would like to show that $g\varphi(v) = \varphi(gv)$ for any $g \in G$ and $v \in V$. Well, for any sufficiently small compact open subgroup $K$ of $g$, we see that $e_K v = v$, so the given equation is equivalent to $\varphi$ preserving the action by the translated measure $ge_K$. ■

**Remark 2.39.** Further, by shrinking $K$, we may assume that $v \in V^K$ and that $\mu = f\,dg$ where $f$ is $K$-invariant. It follows that

$$\mu \cdot v = \sum_{g \in G/K} f(g)\operatorname{vol}(K; dg)(gv)$$

by factoring the integral.

We use this to give a definition of the "smooth" $\mathcal{H}$-modules.

**Definition 2.40** (smooth)**.** Fix a totally disconnected group $G$. Then an $\mathcal{H}$-module $M$ is *smooth* if and only if

$$M = \bigcup_{K \subseteq G} e_K M.$$

**Remark 2.41.** If $V$ is a smooth representation of $G$, then one can check that $e_K$ projects $V$ onto $V^K$. Indeed, $e_K$ certainly fixes $V^K$, and for any $k \in K$ and $v \in V$, we see that $k(e_K v) = e_K v$ by pushing the linear map $k$ through the integral

$$e_K v = \frac{1}{\operatorname{vol}(K; dg)} \int_K gv \, dg.$$

Thus, smooth representations of $G$ go to smooth $\mathcal{H}$-modules.

**Remark 2.42.** Conversely, given a smooth $\mathcal{H}$-module $M$, we can write

$$M = \operatorname*{colim}_{K \subseteq G} e_K M.$$

Then each $e_K M$ upgrades from an $\mathcal{H}$-module to a representation of $G/K$ and thus by inflation to a representation of $G$. Thus, the colimit upgrades to a smooth representation of $G$ because the category of smooth representations is closed under such colimits. This construction provides an inverse functor to the one in Lemma 2.38, thereby providing an equivalence of categories from smooth representations to smooth modules.

It is notable that all of our discussion can also be made to work globally for function fields.

**Example 2.43.** If $G$ is an affine algebraic group over a function field $F$, then $\mathbb{A}_F$ is totally disconnected, so $G(\mathbb{A}_F)$ is a totally disconnected. For example, it turns out that

$$\mathrm{GL}_n(\mathbb{A}_F) = \prod_v (\mathrm{GL}_n(F_v), \mathrm{GL}_n(\mathcal{O}_v)),$$

which one can see directly from the construction of the topology on $G(\mathbb{A}_F) \subseteq \mathbb{A}_F^{n^2+1}$. The basic open subsets of $\mathrm{GL}_n(\mathbb{A}_F)$ can thus be indexed by effective divisors $D$ on $X$: indeed, given such a $D = \sum_v n_v v$, we can define $K_D$ as the kernel of the natural projection

$$\prod_v \mathrm{GL}_n(\mathcal{O}_v) \twoheadrightarrow \prod_{n_v \neq 0} \mathrm{GL}_n(\mathcal{O}_v / \varpi_v^{n_v}).$$

In particular, because this map is surjective, we see that $K_D$ has finite index in $\prod_v \mathrm{GL}_n(\mathcal{O}_v)$, where the index is given by the size of the right-hand group.

**Remark 2.44.** Fix an affine algebraic group $G$ over a global field $F$. Then one may want to write down a decomposition $G(\mathbb{A}_F)$ as a restricted product, but this requires us to make sense of $G(\mathcal{O}_v)$ for all but finitely many places $v$. Luckily, this is not too hard: one simply can choose any model $\mathcal{G}$ over $\operatorname{Spec} \mathcal{O}_F$, and then we pass to the open subset of $\operatorname{Spec} \mathcal{O}_F$ where $G$ remains an affine algebraic group. Because any two models $\mathcal{G}$ and $\mathcal{G}'$ are generically isomorphic to $G$ (namely, over $F$), one can spread out the composite isomorphism $\mathcal{G}_F \to G \leftarrow \mathcal{G}'_F$ to show that $\mathcal{G}$ and $\mathcal{G}'$ are isomorphic at all but finitely many places. Thus, the restricted product $\prod_v (G(F_v), \mathcal{G}(\mathcal{O}_v))$ does not depend on the choice of model $\mathcal{G}$.

Thus, for an affine algebraic group $G$ over a function field $F$, one has a Hecke algebra $\mathcal{H}_{G(\mathbb{A}_F)}$. We will later want a Hecke algebra also for number fields, which requires us to find a Hecke algebra for archimedean

groups. We will return to this later.

Let's do some more representation theory.

**Proposition 2.45** (Schur's lemma)**.** Fix admissible irreducible representations $V_1$ and $V_2$ of a totally disconnected group $G$. Then

$$\dim_{\mathbb{C}} \operatorname{Hom}_G(V_1, V_2) = 1_{V_1 \cong V_2}.$$

*Proof.* Note that any nonzero map $V_1 \to V_2$ has neither full kernel nor full cokernel, so the irreducibility implies that any nonzero map is an isomorphism. Thus, if $V_1$ and $V_2$ are not isomorphic, $\dim \operatorname{Hom}_G(V_1, V_2) = 0$.

It remains to work in the case $V_1 \cong V_2$. We will write $V$ for both representations, for brevity, so we want to show that $\operatorname{End}_G(V) = \mathbb{C}$. The previous paragraph has shown that any nonzero element of $D := \operatorname{End}_G(V)$ is invertible, so this is a division algebra. Furthermore, $D$ contains the scalars in $\mathbb{C}$, so $D$ is a division algebra over $\mathbb{C}$.

The rest of the argument will be by counting, using the bound that $V$ has countable dimension (because it is a countable union of finite subspaces).

- On one hand, suppose for the sake of contradiction that $D \setminus \mathbb{C}$ is nonempty. Because $\mathbb{C}$ is algebraically closed, this means that $D$ has a transcendental element $T$ over $\mathbb{C}$, so $D$ contains $\mathbb{C}(T)$. But the elements $\{1/(T-a) : a \in \mathbb{C}\}$ are linearly independent, so we conclude that $\dim_{\mathbb{C}} D$ is uncountable.
- On the other hand, $\dim_{\mathbb{C}} D$ is countable because $\dim V$ is countable: indeed, any nonzero vector $v \in V$ generates $V$ under the $G$-action, so a $G$-equivariant map $V \to V$ is uniquely determined by the image of $v$. Thus, the map $\operatorname{ev}_v \colon D \to V$ is injective, so $\dim_{\mathbb{C}} D$ is countable.

The above two points produce contradiction if $D \neq \mathbb{C}$, so we conclude that $D = \mathbb{C}$. ■

**Remark 2.46.** Suppose that $G$ is a reductive group over a nonarchimedean local field $F$. It turns out that all irreducible smooth representations $G(F)$ are automatically admissible. This is difficult to prove, so we will not.

**Remark 2.47.** Again, suppose that $G$ is a reductive group over a nonarchimedean local field $F$. Then Proposition 2.45 remains true for representations over $\overline{\mathbb{Q}}$, but a different argument is required. In short, one can still make a size argument by using the fact that the relevant universal enveloping algebra of the Lie algebra is filtered and has finitely generated associated graded algebra.

### 2.2.2 Parabolic Induction

We now fix a reductive group $G$ over a nonarchimedean local field $F$.

**Definition 2.48** (parabolic)**.** Fix a reductive group $G$ over a field $F$. Then a subgroup $P \subseteq G$ is *parabolic* if and only if $G/P$ is proper.

**Remark 2.49.** It is equivalent for $P_{\overline{F}}$ to contain a Borel subgroup $B_{\overline{F}}$, which is a maximal connected solvable subgroup of $G$. For example, $G_{\overline{F}}/B_{\overline{F}}$ can be seen to be proper because this is some sort of flag variety. Indeed, $G_{\overline{F}}/B_{\overline{F}}$ is the space of Borel subgroups, which are all conjugate to each other.

**Remark 2.50** (Levi decomposition)**.** Given a parabolic subgroup $P$ with unipotent radical $U$, the quotient $L := P/U$ is reductive. It turns out that the quotient map $P \twoheadrightarrow L$ admits a (non-canonical) splitting, so we may view $L$ as a subgroup of $P$.

**Example 2.51.** A Borel subgroup $B$ of $\mathrm{GL}_n$ is given by the upper-triangular matrices, and this is the only one up to conjugation. Thus, one can see that parabolic subgroups of $\mathrm{GL}_n$ look like

$$(\mathrm{GL}_{n_1} \times \cdots \mathrm{GL}_{n_d})\, B,$$

where $(n_1, \ldots, n_d)$ is some sequence of positive integers with sum $n$.

Let's attempt to define parabolic induction. Choose a smooth representation $V$ of the Levi quotient $L$. Then we may inflate $V$ into a representation of $P$ via the quotient map $P \twoheadrightarrow L$. We could attempt to define the induction to $G$ as consisting of those functions $f\colon G \to V$ such that

$$f(px) = pf(x),$$

where the $G$-action is given by $(gf)(x) := f(xg)$. This is certainly a well-defined representation, but it has a couple defects, which we now fix.

- This representation has no reason to be smooth. To fix this, we will want to work with those functions $f\colon G \to V$ admitting an open compact subgroup $K \subseteq G$ for which $kf = f$ for all $k \in K$.[1]
- It is useful to normalize by the modular character.

We thus have the following definition.

**Definition 2.52** (parabolic induction)**.** Fix a reductive group $G$ over a nonarchimedean field $F$, and let $P$ be a parabolic subgroup with Levi quotient $L$. Given a smooth representation $V$ of $L(F)$, we define the *parabolic induction* $\mathrm{Ind}_P^G V$ to consist of those functions $f\colon G(F) \to V$ with the following two properties.

- Smooth: there is an open compact subgroup $K \subseteq G(F)$ such that $f(xk) = f(x)$ for all $x \in G(F)$ and $k \in K$.
- Induction: for all $p \in P(F)$ and $x \in G(F)$, we have $f(px) = \delta_P(p)^{1/2}\tau(p)f(x)$, where $\delta_P$ is the modular character.

The $G$-action on $\mathrm{Ind}_P^G V$ is given by $(gf)(x) := f(xg)$.

Let's explain the presence of the modular character $\delta_P$, and thereby also recall its definition. (The following discussion works for all local fields $F$.) Note that $P(F)\backslash G(F)$ is a (nonarchimedean) manifold, so it admits an integration theory. Indeed, any manifold $X$ of dimension $n$ admits an integration theory as follows: if a top differential form $\theta \in \omega_X$ is supported on a single chart $U$ with local coordinates $\theta = f\, dx_1 \wedge \cdots \wedge dx_n$, then we can define

$$\int_X |\theta| := \int_U |f|\, dx_1 \ldots dx_n.$$

This expression is independent of the choice of chart, which one can check by some Jacobian calculation. From here, one can use a partition of unity to compute $\int_X |\theta|$ even if $\theta$ is not supported on a single chart. (The expression will further be independent of a choice of partition of unity, which we can see by further refining the partition of unity given two such choices.)

We now apply our integration theory to $X := P(F)\backslash G(F)$. The action of $P$ on $G$ is free, so we see that $T_e(P(F)\backslash G(F)) = \mathfrak{p}\backslash\mathfrak{g}$. It follows that

$$TX = (\mathfrak{p}\backslash\mathfrak{g}) \times^P G.$$

In particular, $TX$ admits a global frame (by choosing a basis of $\mathfrak{p}\backslash\mathfrak{g}$), so the line bundle of top differential forms is trivial, which we see by writing

$$\omega_X = \wedge^{\dim X}(\mathfrak{p}\backslash\mathfrak{g})^* \times^P G.$$

[1] This really is extracting those locally constant maps on $P(F)\backslash G(F)$ because $P(F)\backslash G(F)$ is a compact space.

But now $P$ acts on $\wedge^{\text{top}}(\mathfrak{p}\backslash\mathfrak{g})^*$ by the scalar

$$\det(p;(\mathfrak{p}\backslash\mathfrak{g})^*) = \frac{\det(p;\mathfrak{p})}{\det(p;\mathfrak{g})}.$$

The absolute value of this is denoted $\delta_P(p)$.

The relevance of $\delta_P$ to our integration theory is as follows: our discussion above explained how to integrate expressions of the form $|\theta|$ where $\theta \in \omega_X$. But we can merely view $|\omega_X|$ as some real line bundle. By choosing some top differential form, we can then identify sections of $|\omega_X|$ with functions $f\colon G(F) \to \mathbb{R}$ such that

$$f(px) = \delta(p)f(x)$$

for all $p \in P$ and $x \in G$. Thus, $\operatorname{Ind}_P^G V$ consists of sections of the line bundle $|\omega_X|^{1/2}$, which are known as "half-densities." The relevance of this for us is that two half-densities $f, g \in \operatorname{Ind}_P^G V$ will have a product which can be integrated. Eventually, this will imply that the parabolic induction of a unitary representation remains unitary.

## 2.3 March 2

There will be a quiz on Wednesday on some hopefully basic concepts.

### 2.3.1 The Modular Character

Let's explain how induction works in general.

**Definition 2.53** (induction)**.** Let $H$ be a closed algebraic subgroup of an algebraic group $G$ over a nonarchimedean local field $F$. For a representation $\tau$ of $H$, we define the *induction* $\operatorname{Ind}_H^G \tau$ to be the representation consisting of those functions $f\colon G(F) \to V$ with the following properties.

- Smooth: there is an open compact subgroup $K \subseteq G$ such that $f(gk) = f(g)$ for all $g \in G$ and $k \in K$.
- Induction: for all $h \in H(F)$, one has $f(hx) = \delta_{G/H}(h)^{1/2}\tau(h)f(x)$ for all $x \in G$.

The $G$-action is given by $(gf)(x) := f(xg)$.

Once again, we need to explain the presence of the character $\delta_{G/H}$, which is similar to our discussion from last class.

**Remark 2.54.** Top differential forms on the quotient $X := H(F)\backslash G(F)$ are given by sections of the line bundle $\omega_X$. The tangent space at the identity is simply $\det(\mathfrak{g}/\mathfrak{h})^*$, so we conclude that $\omega_X = \det(\mathfrak{g}/\mathfrak{h}) \times^H G$.

**Notation 2.55.** Given a line bundle $\mathcal{L}$ on a $p$-adic manifold $X$ with transition functions $\{g_{ij}\}$, we let $|\mathcal{L}|$ denote the real line bundle, whose transition functions are $\{|g_{ij}|\}$.

**Definition 2.56** (density)**.** Fix a $p$-adic manifold $X$. Then a *density* is a section of the real line bundle $|\omega_X|$.

**Remark 2.57.** For any $p$-adic manifold $X$, one can integrate compactly supported densities, simply by choosing local coordinates and evaluating the integral along some choice of partition of unity. In particular, one can evaluate local integrals against some choice of Haar measure on the base local field $F$.

We are now ready to define $\delta_{G/H}$.

**Definition 2.58.** Let $H$ be a closed algebraic subgroup of an algebraic group $G$ over a local field $F$, and let their Lie algebras be $\mathfrak{h}$ and $\mathfrak{g}$, respectively. Then we define $\delta_{G/H} \colon H \to \mathbb{R}^+$ by

$$\delta_{G/H}(x) \coloneqq \left|\det\left(\mathrm{Ad}_x; (\mathfrak{g}/\mathfrak{h})^*\right)\right|.$$

By undoing the dual and then lifting from the quotient, we see that

$$\det\left(\mathrm{Ad}_x; (\mathfrak{g}/\mathfrak{h})^*\right) = \frac{\det(\mathrm{Ad}_x; \mathfrak{h})}{\det(\mathrm{Ad}_x; \mathfrak{g})}.$$

These individual numerator and denominator characters are interesting in their own right.

**Definition 2.59** (modular character)**.** Fix a locally compact topological group $G$ with left-invariant Haar measure $\mu$. For any $g \in G$, we note that $\mathrm{Ad}_g^* \mu$ continues to be a left-invariant Haar measure, so it is the same as $\mu$ up to a unique scalar. We may thus define the *modular character*

$$\delta_G(g) \coloneqq \frac{\mathrm{Ad}_g^* \mu}{\mu}.$$

Note that $\delta_G$ is independent of the choice of $\mu$.

**Remark 2.60.** If $G$ is an affine algebraic group over a local field $F$ with Lie algebra $\mathfrak{g}$, then one can show that

$$\delta_G(g) = \left|\det(\mathrm{Ad}_g; \mathfrak{g})\right|^{-1}.$$

Indeed, the left-invariant Haar measure is simply given (algebraically!) by a choice of top form in $\wedge^n \mathfrak{g}^*$, where $n$ is the dimension of $G$. Then one sees that the scalar adjusting such a top form (and thus the Haar measure) is given by the adjoint action on $\wedge^n \mathfrak{g}^*$, which is what is given above.

**Remark 2.61.** It turns out that $G$ is unimodular if and only if $\delta_G$ is trivial. Indeed, $\delta_G$ is trivial if and only if $\mu$ is conjugation-invariant. But $\mu$ being conjugation-invariant while already being left-invariant is equivalent to being right-invariant and conjugation-invariant. For example, it follows that abelian groups have trivial modular character.

**Example 2.62.** Let $G$ be an algebraic group over a local field $F$. Note that $g \mapsto \det(\mathrm{Ad}_g; \mathfrak{g})$ is some character $G \to \mathbb{G}_m$, so if $G$ admits no such nontrivial homomorphism, then $\delta_G$ is automatically trivial. For example, this holds if $G$ is unipotent (namely, an extension of $\mathbb{G}_a$s) or semisimple.

**Example 2.63.** Because reductive groups are covered by a product of a semisimple group and an abelian torus, it follows that reductive groups are unimodular.

**Example 2.64.** Consider the algebraic group $B \subseteq \mathrm{GL}_2$ of upper-triangular matrices over a local field $F$. Then $\left[\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right]$ acts on the Lie algebra $\mathfrak{b}$ by

$$\begin{bmatrix} x & y \\ 0 & z \end{bmatrix} \mapsto \begin{bmatrix} x & (a/d)y \\ 0 & z \end{bmatrix},$$

so it follows that $\delta_B\left(\left[\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right]\right) = |a/d|$.

### 2.3.2 Back to Induction

We are still owed an explanation why we would want to include this character $\delta_{G/H}^{1/2}$ in our definition of induction. Here is such an application.

**Proposition 2.65.** Let $P \subseteq G$ be a parabolic subgroup of a reductive group $G$ over a local field $F$. If $\tau$ and $\tau^\vee$ are dual admissible representations of $P(F)$, then $\operatorname{Ind}_P^G \tau$ and $\operatorname{Ind}_P^G \tau^\vee$ are dual admissible representations.

*Proof.* We will merely exhibit the pairing between these representations. Note that $P\backslash G$ is projective, and $P(F)\backslash G(F)$ is compact. The moral is that our integration theory at most allows us to integrate sections of $|\omega_{P(F)\backslash G(F)}|$. Here, note that $|\omega_{P(F)\backslash G(F)}|$ is an equivariant real line bundle $\mathbb{R} \times^P G$, where $P$ acts on $R$ by $\delta_{G/P}$, which in this case agrees with $\delta_P$.

Now, let $\tau$ and $\tau^\vee$ be dual representations of $P$. Then we may define a pairing

$$\operatorname{Ind}_P^G \tau \times \operatorname{Ind}_P^G \tau^\vee \to \mathbb{C}$$

by

$$\langle f, f' \rangle := \int_{P(F)\backslash G(F)} \langle f(g), f'(g) \rangle \, dg.$$

Namely, for this pairing to make sense, we need to be able to realize $\varphi \colon g \mapsto \langle f(g), f(g') \rangle$ as a density on $P(F)\backslash G(F)$. Well, for any $p \in P(F)$, we then see that

$$\langle f(pg), f'(pg) \rangle = \delta_P(p) \langle f(g), f'(g) \rangle,$$

so $\varphi$ is a section of $|\omega_{P(F)\backslash G(F)}|$ on $P\backslash G$.

Of course, it remains to check that we have actually provided a duality. Well, once one shows that $\operatorname{Ind}_P^G \tau$ is actually an admissible representation, one can pass to $K$-finite vectors everywhere (for some open compact subgroup $K \subseteq G(F)$) to check that this is a perfect pairing on finite-dimensional vector spaces. ■

**Remark 2.66.** We have slid the fact that $\operatorname{Ind}_P^G \tau$ is admissible under the rug because it is on the homework. However, this argument crucially uses that $P$ is parabolic: one does not expect general inductions to send admissible representations to admissible representations!

Anyway, let's see some examples.

**Example 2.67.** Note that $\operatorname{Ind}_P^G \delta_P^{-1/2}$ contains the constant functions. Indeed, by definition, functions in $\operatorname{Ind}_P^G \delta_P^{-1/2}$ are simply smooth functions on $P\backslash G$.

**Example 2.68.** Dually, we see that $\operatorname{Ind}_P^G \delta_P^{1/2}$ consists of sections of $|\omega_{P\backslash G}|$. Thus, there is a quotient map to the trivial representation $\mathbb{C}$ given by integration against $\int_{P(F)\backslash G(F)}$.

Let's say something about the representation theory of $\mathrm{GL}_2$.

**Example 2.69.** Parabolic induction produces many irreducible representations of $\mathrm{GL}_2(F)$. Indeed, let $B \subseteq \mathrm{GL}_2$ be the upper-triangular subgroup, and let $L$ be the diagonal subgroup, which is the Levi quotient. Then $L(F) = F^\times \times F^\times$, so representations of $L$ are given by a pair of characters $\chi_1, \chi_2 \colon F^\times \to \mathbb{C}^\times$. With this description, one can calculate that $\delta_B$ descends to the character $\left(|\cdot|, |\cdot|^{-1}\right)$ on $L(F)$. By doing some Mackey theory, it turns out that $\mathrm{Ind}_P^G(\chi_1, \chi_2)$ is irreducible except in the following situations.

- If $\chi_1/\chi_2 = |\cdot|$, then $\mathrm{Ind}_P^G(\chi_1, \chi_2)$ admits a one-dimensional quotient (and has irreducible kernel).
- If $\chi_2/\chi_1 = |\cdot|$, then $\mathrm{Ind}_P^G(\chi_1, \chi_2)$ admits a one-dimensional subrepresentation (and has irreducible quotient).

By finding some intertwining operators, one can show that $\mathrm{Ind}_B^G(\chi_1, \chi_2) \cong \mathrm{Ind}_B^G(\chi_2, \chi_1)$.

**Remark 2.70.** The moral is that one roughly has a map from $(F^\times \times F^\times)/S_2$ to irreducible representations of $\mathrm{GL}_2(F)$. It turns out that there are more representations, which are the "cuspidal" ones. The other ones are parameterized by characters of $E^\times$ (up to an $S_2$-symmetry), where $E$ is a quadratic extension of $F$. One way to construct these representations is by the Weil representation; another way is to do (compact) induction from general compact open subgroups.

**Remark 2.71** (base change)**.** Let $\sigma$ be the automorphism of a quadratic extension $E$ of a local field $F$. Given a smooth irreducible representation $\pi$ of $\mathrm{GL}_2(E)$, one may hope that $\pi \cong \pi^\sigma$ would imply that $\pi$ "comes from" $\mathrm{GL}_2(F)$, but there is no obvious way to make this association. This is one instance of Langlands's functoriality principle. In the modern era, we are able to do this descent and even give the character of the descended representation, but we have no direct construction.

### 2.3.3 Spherical Representations

For most calculations, we will be interested in certain subclasses of representations.

**Definition 2.72** (spherical)**.** Fix an affine algebraic group $G$ over $\mathcal{O}_F$. Then an irreducible smooth representation of $G(F)$ is *spherical* if and only if $V^{G(\mathcal{O}_F)} \neq 0$.

**Theorem 2.73.** The spherical representations of $\mathrm{GL}_n(F)$ are in bijection with $n$-tuples in $(\mathbb{C}^\times)^n$ up to permutation.

To begin, we will have to construct some spherical representations. For the rest of this subsection, $G$ is $\mathrm{GL}_n$, and $B \subseteq G$ is the subgroup of upper-triangular matrices. Then $B$ has Levi quotient $L$ given by the diagonal subgroup of $G$. Now, given any $n$ characters $(\chi_1, \ldots, \chi_n)$ of $F^\times$, we can construct the parabolic induction

$$\mathrm{Ind}_B^G(\chi_1, \ldots, \chi_n).$$

Suppose further that each $\chi_i$ is unramified, in the sense that it factors through $|\cdot|$. In particular, $\chi_i$ is uniquely determined by the value $\chi_i(\varpi) \in \mathbb{C}^\times$, where $\varpi \in \mathcal{O}_F$ is any choice of uniformizer. Then our $n$-tuple of characters $(\chi_1, \ldots, \chi_n)$ is basically given by the vector $(\chi_1(\varpi), \ldots, \chi_n(\varpi)) \in (\mathbb{C}^\times)^n$.

Let's check that the constructed representation in fact admits a vector fixed by $K := \mathrm{GL}_n(\mathcal{O}_F)$. Well, we are on the hunt for functions $f \colon G \to \mathbb{C}$ such that

$$f(bxk) = \delta_B(b)^{1/2}(\chi_1(b) \cdots \chi_n(b)) f(x)$$

for any $b \in B$ and $k \in K$. By doing some explicit calculations, one has the following.

**Proposition 2.74** (Iwasawa decomposition)**.** Continue with $G := \mathrm{GL}_n(F)$. Let $U \subseteq G$ be the subgroup of strictly upper-triangular matrices. Then there is a decomposition

$$\mathrm{GL}_n(F) = \bigsqcup_{\lambda \in \mathbb{Z}^n} U_n(F) \operatorname{diag}\left(\varpi^{\lambda_1}, \ldots, \varpi^{\lambda_n}\right) \mathrm{GL}_n(\mathcal{O}_F).$$

*Sketch.* We would like to show that we can modify any $g \in \mathrm{GL}_n(F)$ by an element of $U(F)$ on the left and by $K$ on the right to make it in the form $\operatorname{diag}\left(\varpi^{\lambda_1}, \ldots, \varpi^{\lambda_n}\right)$. This means that we are allowed column operations which are invertible over $\mathcal{O}$ and row operations which preserve row ordering (without scaling). Then one does some induction with a version of the Gram–Schmidt process. ■

**Remark 2.75.** Over $\mathbb{R}$, one has an analogous decomposition

$$\mathrm{GL}_n(\mathbb{R}) = U(\mathbb{R}) \left(\mathbb{R}^+\right)^n \mathrm{O}(n).$$

This is more or less a Gram–Schmidt reduction to turn any basis of $\mathrm{GL}_n(\mathbb{R})$ into an orthonormal one.

The moral is that the function $f$ that we are hunting for is uniquely determined by its value at $1$, so there is at most one such function $f$. After doing a little unwinding, it remains to check that having $bk = 1$ is consistent with $f(bk) = f(1)$. But having $b \in B(F)$ and $k \in K$ with $bk = 1$, we see that $b, k \in B \cap K$, so $b$ and $k$ are upper-triangular with diagonal entries contained in $\mathcal{O}^\times$. But $\delta_B$ and the $\chi_i$s all vanish on $\mathcal{O}^\times$ by construction!

**Remark 2.76.** In fact, this argument shows that the only parabolic inductions which are unramified must come from unramified characters.

## 2.4 March 4

Here we go.

### 2.4.1 The Satake Isomorphism

Let's continue talking about our parabolic induction. We continue with $G = \mathrm{GL}_n(F)$ with compact open subgroup $K = \mathrm{GL}_n(\mathcal{O}_F)$.

**Remark 2.77.** It turns out that $\operatorname{Ind}_B^G \chi$ always has finite length, and it admits a unique spherical subquotient. In general, permuting the entries of $\chi$ grants the same spherical representation.

**Example 2.78.** For $G = \mathrm{GL}_2(F)$, we saw that $\operatorname{Ind}_B^G \delta^{-1/2}$ admits the subrepresentation $\mathbb{C}$, but $\operatorname{Ind}_B^G \delta^{1/2}$ admits the quotient $\mathbb{C}$. This is unsurprising because $\delta^{1/2}$ and $\delta^{-1/2}$ differ merely by a transposition.

Thus, we have constructed a map from $(\mathbb{C}^\times)^n / S_n$ to the collection of spherical representations.

It remains to show that we have in fact found all the spherical representations.

**Remark 2.79.** For any compact open subgroup $K$ of a totally disconnected group $G$, there is a functor $(-)^K \colon \operatorname{Rep}(G) \to \operatorname{Mod}(\mathcal{H}_K)$. (In fact, this functor is exact because it passes to finite-dimensional spaces, and fixed points by compact subgroups is exact.) Namely, recall $\mathcal{H}_K$ is presented as $e_K \mathcal{H}_G e_K$, which then acts naturally on $V^K = e_K V$.

We want to make the functor in the previous remark into an equivalence. Before we achieve this, we will find some adjoints.

**Definition 2.80.** There is an induction $\mathrm{ind}_! \colon \mathrm{Mod}(\mathcal{H}_K) \to \mathrm{Rep}(G)$ given by $M \mapsto \mathcal{H}e_K \otimes_{\mathcal{H}_K} M$. There is similarly an induction $\mathrm{ind}_*$ given by $M \mapsto \mathrm{Hom}_{\mathcal{H}_K}(\mathcal{H}e_K, M)$.

**Remark 2.81.** Both of these functors output to smooth representations.

**Remark 2.82.** One can check that composing either of these functors with $(-)^K$ recovers the original module, so we have "one-sided inverses."

If we would like for either of these functors be equivalences, we need to restrict the category $\mathrm{Rep}_{\mathrm{sm}}(G)$. For example, $V$ should be generated by $V^K$ if it is to come from $V^K$. In fact, this implies that every quotient of $V$ should have $K$-fixed vectors, so we will want to require the "dual condition" that every nonzero subrepresentation of $V$ should be $K$-fixed vectors.

**Notation 2.83.** Let $G$ be a totally disconnected group, and choose a compact open subgroup $K \subseteq G$. Then $\mathrm{Rep}_{\mathrm{sm}}(G, K)$ is the full subcategory of smooth representations of $G$ such that every quotient and subrepresentation of $V$ admits $K$-fixed vectors.

**Theorem 2.84.** Let $G$ be a totally disconnected group, and choose a compact open subgroup $K \subseteq G$. The functor $(-)^K \colon \mathrm{Rep}_{\mathrm{sm}}(G, K) \to \mathrm{Mod}(\mathcal{H}_K)$ is an equivalence.

*Proof.* Omitted. ■

**Corollary 2.85.** Let $G$ be a totally disconnected group, and choose a compact open subgroup $K \subseteq G$. The irreducible smooth representations $V$ of $G$ with $V^K \neq 0$ are in bijection with finite-dimensional simple $\mathcal{H}_K$-modules.

Thus, returning to $\mathrm{GL}_n(F)$, we are interested in understanding the "spherical" Hecke algebra

$$\mathcal{H}_K = C_c(\mathrm{GL}_n(\mathcal{O}) \backslash \mathrm{GL}_n(F) / \mathrm{GL}_n(\mathcal{O}))\, dg.$$

This ring turns out to be commutative (this is a general fact for split reductive groups $G$), and it is not too hard to describe this ring explicitly.

**Theorem 2.86** (Satake)**.** Set $G := \mathrm{GL}_n(F)$ and $K := \mathrm{GL}_n(\mathcal{O})$. Then

$$\mathcal{H}_K \cong \mathbb{C}\left[t_1^{\pm 1}, \ldots, t_n^{\pm 1}\right]^{S_n}.$$

Thus, we see that the simple $\mathcal{H}_K$-modules are in bijection with the maximal ideals of $\mathcal{H}_K$, which are then in bijection with $(\mathbb{C}^\times)/S_n$ by some argument with the Nullstellensatz. It remains to track through the Satake isomorphism to verify that the produced $\mathcal{H}_K$-modules do in fact come from our principal series.

**Example 2.87.** Let's work with $G = \mathrm{GL}_2(F)$. One may hope that our modules should come from the quotient $\mathrm{GL}_2(F)/\mathrm{GL}_2(\mathcal{O})$. This quotient is in bijection with $\mathcal{O}$-lattices in $F^2$ by sending $g\,\mathrm{GL}_2(\mathcal{O})$ to the lattice $g\mathcal{O}^2$. If we descend to $\mathrm{PGL}_2(F)/\mathrm{PGL}_2(\mathcal{O})$ (which parameterizes lattices up to scalar), there is a bijection to the Bruhat–Tits tree, which is the infinite $(q+1)$-regular tree. Indeed, start with a vertex for the standard lattice $\mathcal{O}^2$, and then the immediate neighbors are given by those lattices $\mathcal{O}^2 + \varpi^{-1}\ell$, where $\ell \in \mathbb{P}^1(\mathbb{F}_q)$ is some line; this "addition of a line" edge rule can then be iterated.

**Remark 2.88.** Let's explain how this Bruhat–Tits tree can help us: the Hecke algebra can be viewed as $\mathrm{GL}_n(F)$-invariant functions on

$$\mathrm{GL}_n(F)/\mathrm{GL}_n(\mathcal{O}) \times \mathrm{GL}_n(F)/\mathrm{GL}_n(\mathcal{O}).$$

Thus, for $n = 2$, we are basically looking at lattices up to an action by $\mathrm{GL}_2(F)$; it turns out that the only invariant for such pairs is the distance between the two points in the tree!

**Remark 2.89.** For $\mathrm{GL}_n$ in general, there is a Bruhat–Tits building.

## 2.5 March 9

Here we go.

### 2.5.1 The Bruhat–Tits Tree

Let's recall our story in some level of generality. Let $G$ be a split reductive group over a nonarchimedean local field $F$, and choose a maximal torus $T \subseteq G$ contained in a Borel $B \subseteq G$. For any unramified character $\chi$, there is a unique spherical subquotient of $\mathrm{Ind}_B^G \chi$, so one accordingly finds that spherical representations are parameterized such unramified characters up to the $W$-action.

On the other hand, spherical representations are simply given by simple $\mathcal{H}_K$-modules. Thus, we would like to understand our Hecke algebra $\mathcal{H}_K = C_c(K\backslash G/K)\,dg$.

**Lemma 2.90.** Consider the group $G := \mathrm{GL}_n$ over a nonarchimedean local field $F$. Then $K\backslash G/K$ is in bijection with the set of $G$-orbits of pairs $\mathcal{O}$-lattices $(\Lambda_1, \Lambda_2)$ in $F^n$.

*Proof.* Send $KgK$ to $(\mathcal{O}^n, g\mathcal{O}^n)$; this is of course $K$-invariant on the right, and it is $K$-invariant on the left because $(\mathcal{O}^n, g\mathcal{O}^n)$ and $(\mathcal{O}^n, kg\mathcal{O}^n)$ are in the same $G$-orbit. Conversely, given any pair $(\Lambda_1, \Lambda_2)$, one can find a distinguished element in the $G$-orbit with $\Lambda_1 = \mathcal{O}^n$ so that $(\Lambda_1, \Lambda_2) = (\mathcal{O}^n, g\mathcal{O}^n)$ for some $g \in G$, and the element $g \in G$ is well-defined in $K\backslash G/K$. ■

Thus, we are motivated to understand lattices.

**Proposition 2.91.** Fix two lattices $\Lambda_1, \lambda_2 \subseteq F^n$. Then there is a basis $\{e_1, \ldots, e_n\} \subseteq F^n$ of $\Lambda_1$ so that

$$\Lambda_2 = \varpi^{d_1}\mathcal{O}e_1 \oplus \cdots \oplus \varpi^{d_n}\mathcal{O}e_n$$

for some decreasing sequence $\{d_1, \ldots, d_n\}$ of integers. In fact, the sequence of integers does not depend on the choice of basis.

*Proof.* We induct on $n$. In the case where $n = 0$, the statement is vacuous, and for $n = 1$, the statement has no content. By rescaling the conclusion, we may assume that $\Lambda_1 \subseteq \Lambda_2$ but $\Lambda_1 \not\subseteq \varpi\Lambda_2$. Then the natural inclusion induces a nonzero map

$$\Lambda_1/\varpi\Lambda_1 \to \Lambda_2/\varpi\Lambda_2.$$

As such, we just choose a basis $\{e_1, \ldots, e_r\}$ of the image and lift them to $\Lambda_1$, where they will continue to be linearly independent, and they generate a saturated sublattice $\Lambda_0$ of both $\Lambda_1$ and $\Lambda_2$. Thus, we may set $d_1 = \cdots = d_r = 0$, and the remaining integers are found by induction by passing to the pair of lattices $(\Lambda_1/\Lambda_0, \Lambda_2/\Lambda_0)$ of smaller rank. Indeed, one can lift the basis for $\Lambda_1/\Lambda_0$ to the required remaining basis vectors $\{e_{r+1}, \ldots, e_n\}$; one will need to modify these $e_i$s by a vector in $\Lambda_0$ to ensure that $\varpi^{d_i}e_i \in \Lambda_2$ (because a priori we merely have $\varpi^{d_i}e_i \in \Lambda_0$). ■

**Remark 2.92.** This is basically equivalent to the Smith normal form: we are trying to use integral row and column operations to make a given matrix into a diagonal one.

**Corollary 2.93** (Cartan decomposition)**.** Consider the group $G := \mathrm{GL}_n$ over a nonarchimedean local field $F$. There is a disjoint union

$$G = \bigsqcup_{d_1 \geq \cdots \geq d_n} K \operatorname{diag}\left(\varpi^{d_1}, \ldots, \varpi^{d_n}\right) K.$$

*Proof.* Combine Lemma 2.90 with Proposition 2.91. ■

**Remark 2.94.** The same argument shows that

$$\mathrm{PGL}_2(F) = \bigsqcup_{d \geq 0} K \operatorname{diag}\left(\varpi^d, 1\right) K.$$

The corresponding statement of Proposition 2.91 simply needs to notice that if we replace the lattices with homothety classes, then we are looking at pairs of integers $(d_1, d_2)$ up to shifting, which can be uniquely presented as $(d, 0)$. The Bruhat–Tits tree is now recovered as having vertices given by the homothety classes of lattices and edges which are connected if and only if $([\Lambda], [\Lambda'])$ is represented by the pair of integers $(1, 0)$. Accordingly, general pairs $([\Lambda], [\Lambda'])$ classified by $(d, 0)$ have distance $d$.

**Remark 2.95.** For $\mathrm{PGL}_n$, the Bruhat–Tits tree becomes a Bruhat–Tits building is a simplicial complex of dimension $(n-1)$, where vertices are given by lattices $\Lambda$.

Thus, we see that $\mathcal{H}_K$ admits a basis of indicators of the double cosets of Corollary 2.93.

### 2.5.2 The Spherical Hecke Algebra

Here is the usual application of the Cartan decomposition.

**Lemma 2.96** (Gelfand)**.** Consider the group $G := \mathrm{GL}_n$ over a nonarchimedean local field $F$. Then $\mathcal{H}_K$ is commutative.

*Proof.* We use the Gelfand trick: consider the map $G \to G$ given by the transpose. This operation induces an anti-commutative map $C(G) \to C(G)$ given by sending $f$ to $f^\intercal(g) := f(g^\intercal)$. In particular, we see that $(-)^\intercal$ is linear, but $(f_1 f_2)^\intercal = f_2^\intercal f_1^\intercal$. However, $(-)^\intercal$ fixes the basis elements

$$1_{K \operatorname{diag}\left(\varpi^{d_1}, \ldots, \varpi^{d_n}\right) K}$$

of Corollary 2.93 are fixed by $(-)^\intercal$, so $(-)^\intercal$ is the identity. We conclude that $\mathcal{H}_K$ is commutative. ■

The above miraculous proof turns out to work for general split reductive groups.

**Proposition 2.97** (Cartan decomposition)**.** Fix a split reductive group $G$ over a nonarchimedean local field $F$, and let $T \subseteq G$ be a split maximal torus. For $\lambda \in X_*(T)$, we let $\varpi^\lambda$ denote $\lambda(\varpi) \in T(F)$. Then

$$G(F) = \bigsqcup_{\text{dominant } \lambda \in X_*(T)} G(\mathcal{O}) \varpi^\lambda G(\mathcal{O}).$$

Here, a coweight $\lambda$ is dominant if and only if $\langle \lambda, \alpha_i \rangle \geq 0$ for all simple roots $\alpha_i$.

*Proof.* Omitted! ■

**Theorem 2.98.** Fix a split reductive group $G$ over a nonarchimedean local field $F$. Then the Hecke algebra $\mathcal{H}_{G(\mathcal{O})}$ is commutative.

*Proof.* It is a fact that there is an algebraic group involution $\sigma$ of $G$ for which $\sigma(t) = t^{-1}$ for all $t \in T$; it also sends a choice of Borel to the opposite Borel. (In fact, this $\sigma$ is defined over $\mathbb{Z}$. Notably, the action by the longest Weyl element should work.) Then one replaces $(-)^\intercal$ in the proof of Lemma 2.96 with the function $\tau\colon G(F) \to G(F)$ given by $g \mapsto \sigma(g)^{-1}$, and the entire argument goes through. ■

Of course, we are still interested in the ring structure of the spherical Hecke algebra. For example, with $G = \mathrm{GL}_n$, we claim that

$$1_{K\,\mathrm{diag}\left(\varpi^{d_1},\ldots,\varpi^{d_n}\right)K} * 1_{K\,\mathrm{diag}\left(\varpi^{d_1'},\ldots,\varpi^{d_n'}\right)K} = 1_{K\,\mathrm{diag}\left(\varpi^{d_1+d_1'},\ldots,\varpi^{d_n+d_n'}\right)K} + \text{lower-order terms.}$$

(Here, our collection of decreasing sequences of integers is given by the usual order on partitions.[2]) Indeed, it is enough to compute the left-hand side on matrices of the form $\mathrm{diag}(\varpi^{\ell_1}, \ldots, \varpi^{\ell_n})$. By expanding out the convolution, we see that this value is given by the number of lattices $\Lambda$ which can fit in a sequence

$$\mathcal{O}^{\oplus n} \xrightarrow{d} \Lambda \xrightarrow{d'} \varpi^{\ell_1}\mathcal{O} \oplus \cdots \oplus \varpi^{\ell_n}\mathcal{O},$$

where $\xrightarrow{d}$ means that the pair of consecutive lattices is parameterized by $d$.

**Example 2.99.** We go down to $\mathrm{PGL}_2$. Then we are trying to show

$$1_{K\,\mathrm{diag}(\varpi^d,1)K} * 1_{K\,\mathrm{diag}\left(\varpi^{d'},1\right)K} = 1_{K\,\mathrm{diag}\left(\varpi^{d+d'},1\right)K} + \text{lower-order terms.}$$

Well, on $\mathrm{diag}\left(\varpi^\ell, 1\right)$, we are asking for the number of lattices $\Lambda$ which are a distance of $d$ from $\mathcal{O}^{\oplus 2}$ and a distance $d'$ from $\varpi^\ell\mathcal{O} \oplus \mathcal{O}$. For example, if $d + d' < \ell$, there are no such lattices; if $d + d' = \ell$, there is in fact a distinguished $\Lambda$ for $\ell = d + d'$. However, there are allowed to be "lower-order terms" for $\ell > d + d'$.

Admitting the claim, we conclude that $\mathrm{gr}\,\mathcal{H}_K$ is a polynomial ring on $X_*(T)/W$. This sort of discussion generalizes to prove the following.

**Theorem 2.100.** Fix a split reductive group $G$ over a nonarchimedean local field $F$, and set $K := G(\mathcal{O})$. Then $\mathrm{gr}\,\mathcal{H}_K$ is isomorphic to $\mathbb{C}[X_*(T)^{\mathrm{dom}}]$, where $X_*(T)^{\mathrm{dom}}$ denotes the monoid of dominant weights.

*Proof.* Proceed as above with more effort. ■

**Remark 2.101.** It follows purely formally that $\mathcal{H}_K$ (with some Five lemma argument) that $\mathcal{H}_K$ is a polynomial ring of rank $\mathrm{rank}\,G$.

### 2.5.3 The Satake Isomorphism, Again

It may be desirable to have a more explicit description of $\mathcal{H}_K$, without the need to pass to the associated graded. This is given by the Satake isomorphism.

[2] One can show that having $d \geq d'$ is equivalent to $K\varpi^d K$ containing $K\varpi^{d'}$ in its closure.

**Theorem 2.102** (Satake)**.** Fix a split reductive group $G$ over a nonarchimedean local field $F$, and set $K := G(\mathcal{O})$. Then there is a canonical isomorphism

$$\mathcal{H}_K = \mathbb{C}[X_*(T)]^W.$$

**Example 2.103.** For $G = \mathrm{PGL}_2$, one sees that $\mathcal{H}_K$ is a polynomial ring with generator $t := 1_{K\,\mathrm{diag}(\varpi,1)K}$. On the other hand, $\mathbb{C}[X_*(T)]^W$ is just $\mathbb{C}[\mathbb{Z}x]^{S_2}$, which is $\mathbb{C}\left[x + x^{-1}\right]$. It turns out that the Satake isomorphism sends $t$ to a scalar times $x + x^{-1}$.

*Sketch.* Let's start by defining the map $S\colon \mathcal{H}_K \to \mathbb{C}[X_*(T)]^W$. Thus, given $\varphi \in \mathcal{H}_K$, we are being asked for a coefficient $S\varphi(\lambda)$ for each $\lambda \in X_*(T)$. To this end, we take as definition

$$S\varphi(\lambda) := \delta_B^{1/2}(\varpi^\lambda) \int_{N(F)} \varphi\left(\varpi^\lambda x\right)\, dx,$$

where $B \subseteq G$ is a chosen Borel containing $T$, and $N \subseteq B$ is the unipotent radical. One can check that $S\varphi(w\lambda) = S\varphi(\lambda)$ always; this is not hard to see directly for $G = \mathrm{GL}_2$, and then the case for general $\mathrm{GL}_n$ can probably be checked via transpositions.

It is rather non-obvious that this map has any good properties, so let's give another construction of $S$. Recall that $\mathrm{Ind}_B^G \chi$ basically consists of $\chi$-eigenvectors for functions $N(F)\backslash G(F) \to \mathbb{C}$. Because $\chi$ is unramified, we thus see that our functions in fact descend to $T(\mathcal{O})N(F)\backslash G(F) \to \mathbb{C}$; note that the quotient continues to admit a natural action by $X_*(T) = T(F)/T(\mathcal{O})$. Thus, as $\chi$ varies, we find the "universal" unramified principal series

$$\mathrm{Ind}_{T(\mathcal{O})N(F)}^{G(F)} 1.$$

If we take $K$-invariants, then we retain the $X_*(T)$-action and exchange the $G$-action for an $\mathcal{H}_K$-action, and the space is given by those functions $T(\mathcal{O})N(F)\backslash G(F)/K \to \mathbb{C}$, but the Iwasawa decomposition tells us that this latter set is given by $X_*(T)$. We thus have a bimodule

$$M_{\mathrm{sph}} := C_c(T(\mathcal{O})N(F)\backslash G(F)/K)$$

with commuting actions by $\mathcal{H}_K$ and $\mathbb{C}[X_*(T)]$. Because the two actions commute, we receive a map $\mathcal{H}_K \to \mathrm{End}_{\mathbb{C}[X_*(T)]}(M_{\mathrm{sph}})$, but $M_{\mathrm{sph}}$ is free of rank $1$ over $\mathbb{C}[X_*(T)]$, so we are actually receiving a map

$$\mathcal{H}_K \to \mathrm{End}_{\mathbb{C}[X_*(T)]} M_{\mathrm{sph}} = \mathbb{C}[X_*(T)].$$

We can also use this description to explain the $W$-invariance. For generic $\chi$, one has that $\mathrm{Ind}_B^G \chi = \mathrm{Ind}_B^G w\chi$ by some intertwining operators, which descends to a $W$-action on $C_c(T(\mathcal{O})N(F)\backslash G(F)/K)$.

It remains to show that we actually have an isomorphism. Well, it is enough to show that we have an isomorphism of vector spaces, for which one can pass to some associated graded things, so the calculation is similar to what was done before to show that $\mathcal{H}_K$ is isomorphic to a polynomial ring. ■

Let's put this all together. We see that spherical representations are in bijection with simple $\mathcal{H}_K$-modules, which are in bijection with maximal ideals of $\mathcal{H}_K$, which are in bijection with maximal ideals of $\mathbb{C}[X_*(T)]^W$, which are in bijection with $W$-orbits in $\widehat{T} = \mathrm{Hom}(X_*(T), \mathbb{C}^\times)$. Thus, our spherical representations are in bijection with $\widehat{T}/W$.

**Remark 2.104.** The group $\widehat{T}$ is known as its "Langlands dual." Note that the functor $T \mapsto \widehat{T}$ (from the category of tori) is contravariant, which is one explanation for why this is a dual.

## 2.6 March 11

We start class by saying a little about the Langlands dual.

### 2.6.1 The Langlands Dual

Let's attempt to generalize our construction of $\widehat{T}$. Recall that split reductive groups are classified by their root data.

**Definition 2.105** (root datum)**.** Fix a connected split reductive group $G$ over a field $F$, and choose a split maximal torus $T \subseteq G$. Then we define its *root datum* as the quadruple $(\Lambda, R, \Lambda^\vee, R^\vee)$, where $\Lambda = X^*(T)$ is the weight lattice, $\Lambda^\vee = X_*(T)$ is the coweight lattice, $R$ consists of the roots, and $R^\vee$ consists of the coroots.

**Remark 2.106.** Semisimple groups are classified merely by the root system $(\Lambda, R)$. Remembering the coroots is required to access all split reductive groups.

Here is the main theorem.

**Theorem 2.107.** Isomorphism classes of connected split reductive groups are equivalent to isomorphism classes of root data.

*Proof.* This is similar to the argument for connected semisimple groups, but now we attempt to remember the central torus of the group. We remark that the existence of this group is difficult. ■

We are thus allowed to make the following definition.

**Definition 2.108** (Langlands dual)**.** Fix a connected split reductive group $G$ over a field $F$ with root datum $(\Lambda, R, \Lambda^\vee, R^\vee)$. Then the *Langlands dual* $\widehat{G}$ is the complex connected split reductive group with root datum $(\Lambda^\vee, R^\vee, \Lambda, R)$.

**Remark 2.109.** We can see that the type of the dual group is the one given by reversing the Dynkin diagram.

**Example 2.110.** Take $G := \mathrm{GL}_n$, for which we can take the diagonal torus. Then $X^*(T)$ and $X_*(T)$ both have a canonical basis $\{e_i\}$ (which is self-dual). The root system of $\mathrm{GL}_n$ is $\{e_i - e_j\}$, and the coroots are given by $\{e_i^* - e_j^*\}$, so we conclude that the Langlands dual group is $\mathrm{GL}_{n,\mathbb{C}}$.

**Example 2.111.** Take $G := \mathrm{SL}_n$, for which we can again take the diagonal torus. Then $X^*(T) = \mathbb{Z}^n/\Delta\mathbb{Z}$ (meaning that we are taking a quotient by the "diagonal" copy of $\mathbb{Z}$, arising because determinants must vanish). Then the root system is still $\{e_i - e_j\}$ for the standard basis $\{e_i\}$ of $X^*(T)$. On the other hand, $X_*(T)$ is the trace-zero hyperplane $\mathbb{Z}_0^n$ of $\mathbb{Z}^n$, and the coroot system is again just given by $\{e_i^* - e_j^*\}$. Redoing the argument shows that the dual root data is given by $\mathrm{PGL}_n$, so $\widehat{G} = \mathrm{PGL}_{n,\mathbb{C}}$.

**Example 2.112.** Take $G = \mathrm{Sp}_{2n}$. Using the standard symplectic form, one finds that $X^*(T) = \mathbb{Z}^n$ with dual $X_*(T) = \mathbb{Z}^n$. The root system of $\mathrm{Sp}_{2n}$ is given by $\{e_i - e_j\}_{i \neq j} \sqcup \{2e_i\}$, which only has the long root $2e_i$. On the other hand, the coroot system is given by $\{e_i^* - e_j^*\}_{i \neq j} \sqcup \{e_i^*\}$, which has only the short root $e_i^*$. Redoing the argument shows that the dual root data is given by $\mathrm{SO}_{2n+1}$, so $\widehat{G} = \mathrm{SO}_{2n+1,\mathbb{C}}$.

The moral is that we are able to restate our classification of spherical representations as follows.

**Theorem 2.113.** Fix a split connected reductive group $G$ over a nonarchimedean local field $F$. Then the spherical representations of $G(F)$ are in bijection with the semisimple conjugacy classes in $\widehat{G}(\mathbb{C})$.

*Proof.* In light of our prior discussion, it only remains to show that the semisimple conjugacy classes in $\widehat{G}(\mathbb{C})$ are in bijection with $\widehat{T}/W$. Well, any semisimple element has a conjugate in $\widehat{T}$, which is unique up action by the Weyl group $W = \mathrm{N}_G(T)/T$. ■

**Remark 2.114.** To explain how this generalizes, note that we can think of semisimple conjugacy classes as homomorphisms $W_F \to \widehat{G}(\mathbb{C})$ (up to conjugacy) which factor through $\mathbb{Z}$ and have semisimple image. It is a major area of research to figure out how to parameterize arbitrary representations of $G(F)$ by arbitrary homomorphisms $W_F \to \widehat{G}(\mathbb{C})$.

**Remark 2.115.** There is an analogous statement for real groups. The correct definition of Weil group $W_{\mathbb{R}}$ is fitting into a non-split short exact sequence

$$1 \to \mathbb{C}^\times \to W_{\mathbb{R}} \to \mathrm{Gal}(\mathbb{C}/\mathbb{R}) \to 1.$$

Theme 3

# Archimedean Representation Theory

## 3.1 March 11

We now turn to archimedean theory.

### 3.1.1 Real Lie Groups

We would like to retell our nonarchimedean story for real Lie groups. Thus, we want to define smooth representations, produce a Hecke algebra, and define a notion like spherical representations.

Let's start by explaining what real Lie groups we will consider, and we will discuss some of their structure theory.

**Definition 3.1** (Lie group)**.** A *Lie group* is a group object in the category of smooth manifolds.

**Example 3.2** (classical groups)**.** The classical groups look like $\mathrm{GL}_n(\mathbb{R})$, $\mathrm{O}(p,q)$, $\mathrm{GL}_n(\mathbb{C})$, $\mathrm{U}(p,q)$, $\mathrm{GL}_n(\mathbb{H})$, $\mathrm{Sp}(p,q)$, $\mathrm{Sp}_{2n}(\mathbb{R})$, and a few others. All of these are real points of real algebraic groups. For example, $\mathbb{H}$ embeds into $M_4(\mathbb{R})$, so $\mathrm{GL}_n(\mathbb{H})$ embeds into $\mathrm{GL}_{4n}(\mathbb{R})$.

The plethora of examples is more or less because we have chosen to study our groups over $\mathbb{R}$ instead of over $\mathbb{C}$.

**Remark 3.3.** For any real algebraic group $G$, one can base-change to $\mathbb{C}$, and then $G_{\mathbb{C}}$ is classified by the root datum (by Theorem 2.107).

**Example 3.4.** All the groups $\mathrm{GL}_n(\mathbb{R})$, $\mathrm{U}(p,q)$ with $p+q=n$, and $\mathrm{GL}_{n/2}(\mathbb{H})$ all base-change to $\mathrm{GL}_{n,\mathbb{C}}$.

Lie groups come with a distinguished compact subgroup.

**Theorem 3.5** (Cartan)**.** Fix a connected real Lie group $G$. Then $G(\mathbb{R})$ admits a maximal compact subgroup, and all maximal compact subgroups are conjugate.

**Example 3.6.** The real Lie group $\mathrm{SL}_2(\mathbb{R})$ admits the maximal compact subgroup $\mathrm{SO}_2(\mathbb{R})$.

**Example 3.7.** The maximal compact subgroup $\mathrm{GL}_n(\mathbb{R})$ is $\mathrm{O}(n)$. The maximal compact subgroup of $\mathrm{U}(p,q)$ is $\mathrm{U}(p) \times \mathrm{U}(q)$.

**Example 3.8.** It turns out that $\mathrm{Sp}_{2n}(\mathbb{R})$ admits $\mathrm{U}(n)$ as a maximal compact subgroup. To explain the embedding, note that one can give $\mathbb{R}^{2n}$ a complex structure so that the standard symplectic form $\omega$ is the imaginary part of a positive-definite Hermitian form $h$ on $\mathbb{C}^n$. Then $\mathrm{U}(h)$ embeds into $\mathrm{Sp}(\omega)$.

**Remark 3.9.** These maximal compact subgroups are important because they (roughly) play the role of our $G(\mathcal{O})$ in our nonarchimedean theory. For example, the name "spherical vector" actually comes from this archimedean theory: a spherical vector for $\mathrm{GL}_n(\mathbb{R})$ should be fixed by the rotation subgroup $\mathrm{O}_n(\mathbb{R})$!

Let's explain how we found so many real groups.

**Theorem 3.10** (Cartan)**.** Fix a complex reductive group $G$ over $\mathbb{C}$.

(a) There are finitely many real algebraic groups $G_0$ for which $(G_0)_{\mathbb{C}}$ (up to isomorphism).

(b) The set of isomorphism classes is in bijection with Cartan involutions $\theta\colon G \to G$ (satisfying $\theta^2 = 1$), up to conjugacy by automorphisms.

Roughly speaking, the involution $\theta$ should be treated as complex conjugation.

**Example 3.11.** Consider the group $G = \mathbb{G}_{m,\mathbb{C}}$. There are two involutions $\theta$, which are the identity and the inversion. The identity corresponds to the real form $\mathrm{U}(1)$, and the inverse corresponds to $\mathbb{G}_{m,\mathbb{R}}$.

**Example 3.12.** Consider the group $T = \mathbb{G}_{m,\mathbb{C}}^r$. Then $\theta$ should be chosen among automorphisms of $X_*(T)$. If one can diagonalize the action of $\theta$ on the lattice $X_*(T)$, then it diagonalizes into some copies of $(+1)\,\mathrm{id}$ and $(-1)\,\mathrm{id}$, so the real form is a product of $\mathrm{U}(1)$s and $\mathbb{G}_{m,\mathbb{R}}$s. However, there are more real forms: for example, $\mathrm{Res}_{\mathbb{C}/\mathbb{R}}\,\mathbb{G}_{m,\mathbb{C}}$ is a real form of $\mathbb{G}_{m,\mathbb{C}}^2$, and it corresponds to the involution $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ on the lattice.

**Example 3.13.** Consider the group $G = \mathrm{GL}_{n,\mathbb{C}}$. There are inner involutions, which look like conjugation by

$$\mathrm{diag}(\underbrace{+1,\dots,+1}_{p},\underbrace{-1,\dots,-1}_{q}).$$

These correspond to the real forms $\mathrm{U}(p,q)$. However, $\mathrm{GL}_{n,\mathbb{C}}$ also has an outer automorphism given by $g \mapsto g^{-\intercal}$; equivalently, we may view this as a composite $\mathrm{GL}_n(\mathbb{C}) \cong \mathrm{GL}(V) \cong \mathrm{GL}(V^*) \cong \mathrm{GL}_n(\mathbb{C})$, where the middle isomorphism is given by a perfect pairing $V \times V \to \mathbb{C}$, where $\dim V = n$. (Alternatively, up to inner automorphisms, this outer automorphism is basically given by "flipping" the Dynkin diagram, which is the only nontrivial graph isomorphism.) There are basically two cases for such a perfect pairing.

- Symmetric: this always exists, and it gives the "split" form $\mathrm{GL}_{n,\mathbb{R}}$.
- Alternating: this exists when $n$ is even, and it gives the real form $\mathrm{GL}_{n/2}(\mathbb{H})$.

### 3.1.2 Smooth Representations

Let $G$ be a real algebraic group. Our next task is to figure out what our representations are.

**Definition 3.14.** Fix a real algebraic group $G$, and let $V$ be a topological complex vector space. Then a *continuous representation* of $G$ is one for which the action map

$$G \times V \to V$$

is continuous. If $V$ is further a Hilbert space, and the action preserves the Hermitian inner product, then the representation is *unitary*.

**Remark 3.15.** It is typical to require $V$ to at least be a Fréchet space.

**Example 3.16.** Suppose that $G(\mathbb{R})$ acts smoothly on some real manifold $X$, and let $L^2\left(X, |\omega_X|^{1/2}\right)$ be the collection of half-densities, which we recall is a real line bundle on $X$. Then $G$ acts continuously on $L^2\left(X, |\omega_X|^{1/2}\right)$, and the $G$-action preserves the canonical inner product

$$\langle f, g\rangle := \int_X fg.$$

Here, $fg$ is a global section of $|\omega_X|$ and can therefore be integrated over $X$. However, we remark that $G$ also acts continuously on $L^p(X)$ for any $p \geq 1$, so there are many representations here!

To understand our representations, we must walk the roads lain by Harish–Chandra. We will want to two subspaces of our continuous representations. First, we take smooth vectors.

**Definition 3.17** (smooth)**.** Fix a continuous complex representation $V$ of a real Lie group $G$. Then a representation of a real algebraic group $G$ on $V$ is *smooth* if and only if, for any $v \in V$ and continuous $\ell \in V^*$, the matrix coefficient $g \mapsto \ell(gv)$ is a smooth function on $G$.

**Remark 3.18.** It is not hard to check that $V^{\mathrm{sm}}$ is a subspace of $V$.

**Remark 3.19.** It turns out that $V^{\mathrm{sm}}$ is dense in $V$ as long as $V$ is Fréchet. Indeed, for any $\mu \in C_c^\infty(G)\,dg$, one can check that the integral

$$\mu * v := \int_G gv\,d\mu(g)$$

makes sense in the Fréchet space $V$. It is not hard to check that the smoothness of $\mu$ implies smoothness of $\mu * v$. By having $\mu$ concentrate around the identity, one can show that $\mu * v \to v$, so we see that the smooth vectors are dense.

**Remark 3.20.** The Lie algebra $\mathfrak{g}$ of $G$ acts on $V^{\mathrm{sm}}$: namely, for $X \in \mathfrak{g}$, being Fréchet implies that the limit

$$Xv := \lim_{t\to 0} \frac{\exp(tX)v - v}{t}$$

exists in $V$; in fact, this is a smooth vector!

Second, we take $K$-finite vectors.

**Definition 3.21** ($K$-finite)**.** Fix a continuous complex representation $V$ of a real Lie group $G$, and choose a compact subgroup $K \subseteq G$. A vector $v \in V$ is *$K$-finite* if and only if

$$\dim_{\mathbb{C}} \operatorname{span}\{kv : k \in K\} < \infty.$$

**Remark 3.22.** A similar argument is able to show that $(V^{\mathrm{sm}})_{\mathrm{fin}} \subseteq V$ is smooth when $V$ is a Fréchet space.

## 3.2 March 16

Today, we say something about $(\mathfrak{g}, K)$-modules.

### 3.2.1 $(\mathfrak{g}, K)$-Modules

The moral of passing to smooth and $K$-finite vectors is that $V_{\mathrm{fin}}^{\mathrm{sm}}$ admits an action by $\mathfrak{g}$ and an action by $K$.

**Lemma 3.23.** Fix a continuous representation $V$ of a real Lie group $G$. Then $V_{\mathrm{fin}}^{\mathrm{sm}}$ admits an action by $\mathfrak{g}$ and an action by $K$.

*Proof.* We can see that $K$ sends $K$-finite vectors to $K$-finite vectors, and it also preserves smoothness (because even $G$ sends smooth vectors to smooth vectors). The action by $\mathfrak{g}$ also turns out to send $K$-finite vectors to $K$-finite vectors: the point is that

$$kXv = \left(kXk^{-1}\right) kv = (\mathrm{Ad}_k X)(kv).$$

Thus, $K \cdot \mathfrak{g} \cdot v = \mathfrak{g} \cdot K \cdot v$, and $K \cdot v$ is finite-dimensional, so $K \cdot \mathfrak{g} \cdot v$ is also finite-dimensional. ■

**Remark 3.24.** On the other hand, $V_{\mathrm{fin}}^{\mathrm{sm}}$ does not, in general, admit an action by $G$ because $G$ does not necessarily send $K$-finite vectors to $K$-finite vectors if $K$ fails to be normal.

Let's codify the data in the lemma into a definition.

**Definition 3.25** ($(\mathfrak{g}, K)$-module)**.** Fix a real Lie group $G$, let $K \subseteq G$ be the maximal compact subgroup, and let $\mathfrak{g}$ be the Lie algebra. Then a $(\mathfrak{g}, K)$-*module* $M$ is a complex vector space which is simultaneously a representation of $\mathfrak{g}$ and $K$, satisfying the following conditions.

(i) Locally finite over $K$: for any $m \in M$, the vector space $\mathrm{span}\{km : k \in K\}$ is finite-dimensional.

(ii) Differential compatibility: the differential action by $\mathrm{Lie}\, K$ on $M$ agrees with the action of $\mathfrak{g}$ on $M$.

(iii) Adjoint compatibility: for any $X \in \mathfrak{g}$ and $k \in K$ and $m \in M$, we have

$$kXm = (\mathrm{Ad}_k X)kv.$$

A morphism of $(\mathfrak{g}, K)$-modules is one preserving the $\mathfrak{g}$-action and the $K$-action.

**Remark 3.26.** On the homework, we will check that (iii) follows from (i) and (ii) if $K$ is connected.

**Example 3.27.** The smooth $K$-finite vectors of a continuous Fréchet representation produces a $(\mathfrak{g}, K)$-module by Lemma 3.23, but it is not faithful in general. In particular, it can send non-isomorphic representations of $G$ to isomorphic $(\mathfrak{g}, K)$-modules. For example, if $G$ acts on a variety $X$, then the representations $L^p(X)$ should give the same $(\mathfrak{g}, K)$-module for different values of $p$. On the other hand, this functor surjects onto simple modules, and it is an equivalence if one restricts to representations of moderate growth.

**Remark 3.28.** The notion of a $(\mathfrak{g}, K)$-module is rather algebraic: indeed, the $(\mathfrak{g}, K)$-modules are equivalent to the $(\mathfrak{g}_{\mathbb{C}}, K_{\mathbb{C}})$-modules, and the finite-dimensional continuous representations of $K$ are equivalent to the algebraic representations of $K_{\mathbb{C}}$, so everything has been moved to pure algebra. Let's explain this latter equivalence: a choice of finite-dimensional faithful complex representation $V$ of $K$ allows us to define $K_{\mathbb{C}}$ as the scheme-theoretic closure of $K \to \mathrm{GL}(V)$. It now turns out that the finite-dimensional representations of $K$ can all be found among the tensor powers of $V$ and $V^*$, so because the representation $K \to \mathrm{GL}(V)$ is tautologically algebraic, algebraicity extends to all representations of $K_{\mathbb{C}}$!

For example, let's classify the $(\mathfrak{g}, K)$-modules from the group $G = \mathrm{SL}_2(\mathbb{R})$. Here, $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{R})$ and $K = \mathrm{SO}_2(\mathbb{R})$, which base-changes to $(\mathfrak{sl}_2(\mathbb{C}), \mathbb{C}^\times)$.

**Example 3.29.** We classify the $(\mathfrak{sl}_2(\mathbb{C}), \mathbb{C}^\times)$-modules.

*Proof.* Let $M$ be a $(\mathfrak{g}_{\mathbb{C}}, K_{\mathbb{C}})$-module, and let's start with the $K_{\mathbb{C}}$-action. An algebraic representation of $\mathbb{C}^\times$ amounts to a choice of grading

$$M = \bigoplus_{n \in \mathbb{Z}} M_n,$$

where $M_n$ is the eigenspace of the $\mathbb{C}^\times$-action where $\mathbb{C}^\times$ acts by the $n$th power.

We now turn to the $\mathfrak{sl}_2(\mathbb{C})$-action; give $\mathfrak{sl}_2(\mathbb{C})$ the standard basis $(e, f, h)$ where $[h, e] = 2e$ and $[h, f] = -2f$ and $[e, f] = h$. By changing basis of $\mathbb{C}^2$, we may assume that $\mathbb{C}h = \operatorname{Lie} \mathrm{SO}_2(\mathbb{C})$. Let's be explicit about this: $\mathrm{SO}(2)$ contains the matrices of the form $\left[\begin{smallmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{smallmatrix}\right]$, which diagonalizes to $\mathrm{diag}(e^{i\theta}, e^{-i\theta})$. We then let $h$ be $i^{-1}\left[\begin{smallmatrix} & 1 \\ -1 & \end{smallmatrix}\right]$ so that $h$ lives in $\mathfrak{sl}_2(\mathbb{R})$ (up to the diagonalization!). One also finds that

$$e = \frac{1}{2}\begin{bmatrix} 1 & i \\ i & -1 \end{bmatrix} \qquad \text{and} \qquad f = \frac{1}{2}\begin{bmatrix} 1 & -i \\ -i & -1 \end{bmatrix}.$$

Now, by the compatibility of the $\operatorname{Lie} K$-action with the $\mathfrak{sl}_2(\mathbb{R})$-action, we see that $h$ must act on $M_n$ by $n$: indeed, $e^{i\theta}$ acts on $M_n$ by $e^{in\theta}$, which differentiates to multiplication by $n$. As for $e$ and $f$, the representation theory of $\mathfrak{sl}_2(\mathbb{C})$ shows that the commutator relation $[h, e] = 2e$ is equivalent to having $e(M_n) \subseteq M_{n+2}$ for each $n$, and $[h, f] = 2f$ means that $f(M_n) \subseteq M_{n-2}$ for each $n$. Lastly, the commutator relation $[e, f] = h$ means that $(ef - fe)|_{M_n} = n \,\mathrm{id}_{M_n}$.

One can analyze the simple modules with a little effort, essentially by choosing a basis and then analyzing the possibilities for the $e$-action and $f$-action. Let's just provide the answer.

- Induction: it turns out that most simple $(\mathfrak{sl}_2(\mathbb{C}), \mathbb{C}^\times)$-modules thus look like infinite chains of $\mathbb{C}$s in a single parity, going off infinitely in both directions. These are the ones induced from the subgroup of upper-triangular matrices.

- There are the finite-dimensional $(\mathfrak{sl}_2(\mathbb{C}), \mathbb{C}^\times)$-modules, which are algebraic and thus all lift to finite-dimensional representations of $\mathrm{SL}_2(\mathbb{C})$. The weights look like $\{-m, -m+2, \ldots, m-2, m\}$ for some positive integer $m$.

- Discrete series: there are irreducible Verma modules, which start from a given positive integer $m \geq 2$ and extend off infinitely in the positive direction; the weights are $m + 2\mathbb{Z}$. One can also start from a negative integer $m \leq -2$ and extend infinitely in the negative direction; the weights are $m - 2\mathbb{Z}$. We denote these modules by $D_m^+$ and $D_m^-$, respectively.

- Limits of discrete series: there are irreducible Verma modules starting from $m = 1$ and $m = -1$, as in the previous point. We denote these modules by $D_1^+$ and $D_1^-$, respectively. ■

### 3.2.2 The Harish-Chandra Isomorphism

Recall that a representation of a Lie algebra $\mathfrak{g}$ is equivalent to a representation of the associative algebra $U\mathfrak{g}$, which is the quotient of the free (non-commutative) algebra $\mathbb{C}\langle\mathfrak{g}\rangle$ by the two-sided ideal generated by relations of the form $XY - YX - [X, Y]$.

It is in general difficult to understand non-commutative rings, but sometimes one can try to say something by passing to commutative pieces of it.

**Theorem 3.30** (Harish-Chandra)**.** Fix a finite-dimensional simple complex Lie algebra $\mathfrak{g}$, and choose a maximal Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$. Then there is an isomorphism

$$Z(U\mathfrak{g}) \cong \mathrm{Sym}(\mathfrak{h})^{(W,\cdot)}.$$

Here, the $(W, \cdot)$-action on $\mathfrak{h}$ is $(-\rho)$-shifted, which means that $W$ fixes $(-\rho)$ (instead of the origin).

*Sketch.* Note that $U\mathfrak{g}$ admits an adjoint action by the complex algebraic group $G$ corresponding to $\mathfrak{g}$, defined by extending the adjoint action of $G$ on $\mathfrak{g}$. Thus, $Z(U\mathfrak{g})$ consists exactly of $(U\mathfrak{g})^G$.

Now, if we want to produce a $G$-invariant element of $U\mathfrak{g}$, we could start by noting that $U\mathfrak{g}$ is a semisimple representation of $G$ (indeed, it is a filtered colimit of finite-dimensional representations), so the surjection $\mathbb{C}\langle\mathfrak{g}\rangle \to U\mathfrak{g}$ descends to a surjection $\mathbb{C}\langle\mathfrak{g}\rangle^G \to (U\mathfrak{g})^G$. Again, it may be hard to find invariants in $\mathbb{C}\langle\mathfrak{g}\rangle$, but we can instead find invariants in the subspace $\mathrm{Sym}\,\mathfrak{g} \subseteq \mathbb{C}\langle\mathfrak{g}\rangle$. (Note that $\mathrm{Sym}\,\mathfrak{g}$ is not a subalgebra!)

While we're here, let's explicate this construction to find a useful element of $Z(U\mathfrak{g})$.

**Definition 3.31** (quadratic Casimir)**.** Fix a simple Lie algebra $\mathfrak{g}$. Recall that $\mathfrak{g}$ admits an invariant non-degenerate bilinear Killing form $K\colon \mathfrak{g} \otimes \mathfrak{g} \to \mathbb{C}$, so there is a canonical identification $\mathfrak{g} \cong \mathfrak{g}^*$. We then define the *quadratic Casimir element* $C \in Z(U\mathfrak{g})$ to be the image of the identity along the composite

$$\mathrm{End}(\mathfrak{g}) = \mathfrak{g} \otimes \mathfrak{g}^* \twoheadrightarrow \mathrm{Sym}^2\,\mathfrak{g} \to U\mathfrak{g}.$$

**Remark 3.32.** Note that $C$ lands in $Z(U\mathfrak{g})$ because this composite is $G$-invariant, and $\mathrm{End}(\mathfrak{g})^G$ is just $\mathbb{C}\,\mathrm{id}_{\mathfrak{g}}$ because $\mathfrak{g}$ is simple!

**Example 3.33.** For $\mathfrak{g} = \mathfrak{sl}_2(\mathbb{C})$ with standard basis $\{e, f, h\}$, the Killing form is given by $(X, Y) \mapsto \mathrm{tr}(XY)$. One can use this to calculate that the quadratic Casimir element is given by

$$ef + fe + \frac{1}{2}h^2.$$

Anyway, it turns out that $\mathrm{Sym}(\mathfrak{g})^G$ is isomorphic to $Z(U\mathfrak{g})$: it is a theorem that $\mathrm{gr}\,U\mathfrak{g} \cong \mathrm{Sym}\,\mathfrak{g}$, and in particular, one has that our map $\mathrm{Sym}\,\mathfrak{g} \to U\mathfrak{g}$ becomes an isomorphism (of $G$-representations) on the associated graded. It follows that we have an isomorphism of vector spaces $\mathrm{Sym}\,\mathfrak{g} \to U\mathfrak{g}$, and it gives an isomorphism on $G$-invariants before passing to the associated graded by some kind of Five lemma argument.

We now relate $\mathrm{Sym}(\mathfrak{g})^G$ to $\mathrm{Sym}(\mathfrak{h})^W$. Well, using the Killing form, we note that there is a natural restriction map

$$\mathrm{Sym}(\mathfrak{g}^*)^G \to \mathrm{Sym}(\mathfrak{h}^*)^W,$$

which turns out to be an isomorphism by Chevalley's theorem.

Thus far, we have only provided an isomorphism of vector spaces. It remains to provide an isomorphism of rings. Let $T \subseteq G$ be the maximal torus corresponding to a choice of Cartan $\mathfrak{h} \subseteq \mathfrak{g}$. We also choose a polarization so that we have a decomposition $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$. The moral is that we can start with $Z(U\mathfrak{g}) = (U\mathfrak{g})^G$ and then embed into $(U\mathfrak{g})^T$. These invariants come from a grading: indeed, $U\mathfrak{g}$ can be decomposed as

$$U\mathfrak{g} = \bigoplus_{\chi \in X^*(T)} (U\mathfrak{g})_\chi,$$

where $(U\mathfrak{g})_\chi$ is the $\chi$-eigenspace. Thus, $(U\mathfrak{g})^T$ simply amounts to taking the degree-zero subring. Thus, we may take a quotient by the ideal $I := (\mathfrak{n}_- \cdot U\mathfrak{g})_0 = (U\mathfrak{g} \cdot \mathfrak{n}_+)_0$. (Indeed, one can see that these are the same by pushing all positive or negative root spaces to one side in $U\mathfrak{g}$.) It turns out that $(U\mathfrak{g})_0/I$ is isomorphic to $\mathrm{Sym}(\mathfrak{h})^W$ (indeed, we are just killing all monomials which are not purely in $\mathfrak{h}$).

**Example 3.34.** Let's pass the Casimir element through the Harish-Chandra isomorphism. Well, starting with $\frac{1}{2}h^2 + ef + fe$, we first reorder the expression to $\frac{1}{2}h^2 + h + 2fe$. Then we take a quotient by $I$, which kills $2fe$, so we get $\frac{1}{2}h^2 + h$ under the Harish-Chandra isomorphism.

It is notably not obvious that this map even lands in $(U\mathfrak{h})^{(W,\cdot)}$, but once we have done so, we remark that we can check that our ring homomorphism is an isomorphism of vector spaces by comparing it with the isomorphism of vector spaces produced with Chevalley's theorem.

Let's explain how one shows that $(U\mathfrak{g})_0/I$ maps to $\mathrm{Sym}(\mathfrak{h})^W$. This requires a different construction of the Harish-Chandra isomorphism, using Verma modules. Fix some weight $\lambda \in \mathfrak{h}^*$, and then there is a Verma module $M_\lambda$. It turns out that $Z(U\mathfrak{g})$ acts by a fixed scalar on $M_\lambda$, essentially by using properties of highest weights, and it turns out that this scalar is the value of $\lambda$ under the polynomial produces by the Harish-Chandra isomorphism. To show that this scalar does not depend on the $(W, \cdot)$-orbit of $\lambda$, one can reduce t to the $\mathfrak{sl}_2$-theory with simple reflections. ■

**Example 3.35.** Let's explicate the last paragraph for $\mathfrak{sl}_2$. It turns out that there is a short exact sequence

$$0 \to M_{-\lambda-2} \to M_\lambda \to L_\lambda \to 0$$

of representations. Thus, the Casimir $\Omega$ acts on $M_{-\lambda-2}$ and $M_\lambda$ by the same scalar!

## 3.3 March 18

Here we go.

### 3.3.1 Admissible Representations

For any representation $M$ of $K$, note that there is a canonical map

$$\bigoplus_{\rho \in \mathrm{IrRep}\, K} \rho \otimes \mathrm{Hom}_K(\rho, M) \to M$$

given by composition, and we showed on the homework that this map is an isomorphism if and only if $M$ is locally finite. This decomposition allows us to define admissibility.

**Definition 3.36** (admissible)**.** Fix a real reductive group $G$ with Lie algebra $\mathfrak{g}$ and maximal compact subgroup $K$. Then a $(\mathfrak{g}, K)$-module $M$ is *admissible* if and only if $\dim \mathrm{Hom}_K(\rho, M)$ is finite-dimensional for all irreducible representations $\rho$ of $K$.

**Remark 3.37.** On the homework, we showed that a representation $M$ of a totally disconnected group $G$ is admissible if and only if $\dim \mathrm{Hom}_K(\rho, M)$ is finite-dimensional for all irreducible representations $\rho$ of a choice of compact open subgroup $K$ of $G$.

We also remark that we can bring down the definition of unitary to $(\mathfrak{g}, K)$-modules.

**Definition 3.38** (unitary)**.** Fix a real reductive group $G$ with Lie algebra $\mathfrak{g}$ and maximal compact subgroup $K$. Then a $(\mathfrak{g}, K)$-module $M$ is *unitary* if and only if $M$ admits a positive-definite Hermitian form $\langle -, - \rangle$ which is invariant under $\mathfrak{g}$ and $K$.

**Remark 3.39.** Invariance for $K$ is as expected, but invariance for $\mathfrak{g}$ must be differentiated: by the product rule, it turns out that we are asking for

$$\langle Xu, v\rangle + \langle u, Xv\rangle = 0$$

for all $X \in \mathfrak{g}$ and $u, v \in M$.

### 3.3.2 The Infinitesimal Character

The center of $U\mathfrak{g}$ is interesting to us because it acts by scalars on simple modules, and scalars are potentially easier to understand.

**Definition 3.40** (infinitesimal character)**.** Fix a real reductive group $G$ with Lie algebra $\mathfrak{g}$ and maximal compact subgroup $K$. For a simple $(\mathfrak{g}, K)$-module $M$, we define the *infinitesimal character* $\chi\colon Z(U\mathfrak{g}) \to \mathbb{C}$ by defining

$$\chi(z) := z|_M.$$

Because $M$ is a $\mathfrak{g}$-module, $\chi$ is an algebra homomorphism.

**Remark 3.41.** The fact that $z|_M$ is a scalar is some variant of Schur's lemma; one uses the same argument as in Proposition 2.45.

**Remark 3.42.** A choice of character $\chi\colon Z(U\mathfrak{g}) \to \mathbb{C}$ is equivalent to giving a maximal ideal of $\mathrm{Sym}(\mathfrak{h})^{(W,\cdot)}$, which is equivalent to the data of a (closed) point in $\mathfrak{h}^*/(W,\cdot)$, which is equivalent to the data of a $(W,\cdot)$-orbit in $\mathfrak{h}^*$.

The infinitesimal character turns out to be a very strong invariant.

**Proposition 3.43.** Fix a real reductive group $G$ with Lie algebra $\mathfrak{g}$ and maximal compact subgroup $K$. For any infinitesimal character $\chi\colon Z(U\mathfrak{g}) \to \mathbb{C}$, there are only finitely many simple $(\mathfrak{g}, K)$-modules (up to isomorphism) with this infinitesimal character.

**Example 3.44.** Let's compute some infinitesimal characters of the simple $(\mathfrak{sl}_2, \mathrm{SO}(2))$-modules $M$. Indeed, let $M = \bigoplus_n M_n$ be the weight decomposition. Then $ef$ should act by some scalar $c_{n-1}$ on $M_n$, and $fe$ should act by some scalar $c_{n+1}$ on $M_n$ as well. Because $[ef, fe] = h$, we see that $c_{n-1} - c_{n+1} = n$ for all $n$. On the other hand, the Casimir $\Omega$ must act by a single eigenvalue $\lambda$ on all the weight spaces $M_n$, so

$$\lambda = \frac{1}{2}n^2 + c_{n-1} + c_{n+1}$$

for all $n$. Thus, a choice of $\lambda$ allows us to fully determine the scalars up to a choice of just one scalar; for example, if we want our weights to be bounded below, we could assert that our lowest weight is given at $k$, and then $c_{k-1} = 0$, and then we can solve for the remaining weights.

It will turn out that all simple $(\mathfrak{g}, K)$-modules for $\mathrm{SL}_2(\mathbb{R})$ can be globalized to an irreducible representation. The easiest way to show this is by hand. A large source of our representations will come from the principal series.

**Definition 3.45** (principal series)**.** Let $B$ be the upper-triangular Borel subgroup of $\mathrm{SL}_2(\mathbb{R})$. For $\varepsilon \in \{0, 1\}$ and $s \in \mathbb{C}$, we define the *principal series* as the induced representation $\mathrm{Ind}_B^G\left(\mathrm{sgn}^\varepsilon |\cdot|^s\right)$, where $\mathrm{sgn}^\varepsilon |\cdot|^s$ is viewed as a character of $\mathbb{R}^\times$, identified with the diagonal torus in $B$.

**Remark 3.46.** Note that $\operatorname{diag}(-1,-1)$ acts by $(-1)^\varepsilon$ on the principal series. Thus, if $\varepsilon = 0$, then the corresponding $(\mathfrak{g}, K)$-module has even weights; conversely, if $\varepsilon = 1$, then the corresponding $(\mathfrak{g}, K)$-module has odd weights. One can also compute the action by Casimir; one finds that $\Omega$ acts by $\frac{1-s^2}{2}$.

**Remark 3.47.** If $s$ is purely imaginary, then $\operatorname{sgn}^\varepsilon |\cdot|^s$ is a unitary representation of $B$, so the principal series representation is unitary. It also turns out that the principal series is unitary when $s \in (-1,0) \cup (0,1)$; this complicates the representation theory, but it will not complicate the theory of automorphic forms because they do not appear.

Most of the remaining $(\mathfrak{g}, K)$-modules are found inside the discrete series. By way of motivation, recall the following result from the representation theory of compact groups.

**Theorem 3.48** (Peter–Weyl)**.** Fix a compact Lie group $G$. Then the matrix coefficient map

$$\bigoplus_{V \in \operatorname{IrRep} G} V \boxtimes V^* \to L^2(G)$$

has dense image.

### 3.3.3 The Discrete Series

We would like a similar decomposition to work for our general Lie groups, so we simply define the discrete series representations to be those which appear.

**Definition 3.49** (discrete series)**.** Fix a Lie group $G$. Then we define a *discrete series representation* of $G$ to be an irreducible representation which appears in $L^2(G)$.

**Remark 3.50.** It is equivalent to ask for $V$ to admit nonzero $v \in V$ and $\ell \in V^*$ for which the matrix coefficient $g \mapsto \ell(gv)$ is in $L^2(G)$.

Of course, it is now difficult to actually construct the discrete series. For example, even if $V$ is finite-dimensional, we expect $\dim V^*$ total copies of $V$ in $L^2(G)$, so it is going to be hard to find particular copies of $V$.

**Example 3.51.** We return to $G = \mathrm{SL}_2(\mathbb{R})$. Then the discrete series $D_k^-$ and $D_k^+$ are Verma modules, whose weights are concentrated in some ray. Accordingly, we are on the hunt for a copy of the tensor product of $D_k^-$ with a line of weight $k$ in $L^2(G)$. Let's look for this line in $L^2(G)$. Notably, our line should have weight $k$ for the $\mathrm{SO}(2)$-action, and it should be killed by the weight-lowering operator $f$; these properties generate copies of $D_k^+$. Thus, we are on the hunt for a function $\varphi$ for which

$$\varphi\left(g \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}\right) = e^{ik\theta}\varphi(g),$$

and with $f = \frac{1}{2}\left[\begin{smallmatrix} 1 & -i \\ -i & 1 \end{smallmatrix}\right]$, we see that we are asking to vanish under some differential operator. For explicit calculations, it is helpful to identify $G/K$ with the upper-half plane $\mathcal{H}$ given by the action of fractional linear transformations (where the stabilizer of $i$ is $K$). Once we do this, we see that vanishing under $f$ is equivalent to vanishing under $\overline{\partial}_z$, so we are hunting for holomorphic functions $\varphi$.

The answer to the construction in the previous example is that $V$ should be the space of holomorphic functions $\varphi\colon \mathcal{H} \to \mathbb{C}$ satisfying the functional equation

$$\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} f\right)(z) := (cz+d)^{-k}\varphi\left(\frac{az+b}{cz+d}\right).$$

In order to have a Hilbert space, we should also try to satisfy the (invariant) condition

$$\int_{\mathcal{H}} |\varphi(z)|^2 \, y^k \cdot \frac{dx \, dy}{y^k} < \infty$$

in order to be able to define a norm. These are called the anti-holomorphic discrete series, for some reason.

**Remark 3.52.** It turns out that these discrete series representations exist if and only if $G$ admits a compact maximal torus. (Equivalently, we are asking for $G$ to be an inner form of the compact form.) For example, there are no discrete series representations for $\mathrm{SL}_n(\mathbb{R})$ as soon as $n \geq 3$. On the other hand, the representations $\mathrm{SU}(p, q)$ do have discrete series. When they exist, there is a classification of the discrete series representations.

# Theme 4
# Automorphic Forms

## 4.1 March 30

Today, we start talking about automorphic forms.

### 4.1.1 Modular Forms

We start by reviewing the classical theory of modular forms, which are examples of automorphic forms for the group $\mathrm{SL}_2$. Roughly speaking, a modular form is a symmetric function on the upper-half plane $\mathcal{H}$.

**Definition 4.1** (cusp)**.** We set $\mathcal{H} := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}$ and $\mathcal{H}^* := \mathcal{H}^* \cup \mathbb{P}^1(\mathbb{Q})$. We say that $\mathbb{P}^1(\mathbb{Q}) \subseteq \mathcal{H}^*$ are the *cusps*.

**Example 4.2.** Note that $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H}^*$ is $\mathrm{SL}_2(\mathbb{Z})\backslash\mathcal{H} \sqcup \{\infty\}$ because $\mathrm{SL}_2(\mathbb{Z})$

**Example 4.3.** For any finite-index subgroup $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$, we see that $\Gamma\backslash\mathcal{H}^* = \Gamma\backslash\mathcal{H}\sqcup$

**Definition 4.4** (modular form)**.** Fix a congruence subgroup $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ (meaning that it contains the kernel of the projection $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ for some $N$), and choose an integer $k$ such that $k \geq 2$. Then a *modular form of weight $k$ and level $\Gamma$* is a holomorphic function $f\colon \mathcal{H} \to \mathbb{C}$ satisfying the following.

- Symmetry: for all $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right] \in \Gamma$ and $z \in \mathcal{H}$, we have
$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z).$$
- Cusps: the function $f$ is holomorphic at the cusps, meaning that $f$ extends to a holomorphic function on $\mathcal{H}^*$.

Here, being holomorphic at the cusps means that $f$ can be given a power series expansion around each cusp, which extends to a holomorphic function at the cusp. Notably, $\mathcal{H}^*$ can be given the structure of a complex manifold, so $f$ can be viewed as a holomorphic function in a punctured disk around a given cusp. Thus, we see that being able to extend $f$ to the puncture amounts to requiring that $f$ is bounded in an open neighborhood of the cusp.

**Example 4.5.** Let's explain what it means for $f$ to be holomorphic at $\infty \in \mathbb{P}^1(\mathbb{Q})$. Here, note that $f(z) = f(z+N)$ for some sufficiently large positive integer $N$. Then $f$ descends to a function on $\mathbb{C}/N\mathbb{Z}$, which is isomorphic to $\mathbb{C}^\times$ via the exponential map $z \mapsto \exp(2\pi i z/N)$. Thus, $f$ can be viewed as a holomorphic function on $\mathbb{C}^\times$ in terms of the coordinate $q := \exp(2\pi i z/N)$, and we are requiring that $f$ should be bounded as $q \to 0$, which corresponds to $\operatorname{Im} z \to \infty$.

Once we have extended $f$ to the cusps, it will turn out to be interesting to just look at those modular forms which vanish at the cusps. Indeed, once we have enough modular forms to be non-vanishing at all the cusps (in a linearly independent way), one can always modify $f$ to vanish at the cusps anyway.

**Definition 4.6** (cusp form)**.** A *cusp form* is a modular form which vanishes at the cusps.

Let's try to view this definition from the perspective of representation theory. For simplicity, we take $k = 2$.

**Lemma 4.7.** Modular forms of weight $2$ and level $\Gamma$ correspond to $\Gamma$-invariant meromorphic $1$-forms on $\mathcal{H}^*$ which are holomorphic everywhere, except possibly simple poles at the cusps.

*Proof.* To start, we note that $\mathcal{H}$ is an analytic open subset of $\mathbb{P}^1(\mathbb{C})$, and the transformation law for a modular form $f$ corresponds to making $f(z)\,dz$ into a regular $1$-form on $\mathcal{H}$ which is $\Gamma$-invariant: indeed,

$$f\left(\frac{az+b}{cz+d}\right) d\left(\frac{az+b}{cz+d}\right) = f\left(\frac{az+b}{cz+d}\right)(cz+d)^{-2}\,dz,$$

which is $f(z)\,dz$ by hypothesis on $f$.

It remains to codify the condition that $f$ is holomorphic at the cusps. Well, by translation, we may as well move the cups to $\infty$, where Example 4.5 tells us that we are requiring that $f(z)$ admits an expansion $\varphi(q)$ via the variable $q = \exp(2\pi i z)$. Then $f(z)\,dz = \frac{1}{2\pi i}\varphi(q)\,\frac{dq}{q}$, so the condition that $\varphi$ is holomorphic at $q = 0$ corresponds to $f(z)\,dz$ admitting at worst a simple pole at $q = 0$. Translating this around all the cusps completes the proof of the lemma. ■

**Remark 4.8.** The same argument shows that cusp forms correspond to $\Gamma$-invariant holomorphic $1$-forms on $\mathcal{H}^*$.

### 4.1.2 Double Quotients

To continue, we note that $\mathcal{H}$ comes from representation theory: $\mathrm{SL}_2(\mathbb{R})$ acts transitively on $\mathcal{H}$ by fractional linear transformations, and the stabilizer of $i$ is exactly $\mathrm{SO}_2(\mathbb{R})$. Thus, $\mathcal{H}$ can be viewed as the quotient $\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2(\mathbb{R})$, and $\Gamma\backslash\mathcal{H}$ is a double quotient $\Gamma\backslash\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2(\mathbb{R})$.

It will be desirable to write this double quotient in the form

$$\mathrm{SL}_2(\mathbb{Q})\backslash\mathrm{SL}_2(\mathbb{A}_\mathbb{Q})/K,$$

where $K \subseteq \mathrm{SL}_2(\mathbb{A}_\mathbb{Q})$ is some compact subgroup. To explain how this works, let's pass to a more general setting. Fix a reductive group $G$ over a number field $F$. For each infinite place $v$, let $K_v \subseteq G(F_v)$ be a maximal compact subgroup; and for each finite place $v$, we let $K_v \subseteq G(F_v)$ be an open compact subgroup. We further require that $K_v = G(\mathcal{O}_v)$ for all but finitely many $v$.

**Remark 4.9.** Note that the condition $K_v = G(\mathcal{O}_v)$ for all but finitely many $v$ makes sense because $G$ admits an integral model over all but finitely many $v$, and the group $G(\mathcal{O}_v)$ is the same between such integral models for all but finitely many $v$.

We further define $K_f := \prod_{v<\infty} K_v$ and $\Gamma := G(\mathbb{Q}) \cap K_f$.

**Example 4.10.** If $G = \mathrm{SL}_{2,\mathbb{Q}}$, then $K_f = \mathrm{SL}_2(\widehat{\mathbb{Z}})$,

**Theorem 4.11.** Fix notation as above. For semisimple, simply connected, split groups $G$ over $\mathbb{Q}$, the canonical map

$$\Gamma\backslash G(\mathbb{R}) \to G(\mathbb{Q})\backslash G(\mathbb{A}_{\mathbb{Q}})/K_f$$

is a bijection.

**Remark 4.12.** One can weaken the hypothesis that $G$ is split, but one needs something. For example, if $G(\mathbb{R})$ is compact, then the source is compact while the target may not be, so our map cannot be surjective. As an example of such a $G$, we may let $F$ be a quadratic imaginary field over $\mathbb{Q}$ and choose a Hermitian space $V$ over $F$ which is not definite (once base-changed to $\mathbb{R}$), and then set $G$ to be the corresponding special unitary group $\mathrm{SU}(V)$.

It is useful to have language for the conclusion of Theorem 4.11.

**Definition 4.13** (strong approximation)**.** Fix a reductive group $G$ over a number field $F$. Then $G$ satisfies *strong approximation* for a finite set of place $v$ if and only if $G(F) \cdot G(F_v)$ is dense in $G(\mathbb{A}_F)$.

**Remark 4.14.** It turns out that the conclusion of Theorem 4.11 is equivalent to satisfying strong approximation at the infinite place.

**Remark 4.15.** In general, $G$ satisfies strong approximation at the infinite place if $G$ is semisimple, simply connected, and $G$ is an almost-direct product of $\mathbb{Q}$-simple groups $G_i$ for which $G_i(\mathbb{R})$ is non-compact. There is a similar statement for other places.

We have just spent some time attempting to explain that strong approximation is a fairly common property of groups, but there are counterexamples.

**Example 4.16.** Consider the group $G = \mathbb{G}_m$ over a number field $F$, and take $K_v = \mathbb{G}_m(\mathcal{O}_v)$ for all finite $v$. (Note that $G$ is not semisimple!) Then there is a short exact sequence

$$1 \to \mathcal{O}_F^\times \backslash F_\infty^\times \to F^\times \backslash \mathbb{A}_F^\times / K_f \to \mathrm{Cl}(F) \to 1.$$

Thus, in general, $\mathbb{G}_m(F) \cdot \mathbb{G}_m(F_\infty)$ cannot be dense in $\mathbb{G}_m(\mathbb{A}_F)$ because it generates a finite-index subgroup.

The moral of all of this is that $\mathrm{SL}_2$ does satisfy strong approximation (one can also check this directly), so the natural map

$$(\mathrm{SL}_2(\mathbb{Q}) \cap K_f)\backslash \mathrm{SL}_2(\mathbb{R}) \to \mathrm{SL}_2(\mathbb{Q})\backslash \mathrm{SL}_2(\mathbb{A}_{\mathbb{Q}})/K_f$$

is a bijection. Thus, our modular forms can be viewed as meromorphic $1$-forms on this double quotient.

### 4.1.3 Automorphic Forms

We are now ready to generalize modular forms to automorphic forms. This will require a growth condition, for which we need a notion of height.

**Definition 4.17** (height)**.** Fix an embedded affine variety $\iota\colon X \hookrightarrow \mathbb{A}^n$ over a number field $F$. Then we define the *height* $\|\cdot\|_\iota : X(\mathbb{A}_F) \to \mathbb{R}^+$ to be given by

$$\|x\|_\iota := \prod_v \max\{|\iota(x)_1|_v, \ldots, |\iota(x)_n|_v, 1\}.$$

**Remark 4.18.** By considering the product embedding, we see that any two heights $\|\cdot\|_1$ and $\|\cdot\|_2$ admit some large $N$ for which $\|\cdot\|_1$ is dominated by $\|\cdot\|_2^N$.

**Definition 4.19** (automorphic form)**.** Fix a reductive group $G$ over a number field $F$. An *automorphic form* $\varphi$ for $G$ is a function $\varphi\colon G(\mathbb{Q})\backslash G(\mathbb{A}_F) \to \mathbb{C}$ satisfying the following.

- Smooth: for any $x \in G(\mathbb{A}_F)$, the function $G(F_\infty) \to \mathbb{C}$ given by $g \mapsto \varphi(xg)$ is smooth.
- Smooth: for any $x \in G(\mathbb{A}_F)$, the function $G(\mathbb{A}_F^\infty) \to \mathbb{C}$ given by $g \mapsto \varphi(xg)$ is locally constant.
- Finiteness: the function $\varphi$ is $K_\infty$-finite.
- Finiteness: for each infinite place $v$, smoothness implies that the Lie algebra $\mathfrak{g}_v$ of $G(F_v)$ may act by derivatives. Then $\varphi$ must be $Z\mathfrak{g}_v$-finite.
- The function $\varphi$ is fixed by some open compact subgroup of $G(\mathbb{A}_F^\infty)$.
- Moderate growth: for any height $\|\cdot\|$ on $G$, there are constants $C$ and $r$ for which $|\varphi(g)| \le C\,\|g\|^r$ for all $g \in G(\mathbb{A}_F)$.

We let $\mathcal{A}(G)$ denote the space of all automorphic forms.

Let's see how modular forms give rise to automorphic forms.

**Example 4.20.** Let's explain this growth condition for $G = \mathrm{SL}_2(\mathbb{R})$. Then $\Gamma\backslash \mathrm{SL}_2(\mathbb{R})$ can be covered by matrices of the form

$$\begin{bmatrix} 1 & x \\ & 1 \end{bmatrix} \begin{bmatrix} a & \\ & a^{-1} \end{bmatrix}$$

where $x$ is found in some finite interval, and $a$ is lower-bounded. (One can show this directly for the congruence subgroups.) By the Phragmén–Lindelöf principle, it turns out that $\varphi$ admitting moderate growth simply means that $\varphi(x+iy)$ has at most polynomial growth as $y \to \infty$.

**Example 4.21.** More specifically, if $\varphi$ is a holomorphic function on $\Gamma\backslash \mathrm{SL}_2(\mathbb{R})$, then the moderate growth condition means that the $q$-series expansion cannot feature any negative powers of $q = e^{2\pi i z/N}$ because the function $y \mapsto e^{2\pi y/N}$ grows faster than any polynomial.

By construction, we see that $\mathcal{A}(G)$ is a $(\mathfrak{g}, K_\infty)$-module. (One needs to check that derivatives of functions of moderate growth continue to have moderate growth.) Similarly, it is automatically a smooth representation of the totally disconnected group $G(\mathbb{A}_F^\infty)$, and these two actions commute. Thus, $\mathcal{A}$ is almost a representation of $G(\mathbb{A}_F)$!

We would like for $\mathcal{A}$ to be (suitably) admissible, but this is not the case. Here is our target definition, and we will try to find admissible parts of $\mathcal{A}$.

**Definition 4.22** (admissible)**.** Fix a module $M$ with commuting actions of $(\mathfrak{g}, K_\infty)$ and a smooth action of $G(\mathbb{A}_F^\infty)$. Then $M$ is *admissible* if and only if each irreducible representation of $K_\infty \times K_f$ appears with finite multiplicity.

**Theorem 4.23** (Harish-Chandra)**.** Fix an ideal $J \subseteq Z\mathfrak{g}$ of finite codimension. Then the subrepresentation $\mathcal{A}(J)$ of $\mathcal{A}$ annihilated by $J$ is admissible.

**Example 4.24.** If we further take $K_f$-invariants, then $\mathcal{A}(J)^{K_f}$ is an admissible $(\mathfrak{g}, K_\infty)$-module.

We close class with the main definition of the course.

**Definition 4.25** (automorphic representation)**.** Fix a reductive group $G$ over a number field $F$. An *automorphic representation* of $G$ is an irreducible admissible representation of $(\mathfrak{g}, K_\infty)$ and $G(\mathbb{A}_F^\infty)$ (with commuting actions) which appears as a subquotient of $\mathcal{A}$.

**Remark 4.26.** Note that automorphic representations admit an infinitesimal character, so it comes from some $\mathcal{A}(J)$. It follows that automorphic representations are admissible by Theorem 4.23.

## 4.2 April 1

Here we go.

### 4.2.1 Back to Modular Forms

We now explain how modular forms give rise to automorphic forms.

**Exercise 4.27.** Set $G := \mathrm{SL}_{2,\mathbb{Q}}$, fix a level $K_f \subseteq G(\mathbb{A}_\mathbb{Q}^\infty)$, and set $\Gamma := K_f \cap \mathrm{SL}_2(\mathbb{Q})$. Now, we let $M_k(\Gamma)$ be the holomorphic modular forms of weight $k$ and level $\Gamma$, and we define a map $\varphi_\bullet M_k(\Gamma) \to \mathcal{A}$ as follows. Identify $\mathcal{A}^{K_f} \subseteq C^\infty(G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q})/K_f)$ with $C^\infty(\Gamma\backslash G(\mathbb{R}))$ by strong approximation. Then given $f \in M_k(\Gamma)$, we define $\varphi_f \in C^\infty(\Gamma\backslash G(\mathbb{R}))$ by

$$\varphi_f\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = f\left(\frac{ai+b}{ci+d}\right)(ci+d)^{-k}.$$

We claim that $\varphi_f$ is an automorphic form.

*Proof.* By construction, we see that $\varphi_f$ is $K_f$-invariant. Here are the other checks for $\varphi_f$ to be an automorphic form.

- Because $\varphi$ has weight $k$, we see that $\varphi_f(gs) = s^k \cdot \varphi_f(g)$ for any $s \in \mathrm{SO}_2(\mathbb{R})$ (identified with $S^1$). Thus, $\varphi_f$ is an eigenfunction under $K_\infty$ of weight $k$, so $K_\infty$-finiteness follows.
- Let $L$ be the Cauchy–Riemann operator. Then $f$ being holomorphic means that $Lf = 0$. On the other hand, it turns out that $L$ is the action by $f \in \mathfrak{sl}_2(\mathbb{C})$, so it follows that $\Omega f = \frac{k(k-2)}{2} f$, so $Z\mathfrak{g}$-finiteness follows.
- Given the above, $f$ being holomorphic at the cusps implies that $\varphi_f$ has moderate growth by an examination of the $q$-expansion at $\infty$ in $\mathcal{H}$. ■

**Remark 4.28.** This argument shows that $M_k(\Gamma)$ embeds into the automorphic forms which are killed by the ideal $(\Omega - k(k-2)/2) \subseteq Z\mathfrak{g}$, have $K_\infty$-action given by $\sigma\colon \mathrm{SO}_2(\mathbb{R}) \to S^1$ which is the $k$th power, and they are invariant under $K_f$. This subspace of automorphic forms must in fact be given by $M_k(\Gamma)$: indeed, one can define $f$ by reversing the definition of $\varphi_f$ (which is well-defined due to the weight $k$), and having weight $k$ and having the given action by Casimir implies that $f$ is holomorphic.

**Remark 4.29.** If one changes $\frac{k(k-2)}{2}$ to some other number $\lambda$, then we get Maass forms. They are notably not holomorphic!

While we're here, we restate our admissibility theorem.

**Notation 4.30.** Fix a semisimple group $G$ over a number field $F$. Further, choose $J \subseteq Z\mathfrak{g}$ of finite codimension, an irreducible representation $\sigma$ of $K_\infty$, and a level $K_f \subseteq G(\mathbb{A}_F^\infty)$. Then we define $\mathcal{A}(J, \sigma, K_f)$ to be the space of automorphic forms killed by $J$ with $K_\infty$-type $\sigma$, and fixed by $K_f$.

**Theorem 4.31** (Harish-Chandra)**.** Fix a semisimple group $G$ over a number field $F$. Further, choose $J \subseteq Z\mathfrak{g}$ of finite codimension, an irreducible representation $\sigma$ of $K_\infty$, and a level $K_f \subseteq G(\mathbb{A}_F^\infty)$. Then

$$\dim \mathcal{A}(J, \sigma, K_f) < \infty.$$

By varying $\sigma$ and $K_f$, one recovers Theorem 4.23. Let's work this out for modular forms.

**Example 4.32.** Theorem 4.31 implies that $M_k(\Gamma)$ is finite-dimensional by Remark 4.28. Let's show this directly. Note that $f \in M_k(\Gamma)$ gives rise to a differential form $f(z)\,(dz)^{k/2} \in \omega_{\mathcal{H}/\mathbb{C}}^{\otimes k/2}$. (This makes sense if $k$ is even. If $k$ is odd, then we should view $\omega$ as $\mathcal{O}(-2)|_{\mathcal{H}}$, where $\mathcal{O}(-2)$ comes from the embedding $\mathcal{H} \hookrightarrow \mathbb{P}^1$; then $\omega^{\otimes k/2}$ is just $\mathcal{O}(-k)|_{\mathcal{H}}$.) The same argument as in Example 4.5 with $q$-expansions shows that $f(z)\,(dz)^\ell$ is allowed a pole of order at most $k/2$ at each cusp. Thus, we have embedded $M_k(\Gamma)$ into the global sections of a coherent sheaf (namely, differential forms with bounded poles) on the compact manifold $\Gamma\backslash\mathcal{H}^*$. The compact complex curve is a projective algebraic curve, and this vector bundle is also algebraic (it is some power of $\omega$, twisted appropriately), so its space of global sections is finite-dimensional!

**Remark 4.33.** The proof of Example 4.32 shows that modular forms are algebraic, so we may find a basis defined over a fixed number field.

### 4.2.2 Suddenly Shimura Varieties

It seems difficult to compute the dimension in Theorem 4.31. One approach is to embed $\Gamma\backslash G(\mathbb{R})/K_\infty$ into a Zariski open subset of a complex projective variety and then proceed as in Example 4.32 to compute dimensions via algebraic geometry. For example, this is done in the theory of Shimura varieties, where $G(\mathbb{R})/K_\infty$ is found to be a Hermitian symmetric domain.

**Example 4.34.** For $G = \mathrm{Sp}(V, \omega)$, we have $K_\infty = \mathrm{U}(V)$; set $X := \mathrm{Sp}(V)/K_\infty$. Note that $\dim_{\mathbb{R}} X = n(n+1)$, which is even, so this has a chance of admitting a complex structure. In fact, $X$ can be embedded into the space of Lagrangian subspaces in the standard symplectic space $V_{\mathbb{C}}$: indeed, $\mathrm{Sp}(V)$ certainly acts on isotropic subspaces of $V_{\mathbb{C}}$, and one can check that this action is transitive. To see that we can achieve $\mathrm{U}(n)$ as a stabilizer, recall that $\mathrm{U}(n)$ is embedded as automorphisms of $V$, which we give a complex structure $J$, which then gives rise to a unique Hermitian structure $h$ for which $\operatorname{im} h$ is the symplectic form on $V$. Now, $V$ with the complex structure $J$ allows us to decompose $V_{\mathbb{C}} = V \oplus \overline{V}$ (via the decomposition $\mathbb{C} \otimes \mathbb{C} = \mathbb{C} \oplus \mathbb{C}$). Now, $V \subseteq V_{\mathbb{C}}$ is Lagrangian, and its stabilizer is exactly $K_\infty = \mathrm{U}(V, h)$!

**Example 4.35.** However, this approach cannot work in general: $\mathrm{SL}_3(\mathbb{Z})\backslash \mathrm{SL}_3(\mathbb{R})/\mathrm{SO}_3(\mathbb{R})$ is five-dimensional, so it cannot be turned into a complex projective variety.

**Remark 4.36.** For $G$ simple, one finds that $X := G(\mathbb{R})/K_\infty$ is a Hermitian symmetric domain if and only if $Z(K_\infty)$ contains a copy of $S^1$. Intuitively, what is going on here is that this action of $S^1$ will rotate tangent vectors in $G(\mathbb{R})/K_\infty$, thereby giving a complex structure on $T_1X$.

**Example 4.37.** The real form $G(\mathbb{R}) = \mathrm{SU}(p,q)$ has $K_\infty = \mathrm{S}(\mathrm{U}(p) \times \mathrm{U}(q))$ has a copy of $\mathrm{U}(1)$, so this case admits a Hermitian symmetric domain. This case turns out to embed into the Grassmannian variety of $p$-dimensional subspaces of $\mathbb{R}^{p+q}$.

### 4.2.3 Siegel Domains

For a semisimple group $G$, one can tile the double quotient $G(F)\backslash G(\mathbb{A}_F)/K_f$ by things which look like $\Gamma\backslash G(F_v)$. It may be desirable to put such elements into a "normal form."

**Theorem 4.38** (Iwasawa decomposition)**.** Fix a real split reductive group $G$ over $\mathbb{R}$. Then

$$G(\mathbb{R}) = N(\mathbb{R})A(\mathbb{R})^\circ K_\infty,$$

where $N(\mathbb{R})$ is the unipotent subgroup of a Borel subgroup, $A(\mathbb{R})$ has positive diagonal elements from a maximal split torus, and $K_\infty$ is the maximal compact subgroup. In fact, this decomposition is unique.

*Proof.* If $G = \mathrm{SL}_n$, then one does Gram–Schmidt orthogonalization. The proof in general is hard. ■

The moral is that $\Gamma\backslash G(F_v)/K_\infty$ is covered by the domain $(\Gamma \cap N(\mathbb{R}))\backslash N(\mathbb{R})A(\mathbb{R})^\circ$.

**Example 4.39.** For $G = \mathrm{SL}_2(\mathbb{R})$, we see that $\Gamma \cap N(\mathbb{R})$ consists of translations, and $N(\mathbb{R})A(\mathbb{R})^\circ$ is the upper-half plane. Thus, the quotient is a vertical strip, which has infinite volume.

Having infinite volume is understood to be undesirable, so we want to cut down our covering set more. The idea is to make $A$ smaller.

**Definition 4.40** (Siegel domain)**.** Fix a split semisimple group $G$ over $\mathbb{R}$, and choose an arithmetic subgroup $\Gamma \subseteq G(\mathbb{R})$. Let $N \subseteq G$ be the unipotent radical of a Borel subgroup, and let $A \subseteq G$ be a split torus. Further, fix a compact fundamental domain $\Omega_N \subseteq N(\mathbb{R})$ for the action of $\Gamma \cap N(\mathbb{R})$, and choose some $\varepsilon > 0$ and define $A_\varepsilon \subseteq A(\mathbb{R})^\circ$ by

$$A_\varepsilon := \{a \in A(\mathbb{R})^\circ : \alpha_i(a) \geq \varepsilon\},$$

where $\alpha_i$ is lifted to a character $A(\mathbb{R}) \to \mathbb{R}^\times$. Then a *Siegel domain* is the image of $\Omega_N A_\varepsilon$ in $G(\mathbb{R})/K_\infty$.

**Remark 4.41.** It turns out that Siegel domains always exist so that $\Omega_N A_\varepsilon$ covers $\Gamma\backslash G(\mathbb{R})/K_\infty$. We show this on the homework for $\mathrm{SL}_n$.

Here is our application.

**Theorem 4.42.** Fix a semisimple group $G$ over a number field $F$. Given $G(F)\backslash G(\mathbb{A}_F)$ a measure by pushing forward the Haar measure on $G(\mathbb{A}_F)$. Then the volume of $G(F)\backslash G(\mathbb{A}_F)$ is finite.

*Proof.* We will only prove this in the case where $G$ is split and $F = \mathbb{Q}$, but the difficulties to remove these assumptions are not important. Because $K_f$ is open and compact, it is enough to show that $G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q})/K_f$

has finite volume, for any level $K_f \subseteq G(\mathbb{A}_\mathbb{Q})$. Recall that $G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q})$ can be partitioned into pieces which look like $\Gamma\backslash G(\mathbb{R})$, for possibly varying $\Gamma$.

Thus, we would like to show that $\Gamma\backslash G(\mathbb{R})$ has finite volume. By Remark 4.41, it is enough to show that $\Omega_N A_\varepsilon K_\infty$ has finite volume, where $\Omega_N A_\varepsilon$ is a Siegel domain. Now, because $G$ is unimodular, Theorem 4.38 allows us to decompose the Haar measure $dg$ into $d_\ell(na)d_r k$ (indeed, this is a measure, and it has enough invariance requirements). Further, $N$ is normal and unimodular in $AN$, so it turns out that

$$d_\ell(na) = da \cdot dn = a^{-2\rho}\, dn\, da.$$

Namely, $a^{-2\rho}$ comes out as the modular character from $A$ acting on $N$.

**Example 4.43.** Take $G = \mathrm{SL}_2(\mathbb{R})$. With $x$ as the coordinate of $\left[\begin{smallmatrix} 1 & x \\ & 1 \end{smallmatrix}\right]$ in $N(\mathbb{R})$, and $y$ as the coordinate of $\operatorname{diag}(\sqrt{y}, 1/\sqrt{y})$ in $A(\mathbb{R})^\circ$, one finds that the Haar measure on $G$ is

$$\frac{dx\, dy}{y^2}\, dk,$$

where $dk$ is the Haar measure on $K_\infty$.

Thus, the volume of $\Omega_N A_\varepsilon K_\infty$ is the product of the volumes of $\Omega_N$, $A_\varepsilon$, and $K_\infty$, all considered separately, along with some factor of $a^{-2\rho}$. Notably, the volumes of $\Omega_N$ and $K_\infty$ are finite because these are compact, so it only remains to show that

$$\int_{A_\varepsilon} a^{-2\rho}\, da$$

is finite. Now, $A$ is split, so this turns into a product of integrals over $\mathbb{G}_m$s, so we reduce to the case of $\mathrm{SL}_2(\mathbb{R})$, where the integral is

$$\int_\varepsilon^\infty a^{-1}\, \frac{da}{a} < \infty,$$

so we are done! ■

**Example 4.44.** One can actually compute that

$$\operatorname{vol}(\mathrm{SL}_n(\mathbb{Z})\backslash \mathrm{SL}_n(\mathbb{R})) = \zeta(2)\zeta(3)\cdots\zeta(n),$$

provided that the left-hand volume is normalized appropriately (given by taking a top form of $\mathfrak{sl}_n$).

## 4.3 April 6

Here we go. The quiz on Wednesday will have two problems, and we will be able to use the entire class period.

### 4.3.1 Cusp Forms

Here is our definition.

**Definition 4.45** (cuspidal)**.** Fix a reductive group $G$ over a number field $F$. Then an automorphic form $f$ is *cuspidal* if and only if

$$\int_{N(F)\backslash N(\mathbb{A}_F)} f(ng)\, dn = 0$$

for any $g \in G(\mathbb{A}_F)$ and proper parabolic subgroup $P \subseteq G$ (defined over $F$) with unipotent radical $N$. We let $\mathcal{A}_{\mathrm{cusp}}$ denote the subspace of cusp forms.

**Remark 4.46.** The integral converges because $f$ is smooth, and the space $N(F)\backslash N(\mathbb{A}_F)$ is compact. Indeed, $N$ is an iterated extension of $\mathbb{G}_a$s, so $N(F)\backslash N(\mathbb{A}_F)$ is an iterated extension of the compact spaces $F\backslash\mathbb{A}_F$.

**Remark 4.47.** It is enough to only consider the maximal parabolic subgroups: a larger parabolic subgroup produces a smaller unipotent radical, so the integral for the larger parabolic can be tiled by integrals for the larger parabolic. It is also enough to even check a single representative in each conjugacy class. For $\mathrm{GL}_n$, these look like the block-upper triangular matrices

$$\begin{bmatrix} * & * \\ & * \end{bmatrix}.$$

A non-maximal such parabolic subgroup may have more blocks.

**Exercise 4.48.** Choose a modular form $f$ of weight $k$ and level $\mathrm{SL}_2(\mathbb{Z})$, and let

$$\varphi_f\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) := f\left(\frac{ai+b}{ci+d}\right)(ci+d)^{-k}$$

be the corresponding automorphic form. We will check that $f$ is cuspidal if and only if $\varphi_f$ is.

*Proof.* The only parabolic to check (up to conjugacy) is the Borel subgroup, so we let $N$ be the subgroup $\left[\begin{smallmatrix} 1 & * \\ & 1 \end{smallmatrix}\right]$, and we would like to check if the integral

$$\int_{N(\mathbb{Q})\backslash N(\mathbb{A}_\mathbb{Q})} \varphi_f(ng)\,dn = \int_{\mathbb{Z}\backslash\mathbb{R}} f(z+x)\,dx$$

vanishes, where $z = gi$. Namely, one can see that the integral on the left descends to $\mathbb{Q}\backslash\mathbb{A}_\mathbb{Q}/\widehat{\mathbb{Z}} = \mathbb{Z}\backslash\mathbb{R}$, which is the right-hand integral. Writing $f(z) = \sum_n a_n q^n$ with $q = e^{2\pi i z}$, we thus see that this integral computes the constant term $a_0$ in this Fourier expansion: $a_0 = \frac{1}{2\pi i}\oint_\gamma f(q)\,\frac{dq}{q}$, where $\gamma$ is the corresponding loop around $0$. Thus, we see that $\varphi_f$ is cuspidal if and only if $a_0 = 0$, which is what it means for $f_0$ to be cuspidal. ■

Cusp forms are important for us because they have an application to the spectral decomposition. In particular, we are interested in decomposing $L^2(G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q}))$.

**Lemma 4.49.** Fix a semisimple group $G$ over a number field $F$. Then the space of automorphic forms embeds as a dense subspace of $L^2(G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q}))$.

*Proof.* We will not show this. Note that the embedding exists because $G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q})$ has finite volume. ■

**Remark 4.50.** A similar statement works for $G$ reductive. There is a slight caveat:

**Notation 4.51.** Fix a semisimple group $G$ over $\mathbb{Q}$. Then we let $L^2_{\mathrm{cusp}}(G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q}))$ be the completion of $\mathcal{A}_{\mathrm{cusp}}$ in $L^2(G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q}))$.

Now, $G(\mathbb{A}_\mathbb{Q})$ acts by right translation on $L^2(G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q}))$, and it preserves the integration pairing given by the Haar measure. Thus, we are basically looking at some operators on a Hilbert space, where one expects to have a discrete spectrum and a continuous spectrum. This cuspidal subspace turns out to give us a lot of information about the discrete spectrum.

**Proposition 4.52.** Fix a semisimple group $G$ over $\mathbb{Q}$. Then the action of $G(\mathbb{A}_\mathbb{Q})$ on $L^2_{\text{cusp}}(G(\mathbb{Q})\backslash G(\mathbb{A}_\mathbb{Q}))$ decomposes into a direct sum of irreducible representations with finite multiplicity.

**Example 4.53.** For $G = \mathrm{GL}_n$, irreducible representations appear at most once.

**Remark 4.54.** There may be other subspaces in the "discrete spectrum." For example, the constant functions are certainly in the discrete spectrum. For $G = \mathrm{SL}_2$, it turns out that the discrete spectrum is the sum of the cuspidal part and the constant functions.

We now turn to the continuous spectrum. Here, these are constructed by parabolic induction, using Eisenstein series. The idea, roughly speaking, is to take a parabolic subgroup $P \subseteq G$ with Levi subgroup $M$. Then choose a discrete irreducible automorphic representation $\pi$ of $M$, which embeds into $L^2(M(\mathbb{Q})\backslash M(\mathbb{A}_\mathbb{Q}); \chi)$, where $\chi$ is some fixed central character. From here, there is a process which allows us to induce from $L^2(M(\mathbb{Q})\backslash M(\mathbb{A}_\mathbb{Q}); \chi)$ back to $G$.

**Example 4.55.** Suppose that $P$ is a Borel subgroup so that $M$ is a torus $T$. Then the choice of central character amounts to a choice of character of $T$. If $T$ is split of rank $n$, then we are basically hunting for $n$ idéle class characters, so there are $n$ continuos parameters in this "continuous spectrum."

### 4.3.2 Automorphic Forms for Function Fields

Let's work with function fields. For convenience, let $G$ be a split semisimple group over a finite field $\mathbb{F}_q$, and we let $X$ be a smooth projective geometrically integral curve, and we set $F := \mathbb{F}_q(X)$.[1] Our automorphic forms should approximately come from $C^\infty(G(F)\backslash G(\mathbb{A}_F))$, which already has an action by $G(\mathbb{A}_F)$ by translation.

Here is our definition of automorphic form, which notably does not have to fight with analytic difficulties to get off the ground.

**Definition 4.56** (automorphic form)**.** Fix a function field $F$, and let $G$ be a split reductive group. Let $G(\mathbb{A}_F)$ act on $C^\infty(G(F)\backslash G(\mathbb{A}_F))$ by right translations, denoted $R$. Then $f \in C^\infty(G(F)\backslash G(\mathbb{A}_F))$ is an *automorphic form* if and only if

$$\operatorname{span}\{R(g)f : g \in G(\mathbb{A}_F)\}$$

is an admissible representation of $G(\mathbb{A}_F)$. We let $\mathcal{A}(G, K)$ be the space of automorphic forms of level $K$.

Unsurprisingly, we have a notion of cusp forms.

**Definition 4.57** (cuspidal)**.** Fix a function field $F$, and let $G$ be a split reductive group. An automorphic form $f$ is *cuspidal* if and only if

$$\int_{N(F)\backslash N(\mathbb{A}_F)} f(ng)\, dn = 0$$

for all $g \in G(\mathbb{A}_F)$ and all proper parabolic subgroups $P \subseteq G$, whose unipotent radical is denoted $N$. We let $\mathcal{A}_{\text{cusp}}(G, K)$ be the space of cusp forms of level $K$.

The benefit of working over function fields is that we can do geometry. To understand automorphic forms, we may as well start by fixing an invariant compact open subgroup. For example, for a divisor $D = \sum_x n_x[x]$,

[1] Because we are assuming that $G$ is split, it already has a model over $\mathbb{Z}$. Thus, we do not lose anything by descending $G$ to $\mathbb{F}_q$ instead of over $X$.

we may take

$$K_D := \prod_{x \in |X|} \Gamma_x(n_x),$$

where $\Gamma_x(n_x)$ is the kernel of the natural projection $G(\mathcal{O}_x) \to G(\mathcal{O}_x/\mathfrak{m}_x^{n_x})$. Thus, we are now interested in studying functions on the space $G(F)\backslash G(\mathbb{A}_F)/K_D$.

On the homework, we have already seen a version of this interpretation for number fields.

**Example 4.58.** For $\mathrm{GL}_n$ over $\mathbb{Q}$, one can interpret $\mathrm{GL}_n(\mathbb{Q})\backslash \mathrm{GL}_n(\mathbb{A}_F)/(\mathrm{O}_n(\mathbb{Q}) \times \mathrm{GL}_n(\widehat{\mathbb{Z}}))$ as a pair $(\Lambda, q)$ of a projective $\mathbb{Z}$-module $\Lambda$ of rank $n$ equipped with a positive-definite quadratic form $q$ of $\Lambda_{\mathbb{R}}$.

One can do the same thing over function fields.

**Exercise 4.59.** Fix a function field $F = \mathbb{F}_q(X)$. Then $\mathrm{GL}_n(F)\backslash \mathrm{GL}_n(\mathbb{A}_F)/K_0$ is equivalent (as a groupoid) to the category of vector bundles of rank $n$ on $X$.

*Proof.* It is enough to show $\mathrm{GL}_n(\mathbb{A}_F)/K_0$ is equivalent to the category of pairs $(\mathcal{E}, \tau)$ where $\mathcal{E}$ is a vector bundle of rank $n$ on $X$, and $\tau$ is a trivialization $\mathcal{E}|_{\operatorname{Spec} F} \cong F^n$ (at the generic fiber). (Here, $\mathrm{GL}_n(F)$ acts on the target category by acting on the target $F^n$ of $\tau$.) To define our map, we note that $\mathrm{GL}_n(\mathbb{A}_F)/K_0$ is

$$\bigoplus_x \mathrm{GL}_n(F_x)/\mathrm{GL}_n(\mathcal{O}_x),$$

and we know that $\mathrm{GL}_n(F_x)/\mathrm{GL}_n(\mathcal{O}_x)$ classifies $\mathcal{O}_x$-lattices embedded in $F_x^n$. Thus, we may define our map by sending $(\mathcal{E}, \tau)$ in the target to the lattices $\tau(\mathcal{E}_{\mathcal{O}_x}) \subseteq F_x^n$. Here are our checks on this map.

- A determinant calculation shows that $\tau$ spreads out as an isomorphism to $\mathcal{O}_x^n$ to all but finitely many places (namely, the determinant only needs finitely many denominators). Thus, $\tau(\mathcal{E}_{\mathcal{O}_x}) \subseteq F_x^n$ is the standard lattice $\mathcal{O}_x^n$ for all but finitely many $x$.

- We will only sketch the remaining checks. To go backward, one needs to be able to construct a vector bundle from the data of our local lattices. The point is to start from ■

**Remark 4.60.** The same argument can show that $\mathrm{GL}_n(F)\backslash \mathrm{GL}_n(\mathbb{A}_F)/K_D$ is equivalent (as a groupoid) to the category of vector bundles $\mathcal{E}$ of rank $n$ along with a level structure isomorphism $\tau\colon \mathcal{E}|_{\operatorname{Spec} \mathcal{O}_D} \cong \mathcal{O}_D^n$.

**Remark 4.61.** One can give similar interpretations for other classical groups. For example, for $G = \mathrm{Sp}_{2n}$, one replaces vector bundles with "symplectic vector bundles," which are vector bundles equipped with a symplectic pairing.

The moral is that automorphic forms can be interpreted as functions on a space of "$G$-bundles."

While we're here, let's also give a geometric interpretation of cusp forms of level $K_0$. Thus, we would like a geometric interpretation of the integral

$$\int_{N(F)\backslash N(\mathbb{A}_F)} f(ng)\, dn,$$

where $N$ is the unipotent radical of some proper parabolic subgroup $P$.

**Example 4.62.** Take $G = \mathrm{GL}_2$. Then $B(F)\backslash B(\mathbb{A}_F)/(K_0 \cap B(\mathbb{A}_F))$ embeds into $G(F)\backslash G(\mathbb{A}_F)/K_0$, and one can show that the former classifies vector bundles $\mathcal{E}$ of rank $2$ along with a vector bundle $\mathcal{L} \subseteq \mathcal{E}$ of rank $1$ for which $\mathcal{E}/\mathcal{L}$ is also a vector bundle. Now, let $B = TN$ be the Levi decomposition, and we see that $N(F)\backslash N(\mathbb{A}_F)/(K_0 \cap N(\mathbb{A}_F))$ can be seen as the fiber of the map

$$B(F)\backslash B(\mathbb{A}_F)/(K_0 \cap N(\mathbb{A}_F)) \to T(F)\backslash T(\mathbb{A}_F)/(K_0 \cap T(\mathbb{A}_F)).$$

It thus turns out that $N(F)\backslash N(\mathbb{A}_F)/(K_0 \cap N(\mathbb{A}_F))$ parameterizes pairs short exact sequences $0 \to \mathcal{O}_X \to \mathcal{E} \to \mathcal{O}_X \to 0$. Indeed, we are asking for the pair $(\mathcal{E}, \mathcal{L})$ from $\mathcal{E}$ to have a given trivialization of both $\mathcal{L}$ and $\mathcal{E}/\mathcal{L}$.

**Example 4.63.** We continue with $G = \mathrm{GL}_2$. Then, we see that $\int_{N(F)\backslash N(\mathbb{A}_F)} f(n)\, dn$ is a suitable average of $f(\mathcal{E})$, where $\mathcal{E}$ varies over short exact sequences

$$0 \to \mathcal{O}_X \to \mathcal{E} \to \mathcal{O}_X \to 0.$$

Notably, a single $\mathcal{E}$ (up to isomorphism of vector bundles) may appear multiple times in the sum because it may be in different isomorphism classes of short exact sequence. In general, for $\int_{N(F)\backslash N(\mathbb{A}_F)} f(ng)$, we may use the Iwasawa decomposition $G(\mathbb{A}_F) = N(\mathbb{A}_F)T(\mathbb{A}_F)K_0$ to force $g$ to be in $T(\mathbb{A}_F)$. In fact, we even only care about $g$ as a class in $T(F)\backslash T(\mathbb{A}_F)/(K_0 \cap T(\mathbb{A}_F)) = \mathrm{Pic}(X)^2$, which means that $g$ is represented by a pair $(\mathcal{L}_1, \mathcal{L}_2)$ of line bundles. It follows that $f$ is a cusp form if and only if we always have

$$\sum_{\mathcal{E} \in \mathrm{Ext}^1(\mathcal{L}_1, \mathcal{L}_2)} f(\mathcal{E}) = 0.$$

**Remark 4.64.** For $G = \mathrm{GL}_n$, a similar argument shows that $f$ is a cusp form if and only if we have

$$\sum_{\mathcal{E} \in \mathrm{Ext}^1(\mathcal{E}_1, \mathcal{E}_2)} f(\mathcal{E}) = 0$$

for all pairs $(\mathcal{E}_1, \mathcal{E}_2)$ of vector bundles of rank summing to $n$.

Next time, we will sketch the following theorem for $\mathrm{GL}_n$.

**Theorem 4.65.** Fix a split semisimple group $G$ over a function field $F$, and let $K \subseteq G(\mathbb{A}_F)$ be a compact open subgroup. Then $C_{\mathrm{cusp}}(G(F)\backslash G(\mathbb{A}_F)/K)$ is finite-dimensional.

This is some version of the admissibility for this cuspidal subspace.

## 4.4 April 13

Welcome back. We continue discussing automorphic forms for function fields.

### 4.4.1 The Central Character

We did not bother showing that the space of cusp forms was finite-dimensional over number fields. Today, we will content ourselves with sketching the result over function fields.

**Theorem 4.66.** Fix a semisimple group $G$ over a function field $F$, and let $K \subseteq G(\mathbb{A}_F)$ be a compact open subgroup.

(a) The space $\mathcal{A}_{\text{cusp}}(G, K)$ is finite-dimensional.

(b) There is a finite subset $U_K \subseteq G(F)\backslash G(\mathbb{A}_F)/K$ such that all cups forms in $\mathcal{A}_{\text{cusp}}(K)$ vanish outside $U_K$.

**Remark 4.67.** It follows that any cusp form for $G$ has support with compact image in $G(F)\backslash G(\mathbb{A}_F)$.

**Remark 4.68.** Over number fields, the analog of this result is that cusp forms have rapid decay at the cusps; this strengthens the "moderate growth" at the cusps condition required for general automorphic forms. For example, for $\mathrm{GL}_{2,\mathbb{Q}}$, this implies that the cuspidal holomorphic modular forms $f(x+iy)$ are $o\left(y^{-N}\right)$ for all $N$. We can see this directly: using the $q$-expansion, one actually sees that $f(x+iy)$ is $O(\exp(-y))$.

We will sketch this result only for $G = \mathrm{GL}_n$ and $K = K_0$. For this, we should figure out what the correct statement is when $G$ is not semisimple. The right thing to do is to fix the central character, which roughly speaking kills the torus.

**Definition 4.69** (central character)**.** Fix a reductive group $G$ over a function field $F$ with center $Z$. Given a continuous character $\omega\colon Z(F)\backslash Z(\mathbb{A}_F) \to \mathbb{C}^\times$, we see that $f \in \mathcal{A}(G)$ has *central character* $\omega$ if and only if $f$ is a $Z(\mathbb{A}_F)$-eigenvector with eigenvalue $\omega$. We let $\mathcal{A}(\omega)$ denote the subspace of automorphic forms with central character $\omega$.

**Remark 4.70.** Morally, fixing the central character passes the automorphic form from $G(F)\backslash G(\mathbb{A}_F)$ to the quotient $G(F)\backslash G(\mathbb{A}_F)/Z(\mathbb{A}_F)$. The former has no chance of being finite volume when $Z$ has positive dimension because the norm map surjects onto $q^{\mathbb{Z}}$, but it turns out that the latter has finite volume.

**Remark 4.71.** Because $Z$ is central, we see that

$$\mathcal{A}(G) = \bigsqcup_{\omega} \mathcal{A}(\omega).$$

**Example 4.72.** Take $G = \mathrm{GL}_n(F)$ with level $K_0 = \prod_{x \in |X|} \mathrm{GL}_n(\mathcal{O}_x)$. Using Exercise 4.59, we see that an automorphic form $f$ with central character $\omega$ is required to have

$$f(\mathcal{E} \otimes \mathcal{L}) = \omega(\mathcal{L}) f(\mathcal{E}).$$

Thus, by letting $\mathcal{L}$ vary over various degrees, we see that $f$ is determined by its values on vector bundles of degrees in $\{0, 1, \ldots, n-1\}$.

The analog of Theorem 4.66 which we will show is the following.

**Notation 4.73.** Fix a curve $X$. Then we let $\mathrm{Vect}_n^d(X)$ be the moduli stack of vector bundles of rank $n$ and degree $d$.

**Remark 4.74.** There is an equivalence

$$\mathrm{Vect}_n(X) = \bigsqcup_{d\in\mathbb{Z}} \mathrm{Vect}_n(X).$$

**Proposition 4.75.** Fix a function field $F = \mathbb{F}_q(X)$, and set $K_0 := \prod_{x\in|X|} \mathrm{GL}_n(\mathcal{O}_x)$. Then there is a finite subset

$$U \subseteq \bigsqcup_{i=0}^{n-1} \mathrm{Vect}_n^i(X)$$

such that $f$ vanishes outside $\mathrm{Pic}(X)\cdot U \subseteq \mathrm{Vect}_n(X)$.

### 4.4.2 Cusp Forms for Function Fields

Let's start by explaining what it means for an automorphic form to be cuspidal.

**Lemma 4.76.** Fix a function field $F = \mathbb{F}_q(X)$, and set $K_0 := \prod_{x\in|X|} \mathrm{GL}_n(\mathcal{O}_x)$. Then an automorphic form $f$ is cuspidal if and only if

$$\sum_{\mathcal{E}\in\mathrm{Ext}^1(\mathcal{E}'',\mathcal{E}')} f(\mathcal{E}) = 0$$

for all vector bundles $\mathcal{E}'$ and $\mathcal{E}''$ of ranks $m$ and $n-m$, respectively.

*Proof.* It is enough to check cuspidality on maximal parabolic subgroups $P \subseteq \mathrm{GL}_n(F)$. Once $P$ contains the standard Borel of upper-triangular matrices, $P$ looks like block-diagonal matrices $\left[\begin{smallmatrix} A & B \\ 0 & D\end{smallmatrix}\right]$, where $A$ is $m\times m$ and $D$ is $(n-m)\times(n-m)$. Now, let $P = MN$ be the Levi decomposition, and we are interested in checking when

$$\mathrm{CT}_P(g) := \int_{N(F)\backslash N(\mathbb{A}_F)} f(ng)\,dn$$

vanishes for all $g \in G(\mathbb{A}_F)$. Well, the Iwasawa decomposition at each place turns out to induce an Iwasawa decomposition

$$G(\mathbb{A}_F) = N(\mathbb{A}_F)M(\mathbb{A}_F)K_0,$$

so the function $\mathrm{CT}_P$ descends to the quotient $M(F)\backslash M(\mathbb{A}_F)/K_{M,0}$, where $K_{M,0} := K_0 \cap M(\mathbb{A}_F)$. (Here, $M(F)$ comes out on the left because $M$ normalizes $N$.) To compute $\mathrm{CT}_P$, we note that $M = \mathrm{GL}_m \times \mathrm{GL}_{n-m}$, so the source $M(F)\backslash M(\mathbb{A}_F)/K_{M,0}$ simply parameterizes pairs $(\mathcal{E}', \mathcal{E}'')$ of vector bundles of ranks $m$ and $n-m$, respectively. Then we claim that

$$\mathrm{CT}_P(\mathcal{E}', \mathcal{E}'') = \sum_{\mathcal{E}\in\mathrm{Ext}^1(\mathcal{E}'',\mathcal{E}')} f(\mathcal{E}) \cdot \frac{1}{\#\,\mathrm{Aut}(\mathcal{E})},$$

where $f(\mathcal{E})$ refers to the value of $f$ on the underlying vector bundle of $\mathcal{E}$. Indeed, for $g \in M(\mathbb{A}_F)$, we see that the choice of $n \in N(\mathbb{A}_F)$ grants us some filling in for a short exact sequence

$$0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0,$$

and then the integral simply sums the value of $f(\mathcal{E})$ as the short exact sequence varies.

Lastly, we have to remove the automorphism group from our sum. This follows because $\mathrm{Aut}(\mathcal{E})$ actually has constant size: an automorphism of the short exact sequence is an automorphism $\alpha$ of the underlying vector bundle $\mathcal{E}$ which restricts to the identity on the sub-bundle $\mathcal{E}'$ and the quotient $\mathcal{E}''$. But then $\alpha - \mathrm{id}_{\mathcal{E}}$ gives rise to a projection $\mathcal{E}'' \to \mathcal{E}'$, which splits and thus trivializes the sequence, so $\alpha - \mathrm{id}_{\mathcal{E}}$ is then parameterized by $\mathrm{Hom}(\mathcal{E}'', \mathcal{E}')$. ■

### 4.4.3 Reduction Theory for Function Fields

Next, we need some reduction theory. Over number fields, reduction theory for $\mathrm{GL}_n$ arises from choosing some basis of shortest vectors of the associated metrized lattices. The analog of a vector in a vector bundle $\mathcal{E}$ is a line sub-bundle $\mathcal{L} \subseteq \mathcal{E}$, and the analog of being shortest is having large degree.[2] This motivates the following discussion.

**Definition 4.77** (slope)**.** Fix a vector bundle $\mathcal{E}$ on a curve $X$. Then the *slope* is

$$\mu(\mathcal{E}) := \frac{\deg \mathcal{E}}{\operatorname{rank} \mathcal{E}}.$$

**Definition 4.78** (semistable)**.** Fix a vector bundle $\mathcal{E}$ on a curve $X$. Then $\mathcal{E}$ is *semistable* if and only if all nonzero sub-bundles $\mathcal{F} \subseteq \mathcal{E}$ have $\mu(F) \leq \mu(E)$.

**Example 4.79.** Suppose that $\mathcal{E}$ is a direct sum of two line bundles $\mathcal{L}_1$ and $\mathcal{L}_2$. If $\deg \mathcal{L}_1 > \deg \mathcal{L}_2$, then $\mathcal{E}$ is not semistable (which we can see by inspection). However, if the degrees are equal, then $\mathcal{E}$ is semistable. Indeed, for any $\mathcal{L} \subseteq \mathcal{L}_1 \oplus \mathcal{L}_2$, one can show that

$$\deg \mathcal{L} \leq \max\{\deg \mathcal{L}_1, \deg \mathcal{L}_2\}.$$

**Theorem 4.80** (Harder–Narasimhan)**.** Fix a vector bundle $\mathcal{E}$ on a curve $X$. Then $\mathcal{E}$ admits a unique filtration

$$0 = \mathcal{F}_0 \subsetneq \mathcal{F}_1 \subsetneq \mathcal{F}_2 \subsetneq \cdots \subsetneq \mathcal{F}_r = \mathcal{E}$$

for which $\mathcal{F}_i/\mathcal{F}_{i-1}$ is semistable for each $i$, and $\{\mu(\mathcal{F}_i/\mathcal{F}_{i-1})\}_i$ is decreasing.

*Sketch.* The main point is to choose $\mathcal{F}_1$, from which the rest of the filtration can be found inductively by passing to the quotient. To this end, we simply choose $\mathcal{F}_1$ to have maximal slope, and among those sub-bundles with maximal slope, we choose $\mathcal{F}_1$ to have maximal rank. It is nontrivial to verify uniqueness. ■

This filtration allows us to define an interesting invariant.

**Definition 4.81** (polygon)**.** Fix a vector bundle $\mathcal{E}$ on a curve $X$ with filtration. Then its *polygon* $\nu$ is the line connecting $(0,0)$ to the points $(\operatorname{rank} \mathcal{F}_i, \deg \mathcal{F}_i)$ in order. We let $\mathrm{Vect}_n^\nu(X)$ denote the collection of vector bundles of rank $n$ and polygon $\nu$.

**Remark 4.82.** The construction of the filtration implies that the polygon is concave down.

**Remark 4.83.** It turns out that each $\mathrm{Vect}_n^\nu(X)$ is finite. Roughly speaking, this follows because there are only finitely many semistable bundles of given rank.

Here is our argument.

*Proof of Proposition 4.75.* It suffices to show that there is a polygon $\nu_0$ with endpoints $(n,d)$ such that all cusp forms on $\mathrm{Vect}_n^d(X)$ must vanish on $\mathrm{Vect}_n^\nu(X)$ when $\nu \not\leq \nu_0$. Namely, we recover the result by letting $d$ vary over our finitely many degrees, and the finite set is simply given by those vector bundles with polygon bounded above by $\nu_0$.

[2] For example, one can expect to find line bundles of smaller degree by appropriately twisting, so it is only ever surprising to have sub-bundles of large degree.

For simplicity, we now take $G = \mathrm{GL}_2$ and $d = 0$ so that the choice of polygon $\nu_0$ amounts to a single choice of point $(1, e_0)$; the rest of the polygon comes from the points $(0,0)$ and $(2,0)$. Now, via the filtration, we see that $\mathrm{Vect}_2^{(1,e)}(X)$ is parameterized by short exact sequences

$$0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0$$

where $\mathcal{E}'$ has degree $e$, $\mathcal{E}$ has degree zero, and $\mathcal{E}''$ has degree $-e$. Now, for our cuspidality check, we should fix line bundles $\mathcal{E}'$ and $\mathcal{E}''$ of degrees $e$ and $-e$, and we are given that

$$\sum_{\mathcal{E} \in \mathrm{Ext}^1(\mathcal{E}'', \mathcal{E}')} f(\mathcal{E}) = 0.$$

Now, for $e$ sufficiently large, we note that the short exact sequence splits: indeed, $\mathrm{Ext}^1(\mathcal{E}'', \mathcal{E}')$ is some cohomology group in $\mathcal{E}'' \otimes (\mathcal{E}')^\vee$, so high degrees induces vanishing by Serre vanishing. Thus, this cuspidality condition directly implies that $f(\mathcal{E}) = 0$ for $e$ large enough. This completes the argument in this case. ■

**Remark 4.84.** For $\mathrm{GL}_n$ for higher $n$, one needs a more complicated induction.

### 4.4.4 The Cuspidal Subspace

Over number fields, the cuspidal subspace decomposes as follows.

**Proposition 4.85.** Fix a semisimple group $G$ over a number field $F$. Let $L^2_{\text{cusp}} \subseteq L^2(G(F)\backslash G(\mathbb{A}_F))$ be the closure of the subspace of cusp forms. Then $L^2_{\text{cusp}}$ is a Hilbert space direct sum of irreducible representations of $G(\mathbb{A}_F)$, and they occur with finite multiplicity.

*Proof.* We will show this in the special case where $G(F)\backslash G(\mathbb{A}_F)$ is compact. In this case, it turns out that all automorphic forms are automatically cusp forms; roughly speaking, this holds because there are no proper parabolic subgroups because $G$ is anisotropic. (Parabolic subgroups can be defined as "attracting subsets" of one-parameter subgroups $\mathbb{G}_m \to G$.)

We start by choosing an approximation to identity $\{f_n\}$ in $C_c(G(\mathbb{A}_F), \mathbb{R})$. By replacing $f_n$ with $g \mapsto \frac{1}{2}\left(f_n(g) + f_n(g^{-1})\right)$, we may assume that $f_n(g) = f_n\left(g^{-1}\right)$ for all $g$. Now, note that the induced action $R(f_n)$ on $C_c^\infty(G(\mathbb{A}_F), \mathbb{R})$ is (by definition) given by a kernel. Thus, they are Hilbert–Schmidt operators and thus compact, and they are self-adjoint by the symmetry property, so we use the spectral theorem of compact operators. Namely, each $R(f_n)$ diagonalizes into a spectrum $\{\lambda_n\}_n$ as

$$L^2(G(F)\backslash G(\mathbb{A}_F))[0] \oplus \bigoplus_n L^2(G(F)\backslash G(\mathbb{A}_F))[\lambda_n],$$

and the sequence $\{\lambda_n\}_n$ (with multiplicities) is discrete, except possibly at zero.

Now, every vector in $L^2(G(F)\backslash G(\mathbb{A}_F))$ has some nontrivial action by some $R(f_n)$ because $\delta_1$ acts nontrivially (indeed, it fixes every vector). Thus, we decompose into nonzero eigenspaces, and it only remains to show that our spaces occur with finite multiplicity. Well, looping over all $f_n$s, one can show a given invariant subspace occurs with finite multiplicity, and the decomposition follows because the ma ■

**Remark 4.86.** This is the same argument as the proof of the Peter–Weyl theorem.

## 4.5 April 15

Here we go.

### 4.5.1 Some Hecke Algebras

We continue in the setting of a reductive group $G$ over a global field $F$. Then $G(\mathbb{A}_F^\infty)$ acts on the space $\mathcal{A}$ of automorphic forms. Over the finite places $v$, we have already explained that this gives rise to the action of a local Hecke algebra $\mathcal{H}_v := C_c^\infty(G(F_v))\,dg$. Let's assemble these Hecke algebras.

**Definition 4.87** (restricted tensor product)**.** Let $\{A_i\}$ be a collection of algebras, and select an element $e_i \in A_i$ which is idempotent for all but finitely many $i$. Then we define the *restricted tensor product*

$$\bigotimes_i (A_i, e_i)$$

to consist of finite linear combinations of pure tensors $\bigotimes_i a_i$, where $a_i \in A_i$ for each $i$, and $a_i = e_i$ for all but finitely many $i$.

One can check that this restricted tensor product forms an algebra, but we will not bother.

**Definition 4.88.** Fix a reductive group $G$ over a global field $F$. For each finite place $v$, let $e_v$ be the idempotent corresponding to $1_{G(\mathcal{O}_v)}$ in $\mathcal{H}_v$. We then define the *nonarchimedean Hecke algebra* as

$$\mathcal{H}_f := \bigotimes_v (\mathcal{H}_v, e_v),$$

where the tensor product is taken over $e_v$s.

**Remark 4.89.** It turns out that $\mathcal{H}_f$ is simply $C_c^\infty(G(\mathbb{A}_F^\infty))\,dg$. In one direction, one sends the pure tensor $\bigotimes_v f_v\,dg_v$ to the measure $(\bigotimes_v f_v)\;dg$. (Here, the local Haar measures $dg_v$ are chosen so that the volume of $\prod_v G(\mathcal{O}_v)$ is unambiguous.)

**Remark 4.90.** As before, we find that smooth $G(\mathbb{A}_F^\infty)$-representations are equivalent to smooth $\mathcal{H}_f$-representations.

We would now like to add on the archimedean places. For simplicity, we work with $\mathbb{Q}$. Here is our Hecke algebra.

**Definition 4.91.** Fix a reductive (real) Lie group $G$ with Lie algebra $\mathfrak{g}$ and maximal compact subgroup $K \subseteq G(\mathbb{R})$. Then we define the *Hecke algebra*

$$\mathcal{H}_{\mathfrak{g},K} := \mathrm{Alg}(K) \otimes_{U\mathfrak{k}_\mathbb{C}} U\mathfrak{g}_\mathbb{C}.$$

Here, $\mathrm{Alg}(K)$ is the subalgebra of $C(K)$ consisting of all matrix coefficients of all finite-dimensional representations of $K$. It is a right module for $\mathfrak{k}$, and the algebra embedding $U\mathfrak{k}_\mathbb{C} \subseteq U\mathfrak{g}_\mathbb{C}$ makes $U\mathfrak{g}_\mathbb{C}$ into a left module.

**Remark 4.92.** It turns out that $(\mathfrak{g}, K)$-modules embed fully faithfully into smooth $\mathcal{H}_{g,K}$-modules. Here, smooth means that the module $M$ is the union of the modules $eM$, where $e \in \mathcal{H}_{g,K}$ is an idempotent, where $e$ the pure tensor given by an idempotent of $\mathrm{Alg}(K)$ times the unit in $U\mathfrak{g}_\mathbb{C}$. Furthermore, the embedding descends to an equivalence of admissible modules.

**Remark 4.93.** By the Peter–Weyl theorem, the natural map

$$\mathrm{Alg}(K) \subseteq L^2(K) \cong \widehat{\bigoplus_{V \in \mathrm{IrRep}(K)}} \mathrm{End}(V)$$

induces an isomorphism

$$\mathrm{Alg}(K) = \bigoplus_{V \in \mathrm{IrRep}(K)} \mathrm{End}(V).$$

Indeed, the point is that $\mathrm{Alg}(K)$ can be checked to be exactly those functions $f \in L^2(K)$ which are $K$-finite. Then one can compute what the $K$-finite vectors on the right-hand side are. Thus, each subset $\Sigma \subseteq \mathrm{IrRep}(K)$ induces an idempotent $e_\Sigma := \sum_{V \in \Sigma} \mathrm{id}_V$.

**Example 4.94.** Take $K = S^1$. Then the matrix coefficients of $K$ are the integer powers of the coordinate $z \in S^1$, so $\mathrm{Alg}(K) = \mathbb{C}[z, 1/z]$. With the convolution algebra structure, each character $z^n$ is idempotent.

**Remark 4.95.** Let's say something about the multiplication on $\mathcal{H}_{\mathfrak{g},K}$. The hard part is defining the multiplication of $X \cdot \varphi$, where $X \in \mathfrak{g}_\mathbb{C}$ and $\varphi \in \mathrm{Alg}(K)$. If we imagine replacing $\varphi$ by a distribution $\delta_k$, then we expect to have $X \cdot \delta_k = \delta_k \cdot \mathrm{Ad}_k X$. In general, we approximate $\varphi$ by linear combination of $\delta$s.

**Remark 4.96.** Here is a more conceptual definition: $\mathcal{H}_{g,K}$ is the space of distributions on $G(\mathbb{R})$ which is supported on $K$ and locally $K$-finite. Here, being supported on $K$ means that the distribution $\mu$ has $\mu(f) = 0$ whenever $f|_K = 0$. This definition also makes the multiplication structure clearer. It turns out that any such distribution is a linear combination of derivatives from $\mathfrak{g}_\mathbb{C}$ and algebraic measures, which gives the above description of $\mathcal{H}_{\mathfrak{g},K}$.

We are now ready to define our global Hecke algebra.

**Definition 4.97** (global Hecke algebra)**.** Fix a reductive group $G$ over a global field $F$. Then we define the *global Hecke algebra* as

$$\mathcal{H} := \bigotimes_{v \text{ infinite}} \mathcal{H}_{\mathfrak{g}_v, K_v} \otimes \mathcal{H}_f,$$

where $\mathfrak{g}_v = \mathrm{Lie}\, G(F_v)$ and $K_v \subseteq G(F_v)$ is a maximal compact subgroup.

By construction, we see that $\mathcal{H}$ admits a natural action on $\mathcal{A}$. Note that $\mathcal{H}$ remembers enough of our representation structure so that an automorphic representation is just an irreducible admissible subquotient of $\mathcal{A}$ (viewed as an $\mathcal{H}$-module). Indeed, the point is that we have all the correct module structures locally, which glue to the global definition.

### 4.5.2 The Tensor Product Theorem

We are now ready to state our theorem.

**Theorem 4.98.** Let $G$ be a reductive group over a global field $F$.

(a) For each $v$, choose a simple $\mathcal{H}_v$-module $M_v$, and suppose that $M_v$ admits a unique spherical line $\mathbb{C}m_v \in M_v^{G(\mathcal{O}_v)}$ for all but finitely many $v$. Then

$$M := \bigotimes_v (M_v, m_v)$$

is an irreducible $\mathcal{H}$-module.

(b) Any admissible irreducible $\mathcal{H}$-module is isomorphic to one constructed in (a).

*Proof.* For (a), let's start with admissibility. For a finite set $S$ of places, choose an idempotent $e \in \mathcal{H}$ by choosing any idempotent $e_S \in \bigotimes_{v \in S} \mathcal{H}_v$ and then selecting $e$ to be $e_v$ at each $v \notin S$. By letting $S$ expand, we see that it is enough to show that $\dim eM$ is finite for all such $e$ (because such idempotents approximate the identity). Then

$$eM = e_S M_S \otimes \bigotimes_{v \notin S} (\mathbb{C}m_v, m_v).$$

Notably, we are only getting $m_v$s on the right because this is our chosen unique spherical vector. Note that $e_S M_S$ is finite-dimensional because $S$ is finite (and the $M_v$s for $v \in S$ are suitably smooth), so the entire space is finite-dimensional.

We now turn to our classification of irreducible objects. Roughly speaking, the main point is to note that $M \otimes N$ is a simple module over $A \otimes B$ when $M$ and $N$ are both simple over $A$ and $B$, respectively, and all simple modules are of this form. We require the following result.

**Definition 4.99** (idempotent algebra)**.** An *idempotent algebra* is an algebra $A$ equipped with a filtered poset of commuting idempotents $\{e_i\}$. Here, we say that $e_i \geq e_j$ if and only if $e_i e_j = e_i$.

**Definition 4.100** (smooth)**.** A module $M$ over an idempotent algebra $A$ is *smooth* if and only if $M = \bigcup_i e_i M$. It is *admissible* if and only if further $e_i M$ is finite-dimensional for each $e_i$.

**Lemma 4.101.** Fix an idempotent algebra $A$ with idempotents $\{e_i\}$. Then a smooth module $M$ is zero or simple if and only if $e_i M$ is zero or simple of finite dimension over $e_i A e_i$ for each idempotent $e_i$.

*Proof.* Omitted. ■

For example, it follows that having two smooth simple modules $M_1$ and $M_2$ over idempotent algebras $A_1$ and $A_2$ provide a smooth simple module $M := M_1 \otimes M_2$ over the idempotent algebra $A_1 \otimes A_2$: indeed, for each idempotent $e_1 \otimes e_2$ of $A \otimes B$, we see that

$$(e_1 \otimes e_2)M = e_1 M_1 \otimes e_2 M_2.$$

We now reduce to finite-dimensional considerations to find this is zero or a simple module for $e_1 A_1 e_1 \otimes e_2 A_2 e_2$.[3]

Conversely, we claim that all simple modules $M$ of $A_1 \otimes A_2$ are a tensor product. Well, we are given that $(e_1 \otimes e_2)M$ is either zero or a finite-dimensional simple module of $e_1 A_1 e_1 \otimes e_2 A_2 e_2$ for each idempotent $e_1 \otimes e_2$. Thus, finite-dimensional considerations again allow us to decompose

$$(e_1 \otimes e_2)M = M_1(e_1) \otimes M_2(e_2)$$

[3] One can basically use the same arguments as one does for representations of finite groups.

for some modules $M_1(e_1)$ and $M_2(e_2)$ unique up to isomorphism. (When $(e_1 \otimes e_2)M$ vanishes, we select these modules to both vanish.) If we choose "larger" idempotents, meaning that $(e_1' \otimes e_2')M$ is larger, then we find that $e_1 N_1(e_1') \otimes e_2 N_2(e_2') \cong N_1(e_1) \otimes N_2(e_2)$. In the application, our idempotents will form an inductive system, so we receive isomorphisms $N_i(e_i) \cong e_i N_i(e_i')$ which are unique up to scaling, but then we can correct these scalars to make all the isomorphisms compatible.

Thus, we may take the colimit of the $N_i(e_1)$s to produce a module $N_i$ of $A_i$. One can then check that the induced map

$$N_1 \otimes N_2 \to M$$

is an isomorphism by smoothness, and one can use the lemma to show that $N_1$ and $N_2$ are irreducible by passing to finite-dimensional considerations.

We now return to $\mathcal{H}$, which is an infinite restricted tensor product. The hard part is (b), so let's focus there. Choose an irreducible admissible module $M$ of $\mathcal{H}$. For each idempotent $e$, we see $eM$ is an irreducible finite-dimensional $e\mathcal{H}e$-module; we may as well take $e$ of the form $\bigotimes_v e_v'$, where $e_v' = e_v$ for all but finitely many places $v$. We collect these bad places $v$ in a finite set $S$. Now,

$$eM = M_S \otimes \chi,$$

where $M_S$ is some module for $e_S \mathcal{H}_S e_S$, and $\chi$ is some one-dimensional module for the commutative unital algebra $\bigotimes_{v \notin S}' e_v \mathcal{H}_v e_v$. One recovers the spherical vectors $m_v$ by looking at the embedding $\chi \hookrightarrow eM$, and one can use the argument above to decompose $M_S$ into a tensor product decomposition. The argument now follows the proof of the toy case with two algebras to take a colimit of the tensor product decompositions as $S$ gets large. ■

Next class, we will say something about $L$-functions.

**Example 4.102.** Fix an automorphic representation $\pi$ of $\mathrm{GL}_2$ over a global field $F$. Then we can define the $L$-function of $\pi$ as

$$L(s, \pi) := \int_{F^\times \backslash \mathbb{A}_F^\times} \varphi\left(\begin{bmatrix} a & \\ & 1 \end{bmatrix}\right) |a|^{s-1/2} \, d^\times a,$$

where $\varphi$ is some chosen element of $\pi$. This generalizes what one expects from holomorphic modular forms. Additionally, this has the desirable property that it unfolds into an Euler product $\prod_v L_v(s, \pi_v)$, where the factors $L(s, \pi_v)$ depends only on the isomorphism class of $\pi_v$. The Euler product factorization depends on a choice of Whittaker functional.

## 4.6 April 22

Today, we discuss Hecke operators.

### 4.6.1 Hecke Operators

Hecke operators come abstractly from our function space.

**Notation 4.103.** Fix a reductive group $G$ over a global field $F$. For a level $K \subseteq G(\mathbb{A}_F^\infty)$, we define $\mathcal{A}_G(K)$ to be the $K$-invariant automorphic forms. We also let $\mathcal{H}_{f,K}$ be the measures on $K \backslash G(\mathbb{A}_F^\infty)/K$.

**Remark 4.104.** Note that $\mathcal{A}_G(K)$ embeds into $C^\infty(G(F) \backslash G(\mathbb{A}_F)/K)$.

**Remark 4.105.** We may identify $\mathcal{H}_{f,K}$ with $C_c(K \backslash G(\mathbb{A}_F^\infty)/K)$ by choosing $dg$ to have $\mathrm{vol}(K; dg) = 1$.

For our Hecke algebras, note that whenever $G(\mathbb{A}_F^\infty)$ acts on the right of a variety $X$, we receive a convolution left action of $\mathcal{H}_f$ on $C^\infty(X)$ by

$$(\mu * f)(x) := \int_{G(\mathbb{A}_F)} \mu(xg)h(g).$$

For example, if we extend from $\mathcal{H}_f$ to all measures so that we can plug in $h = \delta_g$, then $\delta_g * f$ is the right translation action.

**Remark 4.106.** It is possible to write down some explicit formulae when we have fixed a level $K$: then the action of $h\,dg \in \mathcal{H}_{f,K}$ on $f \in \mathcal{A}_G(K)$ is given by

$$(h * f)(x) = \sum_{g \in G(\mathbb{A}_F^\infty)/K} f(xg)h(g).$$

It is not hard to check that $h * f$ continues to be in $\mathcal{A}_G(K)$.

**Example 4.107.** One convenience of Hecke operators is that they admit geometric description: let's describe the action of $1_{KgK} \in \mathcal{H}_{f,K}$. Namely, if $G(\mathbb{A}_F^\infty)$ acts on a variety $X$, note that there is a correspondence as follows.

$$\begin{tikzcd} X/\left(K \cap gKg^{-1}\right) \arrow[r, "R_g"] \arrow[d] & X/\left(g^{-1}Kg \cap K\right) \arrow[d] \\ X/K & X/K \end{tikzcd}$$

It turns out that the Hecke operator is now computed by taking $f$ on $X/K$, pulling back along the projection, pulling back along $R_g$, and then pushing forward along the projection. This last projection is simply summing over the entire fiber, so we can see that we have given the correct operation.

While we're here, we interpret our Hecke operators in the function field setting, so let $X$ be a smooth projective integral curve over $\mathbb{F}_q$ with function field $F$, and we use the group $\mathrm{GL}_2$. Then our automorphic forms are simply some special functions $f$ on the groupoid of vector bundles of rank $2$.

**Notation 4.108.** Fix a smooth projective integral curve $X$ over $\mathbb{F}_q$ with function field $F$. For an effective divisor $D \subseteq X$, we define the Hecke operator $T_D$, defined as the indicator of the set

$$\left\{ g \in M_2(\widehat{\mathcal{O}}_F) : (\det g) = D \right\},$$

**Remark 4.109.** The element $\{T_D\}$ do not form a basis: we are missing the cosets $a\,\mathrm{GL}_2(\widehat{\mathcal{O}})$ as $a \in \mathbb{A}_F^\times$ varies.

**Notation 4.110.** For a divisor $D$ on $X$, we let $\delta_D$ be the indicator of $a\,\mathrm{GL}_2(\widehat{\mathcal{O}})$, where $a \in \mathbb{A}_F^\times$ represents the divisor $D$.

**Remark 4.111.** One can compute that $(\delta_D * f)(\mathcal{E}) = f(\mathcal{E}(D))$. Similarly, using the correspondence diagram, one finds

$$(T_D * f)(\mathcal{E}) = \sum_{\substack{\mathcal{E} \subseteq \mathcal{E}' \\ \mathrm{div}(\mathcal{E}'/\mathcal{E}) = D}} f(\mathcal{E}').$$

### 4.6.2 Hecke Operators on Modular Forms

For some intuition, we return to modular forms.

**Remark 4.112.** In weight $k = 2$, one can find the cusp forms $S_2(\Gamma)$ of level $\Gamma$ inside the cohomology $\mathrm{H}^1(\overline{\mathcal{H}/\Gamma}; \mathbb{C})$ via the Hodge decomposition. Namely,

$$\mathrm{H}^1(\overline{\mathcal{H}/\Gamma}; \mathbb{C}) = S_2(\Gamma) \oplus \overline{S_2(\Gamma)}.$$

**Notation 4.113.** For $\tau \in \mathcal{H}$, we let $E_\tau$ be the elliptic curve $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$.

**Lemma 4.114.** The double quotient $\mathrm{GL}_2(\mathbb{Z}) \backslash \mathrm{GL}_2(\mathbb{R}) / \mathbb{C}^\times$ is in isomorphic with the double quotient of elliptic curves.

*Proof.* We have two steps.

1. We claim that $\mathrm{GL}_2(\mathbb{Z}) \backslash \mathrm{GL}_2(\mathbb{R}) / \mathbb{C}^\times$ is in bijection with pairs $(\Lambda, J)$, where $\Lambda$ is a lattice of rank two, and $J$ is a complex structure on $\Lambda_\mathbb{R}$. Indeed, $\mathrm{GL}_2(\mathbb{R})$ simply classifies $2 \times 2$ matrices, which has a natural complex structure by identifying $\mathbb{R}^2$ with $\mathbb{C}$. Taking a quotient by $\mathbb{C}^\times$ then produces lattices with a basis and some choice of complex structure $J$, and taking a further quotient by $\mathrm{GL}_2(\mathbb{Z})$ produces lattices with some choice of complex structure.

2. We claim that pairs $(\Lambda, J)$ are in bijection with elliptic curve. Indeed, one can take $(\Lambda, J)$ to the complex manifold $\Lambda_\mathbb{R}$ and then take quotient by the lattice $\Lambda$ to get an elliptic curve. Conversely, one can take an elliptic curve $E$ to the lattice $\mathrm{H}_1(E; \mathbb{Z})$ equipped with the natural complex structure on $\mathrm{H}_1(E; \mathbb{R})$ canonically identified with $\operatorname{Lie} E(\mathbb{C})$. ■

**Remark 4.115.** Note that $\mathrm{GL}_2(\mathbb{R})/\mathbb{C}^\times$ has two components (corresponding to positive and negative determinant) which are then swapped by $\mathrm{GL}_2(\mathbb{Z})$. Thus, we could alternatively have worked with the double quotient $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathrm{SL}_2(\mathbb{R}) / \mathrm{SO}_2(\mathbb{R})$ instead, which makes it clearer where the upper-half plane enters the picture.

**Remark 4.116.** It may be useful to notice that one can also interpret the quotient $\mathrm{GL}_2(\mathbb{Z}) \backslash \mathrm{GL}_2(\mathbb{R})$ in terms of elliptic curves. As above, if we undo the quotient by $\mathbb{C}^\times$, this amounts to the data of a triple $(\Lambda, J, v)$ where $J$ is the complex structure on $\Lambda_\mathbb{R}$, and $v \in \Lambda_\mathbb{R}$ is some nonzero vector. The same argument now finds that such triples parameterize pairs $(E, v)$ of an elliptic curve $E$ together with a nonzero vector $v \in \operatorname{Lie} E(\mathbb{C})$.

The moral is that a modular form of level $k$ is now seen to be a function $f$ on pairs $(E, v)$ satisfying the functional equation

$$f(E, v) = c^k f(E, v)$$

for any $c \in \mathbb{C}^\times$ (as well as some analytic conditions, such as being holomorphic and having moderate growth).

We are now ready to say something about Hecke operators. Let's attempt to write down a basis.

**Notation 4.117.** For a positive integer $n$, we let $T_n$ be the indicator of the set

$$\left\{ g \in M_2(\widehat{\mathbb{Z}}) : (\det g) = n\widehat{\mathbb{Z}} \right\}.$$

**Example 4.118.** For prime $p$, we find that

$$\left\{g \in M_2(\widehat{\mathbb{Z}}) : (\det g) = n\widehat{\mathbb{Z}}\right\} = \prod_{\ell \neq p} \mathrm{GL}_2(\mathbb{Z}_\ell) \times \mathrm{GL}_2(\mathbb{Z}_p) \begin{bmatrix} p & \\ & 1 \end{bmatrix} \mathrm{GL}_2(\mathbb{Z}_p).$$

Indeed, we are asking for trivial determinant away from $p$, and at $p$, we are asking for determinant $p$. The given description of the matrices with determinant $p$ in $\mathrm{GL}_2(\mathbb{Q}_p)$ follows from the Cartan decomposition.

**Example 4.119.** For prime $p$, we similarly have

$$T_{p^e} = \sum_{\substack{a+b=e \\ a \leq b}} 1_{\mathrm{GL}_2(\mathbb{Z}_p)\,\mathrm{diag}(p^a, p^b)\,\mathrm{GL}_2(\mathbb{Z}_p)}.$$

Taking the product allows us to compute $T_n$ for any positive integer $n$.

**Remark 4.120.** The elements $\{T_n\}$ are not a basis for the Hecke algebra: one needs to add in the commutative subalgebra of indicators of the double cosets $\mathrm{GL}_2(\widehat{\mathbb{Z}})\,\mathrm{diag}(m,m)\,\mathrm{GL}_2(\widehat{\mathbb{Z}})$, where $m \in \mathbb{Q}^+$. However, these extra operators are not so interesting: if $f$ is a modular form of weight $k$, then

$$1_{m\,\mathrm{GL}_2(\widehat{\mathbb{Z}})} * f(E, v) = f(E, mv) = m^k f(E, v).$$

Anyway, we now see that $\mathcal{H}_{f,K}$ is some commutative ring acting on our space $M_k(\mathrm{GL}_2(\mathbb{Z}))$ of modular forms of weight $k$. It turns out that $M_k(\mathrm{GL}_2(\mathbb{Z}))$ can be given a Hermitian structure, via the Peterson inner product

$$\langle f_1, f_2 \rangle := \int_{\mathcal{H}} f_1 \overline{f_2} y^k \, \frac{dx\,dy}{y^2},$$

and one finds that the Hecke operators are self-adjoint.

**Remark 4.121.** One can compute the adjoint of the Hecke action by $1_{KgK}$ on $M_k(\mathrm{GL}_2(\mathbb{Z}))$ by using the correspondence diagram in Example 4.107: the adjoint is found by going backward along the correspondence diagram, so it is basically $1_{Kg^{-1}\det(g)K}$, but one can show that $KgK = Kg^{-1}\det(g)K$ for $\mathrm{GL}_2$.

Anyway, the point is that the action by the Hecke algebra must diagonalize.

**Theorem 4.122** (Multiplicity one)**.** The action of the Hecke algebra on the space of cusp forms diagonalizes into one-dimensional eigenspaces.

**Definition 4.123** (eigenform)**.** A cusp form $f$ is an *eigenform* if and only if it is a simultaneous eigenvector for the entire Hecke algebra.

**Remark 4.124.** Suppose $f$ is an eigenform with $q$-expansion $f(q) = \sum_n a_n q^n$. Then it turns out that $T_n * f$ is $a_n$ up to the scalar $a_1$. In particular, if $a_1 = 0$, then $f = 0$; otherwise, we may normalize $f$ to have $a_1 = 1$, and then $T_n * f = a_n$.

We are now ready to interpret our Hecke operators.

**Proposition 4.125.** Fix a modular form $f$ of weight $k$ and full level. Then

$$(T_n * f)(E, v) = \sum_{\substack{\pi:\ E' \to E \\ \deg \pi = n}} f(E', v).$$

*Proof.* Quickly, we note that the pair $(E', v)$ makes sense because an isogeny $\pi$ is automatically an isomorphism on the level of the Lie algebras.

Anyway, the main point is to unwind everything to matrices. Given $(E, v)$, the data of an isogeny $E' \to E$ can be translated to lattices as follows: say $E = \mathbb{C}/\Lambda$ and $E' = \mathbb{C}/\Lambda'$, and then the isogeny of degree $n$ means that $\Lambda' \subseteq \Lambda$ with index $n$. However, once $\Lambda$ has been fixed, this collection is now in bijection with

$$\{g \in M_2(\mathbb{Z}) : \det g \in \{\pm n\}\}/\operatorname{GL}_2(\mathbb{Z}),$$

so one can find that everything unwinds as needed. ■

Next time, we will say something about $L$-functions. Here is a preview for modular forms.

**Definition 4.126.** Fix an eigenform $f$ with eigenvalues $\{a_n\}$. Then we define its $L$-function by

$$L(s, f) := \sum_{n=1}^{\infty} \frac{a_n}{n^s}.$$

It is not obvious that this sum has any good properties.

To say anything about this $L$-function, we relate it to the modular form.

**Lemma 4.127.** Fix an eigenform $f$ with eigenvalues $\{a_n\}$. Then

$$\int_{\mathbb{R}^+} f(iy) y^s \frac{dy}{y} = (2\pi)^{-s} \Gamma(s) L(s, f).$$

*Proof.* Direct calculation with $f(q) = \sum_n a_n q^n$ (which holds after normalizing). ■

**Theorem 4.128.** Fix an eigenform $f$. Then $L(s, f)$ extends to an entire function and admits a functional equation.

*Sketch.* One can argue following Riemann: the functional equation $f(i/y) = f(-1/(iy)) = (iy)^k f(iy)$ can be used to relate the integral for $L(s, f)$ with $L(1 - s, f)$, so we receive a functional equation! Writing down the argument carefully enough gives the continuation as well. ■

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# List of Definitions


adéles, 7
admissible, 33, 57, 64
automorphic form, 64, 70
automorphic representation, 65

central character, 73
cocompact, 17
contragredient, 33
cusp, 61
cusp form, 62
cuspidal, 68, 70

Dedeking $\zeta$-function, 21
density, 39
discrete series, 59

eigenform, 83

Fourier transform, 14

general linear group, 8
$(\mathfrak{g}, K)$-module, 54
global field, 5
global Hecke algebra, 78
global integral, 23

Hecke algebra, 34
Hecke character, 9
height, 64

idéle class character, 22
idempotent algebra, 79
induction, 39
infinitesimal character, 58

$K$-finite, 53

Langlands dual, 49
Lie group, 51
local integral, 25, 28

modular character, 40
modular form, 61

parabolic, 37
parabolic induction, 38
place, 6
polygon, 75
Pontryagin dual, 12
principal series, 58

quadratic Casimir, 56

residue, 19
restricted tensor product, 77
root datum, 49

Schwartz, 16, 17
semistable, 75
Siegel domain, 67
slope, 75
smooth, 31, 32, 36, 53, 79
spherical, 42
strong approximation, 63

totally disconnected, 31

unitary, 57
unramified, 9, 23